



FUNDAÇÃO
UNIVERSIDADE
FEDERAL DE
MATO GROSSO DO SUL

Evolução fenotípica e Métodos Filogenéticos Comparativos

Diogo Borges Provete
Prof. Adjunto A

Campo Grande-MS
Agosto 2019

Avisos

- Disciplina no Moodle
 - Slides
 - Scripts
 - Material leitura/links
 - Aula seguinte liberada no fim da tarde (pegar script e materiais)
- Horários (8h30-12h = 14h-17h)
- Locais para almoçar
- Acesso à Internet no campus
- O que fazer na sexta à tarde?
- Apresentação alunos 😊
- Minha apresentação 😊
- Avaliação final => submetida via Moodle
- "Nivelamento" em R e inferência filogenética => evitemos monólogos monótonos: Perguntem!

Programa da disciplina

- Segunda
 - Introdução
 - Revisão rápida sobre métodos de inferência filogenética
 - Métodos para lidar com megaphylogenies
 - Base de dados de atributos e topologias
 - Desenho de estudos comparativos
 - Prática com R
- Terça
 - Modelos de evolução de atributos contínuos e categóricos
 - Sinal filogenético para atributos contínuos e categóricos
- Quarta
 - PGLS
 - Estimativa de estado ancestral caracteres contínuos e categóricos

Programa da disciplina

- Quinta
 - Evolução correlacionada (Pagel's method e threshold)
 - Ajuste matrizes VCV – rate matrix
 - Modelos multi-regime e multi-taxas – prática SURFACE
- Sexta pela manhã
 - Visualização de dados filogenéticos – demonstração (phytools e phylosignal)
 - Métodos para dados multivariados (bem resumido)
 - Dúvidas e discussão de projetos dos participantes

Aula 1

Introdução

Métodos de inferência filogenética

Bases de dados

Desenho de estudos comparativos

Outline

- Breve introdução histórico do desenvolvimento de métodos comparativos;
- métodos para reconstruir hipóteses filogenéticas (incluindo supertree, supermatrix)
- Darwinian e Rauniaerian shortfall.
- Iniciativas recentes parar suprir Darwinian shortfall: SUPERSMART, PASTIS, Open Tree of Life, congruification;
- Como trabalhar com base de dados online de atributos e filogenias.
- Perguntas comuns em estudos de adaptação;
- Como desenhar um estudo comparativo: phyndr e PhyloTargeting

MÉTODOS FILOGENÉTICOS COMPARATIVOS

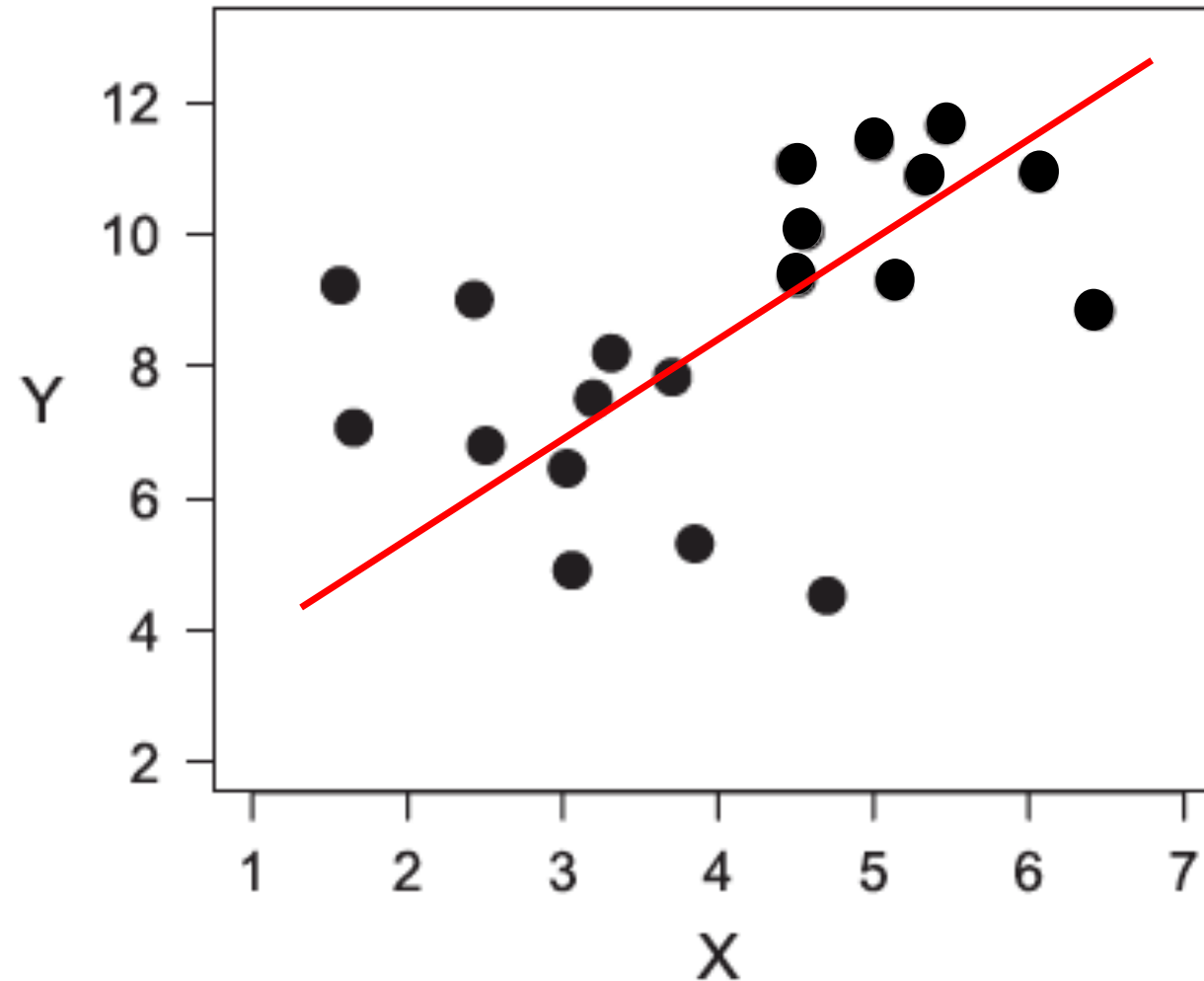
**O QUE SÃO? PARA QUE
SERVEM? DE ONDE VIERAM?**

**VOCÊ VAI VER TUDO AQUI,
HOJE NO GLOBO REP...OPS**

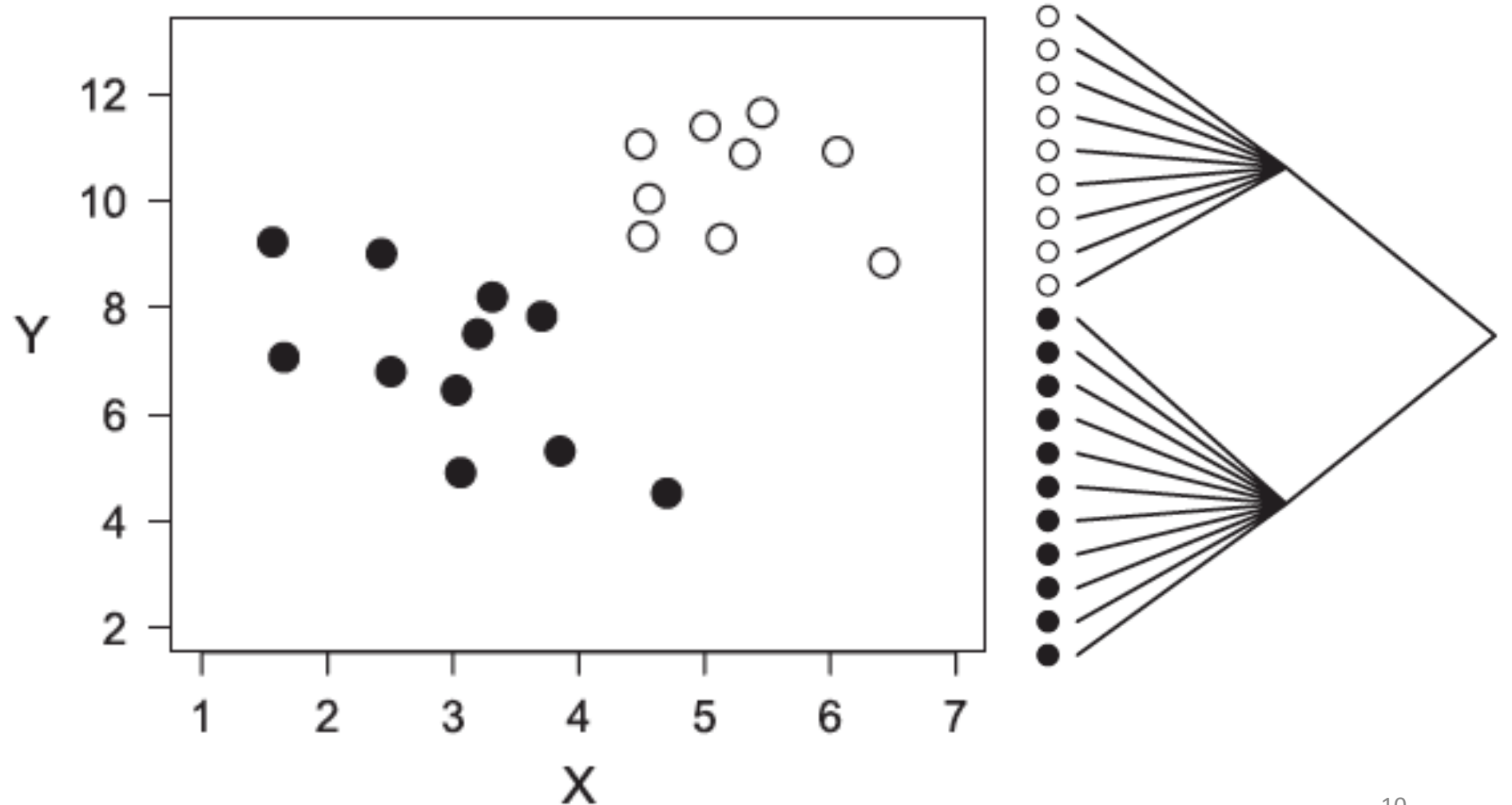
Introdução

- Conjunto de métodos estatísticos que usam a relação de parentesco entre as espécies para testar hipóteses evolutivas sobre seus atributos
- Diferença entre **obter** e **utilizar** uma filogenia
- Perguntas comuns
 - Qual o padrão filogenético de uma dada característica das espécies?
 - Sob que processo e em que velocidade essa característica evolui?
 - Qual a correlação dessa característica com outras ou com a variação ambiental?

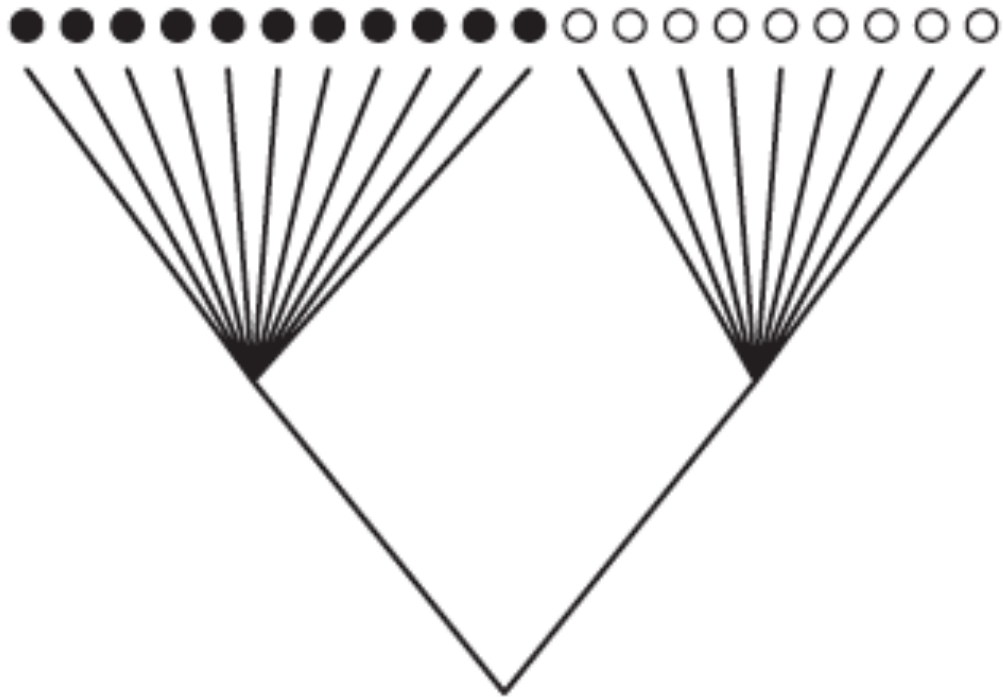
O problema do parentesco comum



O problema do parentesco comum

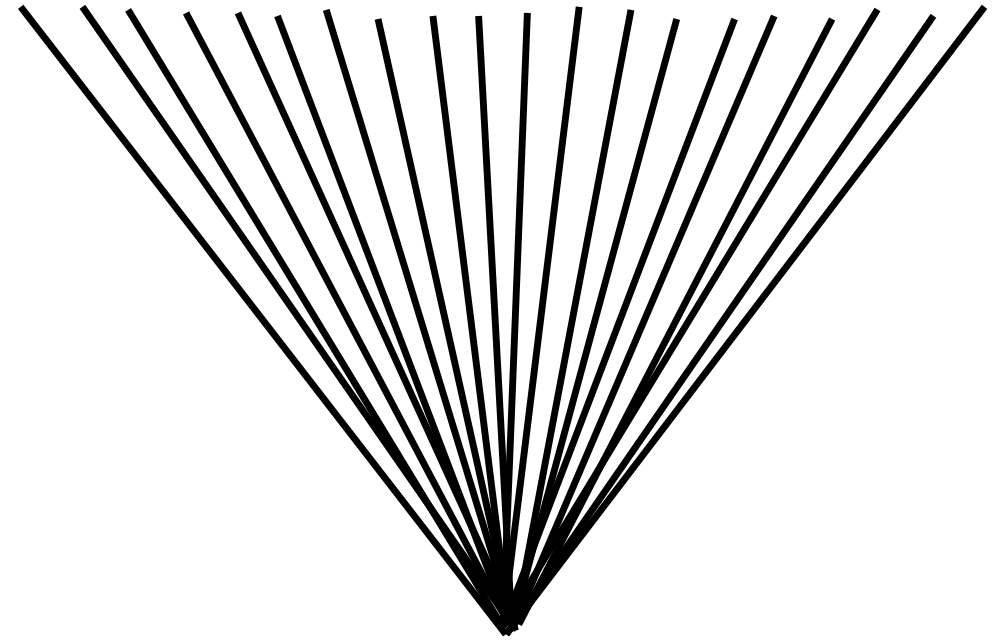


Espécies não são independentes



Filogenia "real"

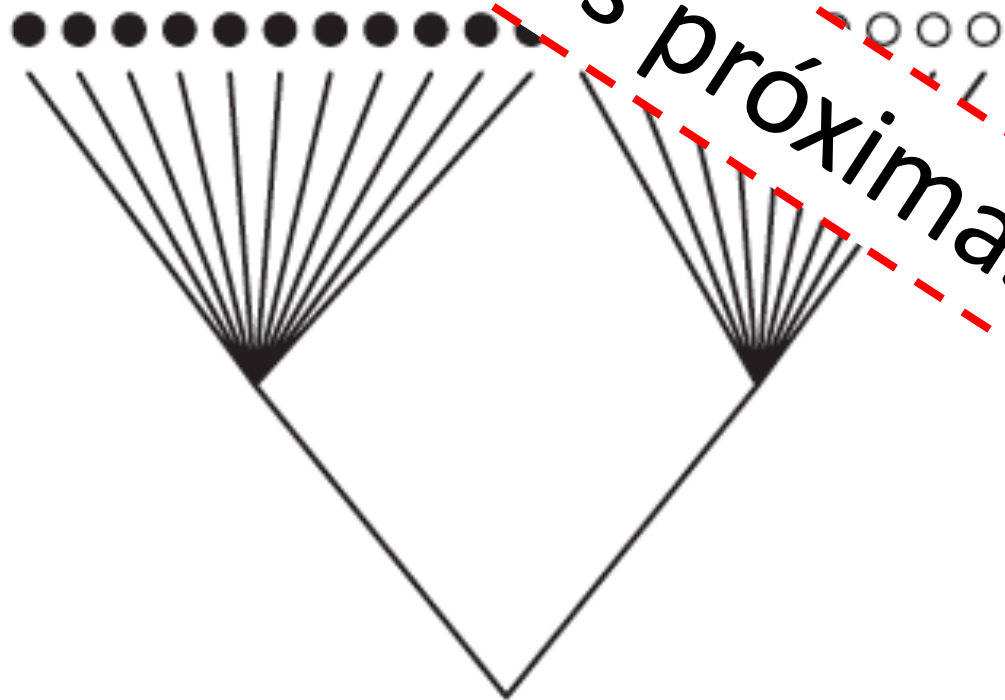
VS.



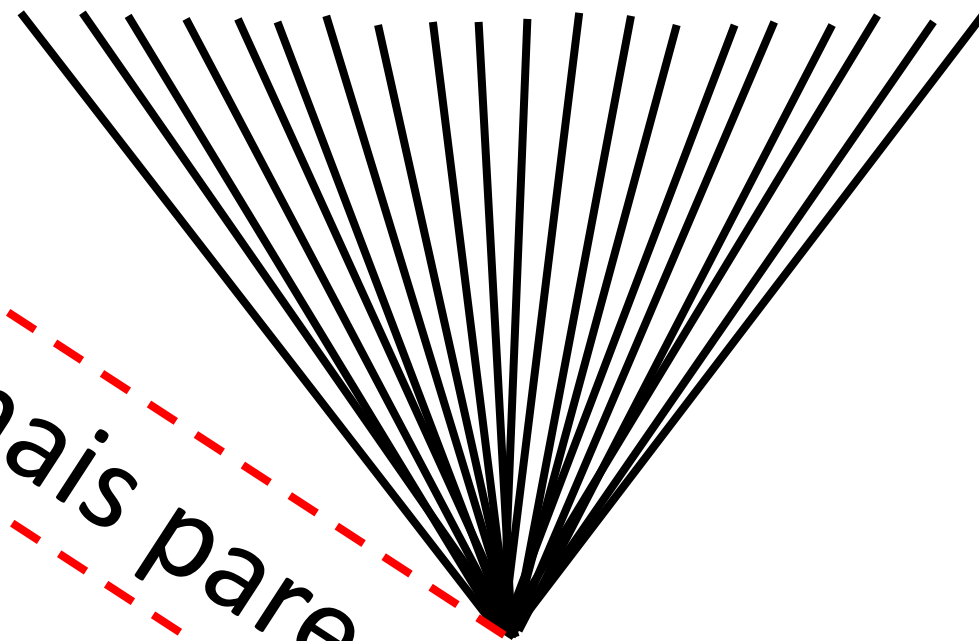
Filogenia em estrela

espécies não são independentes

Espécies próximas são mais parecidas



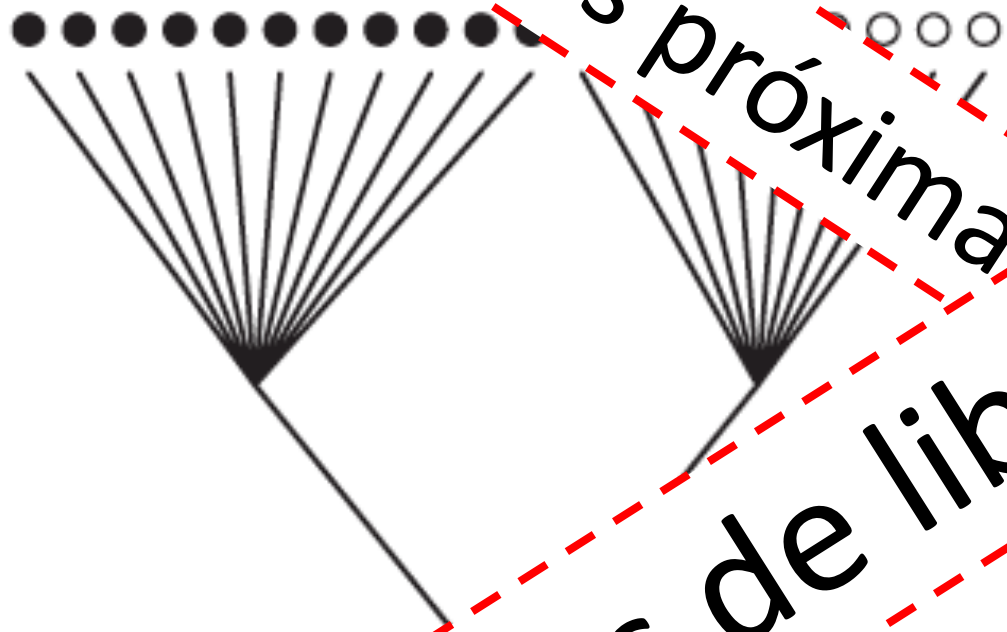
Filogenia "real"



Filogenia estrela

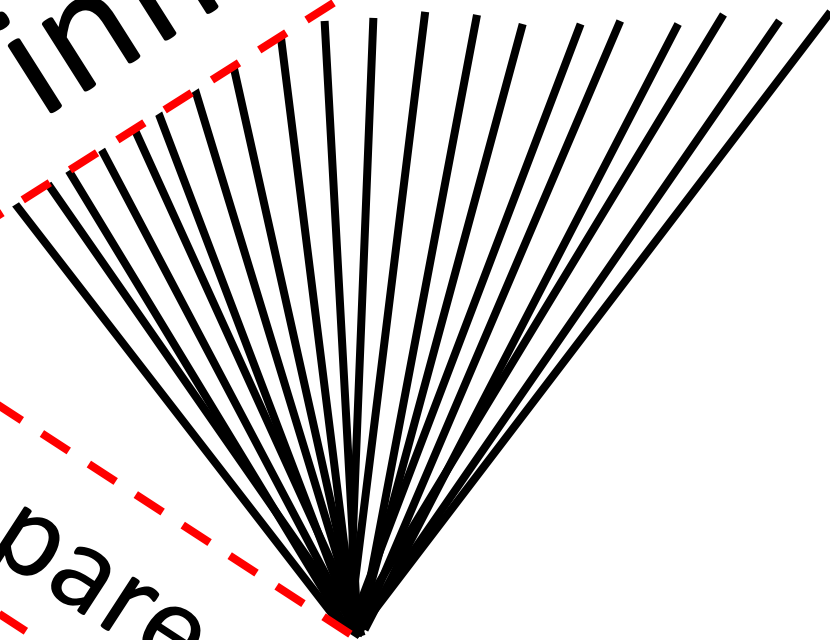
espécies não são independentes

Espécies próximas



Graus de liberdade inflados

mais parecidas



Filoge

estrela



Evolution, 39(6), 1985, pp. 1335–1351

THE QUANTITATIVE ASSESSMENT OF PHYLOGENETIC CONSTRAINTS IN COMPARATIVE ANALYSES: SEXUAL DIMORPHISM IN BODY WEIGHT AMONG PRIMATES

JAMES M. CHEVERUD

*Departments of Anthropology, Cell Biology & Anatomy, and Ecology & Evolutionary Biology,
Northwestern University, Evanston, IL 60201*

MALCOLM M. DOW

*Department of Anthropology and Program in Mathematical Methods in the Social Sciences,
Northwestern University, Evanston, IL 60201*

AND

WALTER LEUTENEGGER

Department of Anthropology, University of Wisconsin, Madison, WI 53706

1985



Vol. 125, No. 1

The American Naturalist

January 1985

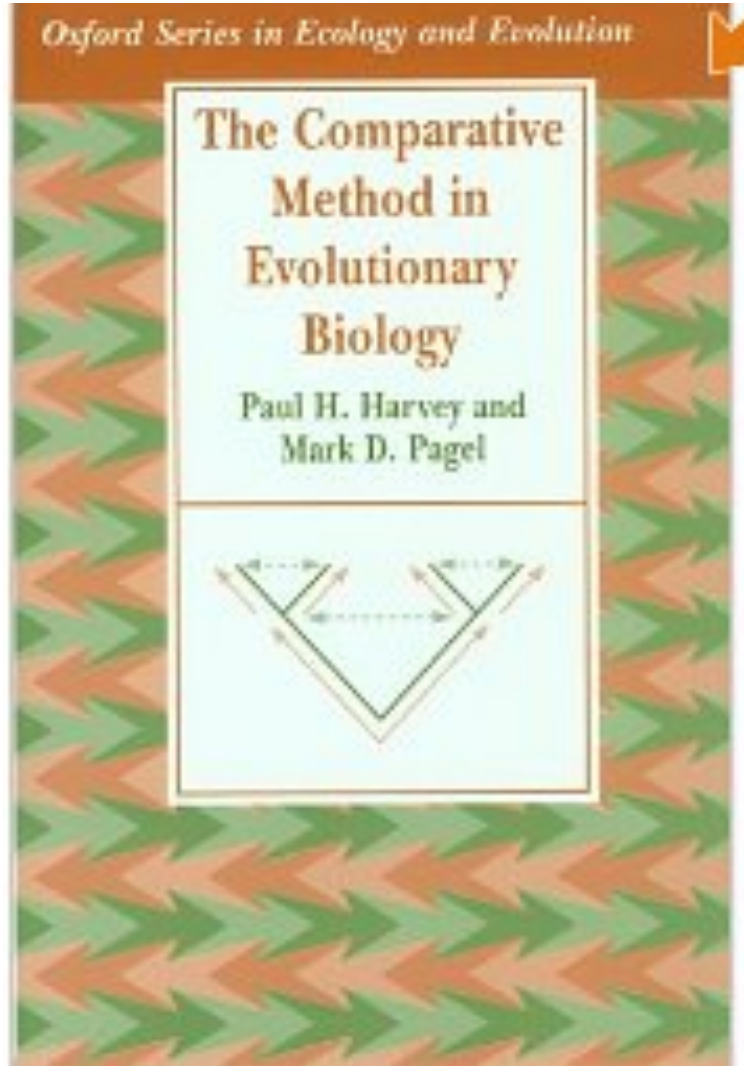
PHYLOGENIES AND THE COMPARATIVE METHOD

JOSEPH FELSENSTEIN

Department of Genetics SK-50, University of Washington, Seattle, Washington 98195

Submitted November 30, 1983; Accepted May 23, 1984

A origem dos métodos filogenéticos comparativos...



1991

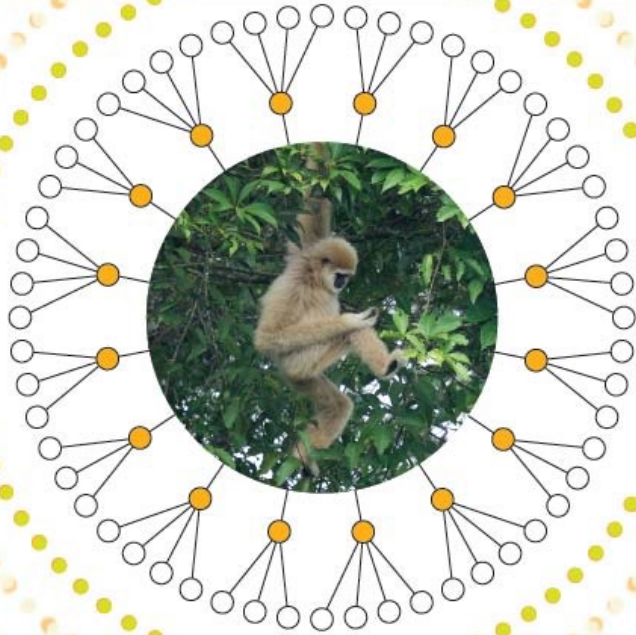


Paul Harvey



Mark Pagel

The **Comparative Approach**
in **Evolutionary Anthropology**
and **Biology**



Charles L. Nunn



Charles Nunn, Duke University

2011

László Zsolt Garamszegi
Editor

Modern Phylogenetic Comparative Methods and Their Application in Evolutionary Biology

Concepts and Practice

 Springer

2015



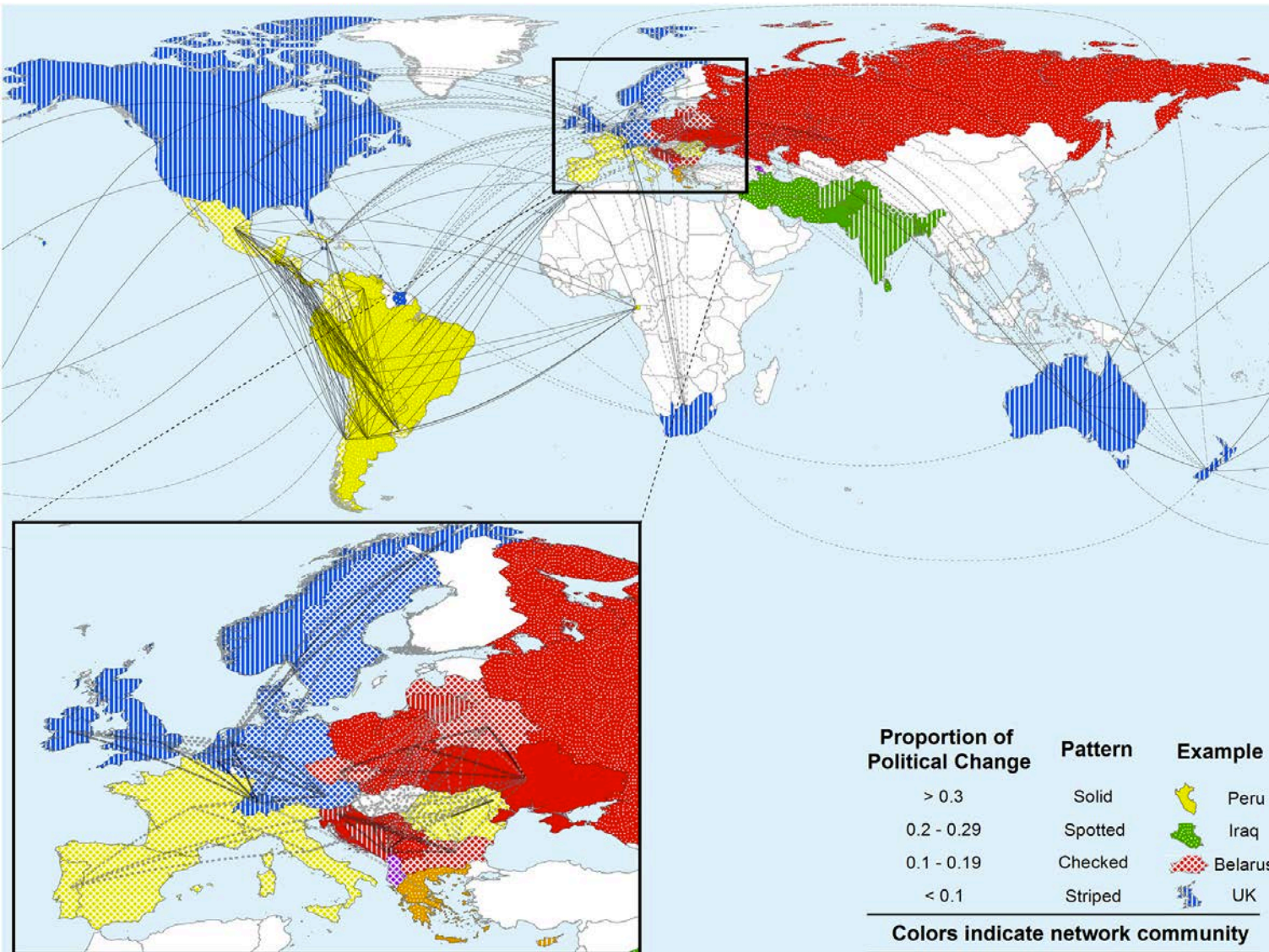
Sevilha, Espanha, Novembro 2014

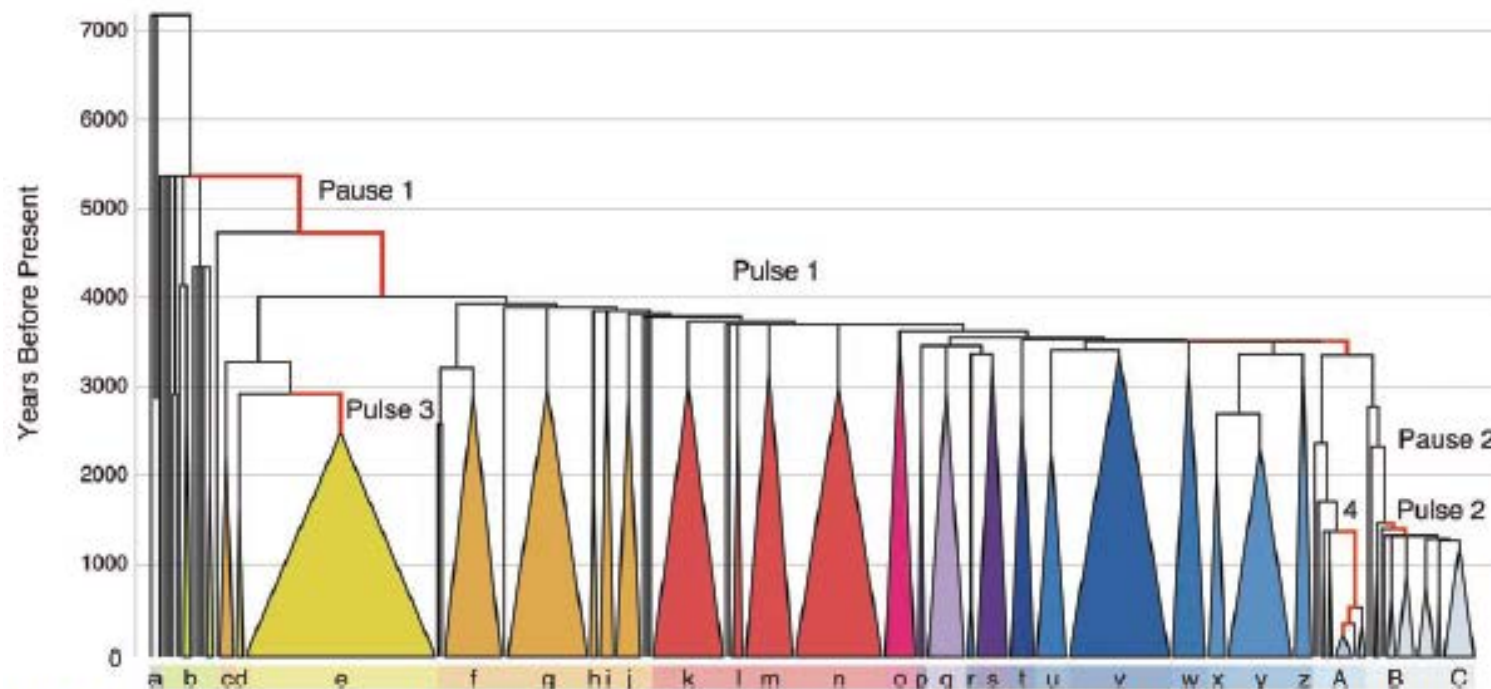
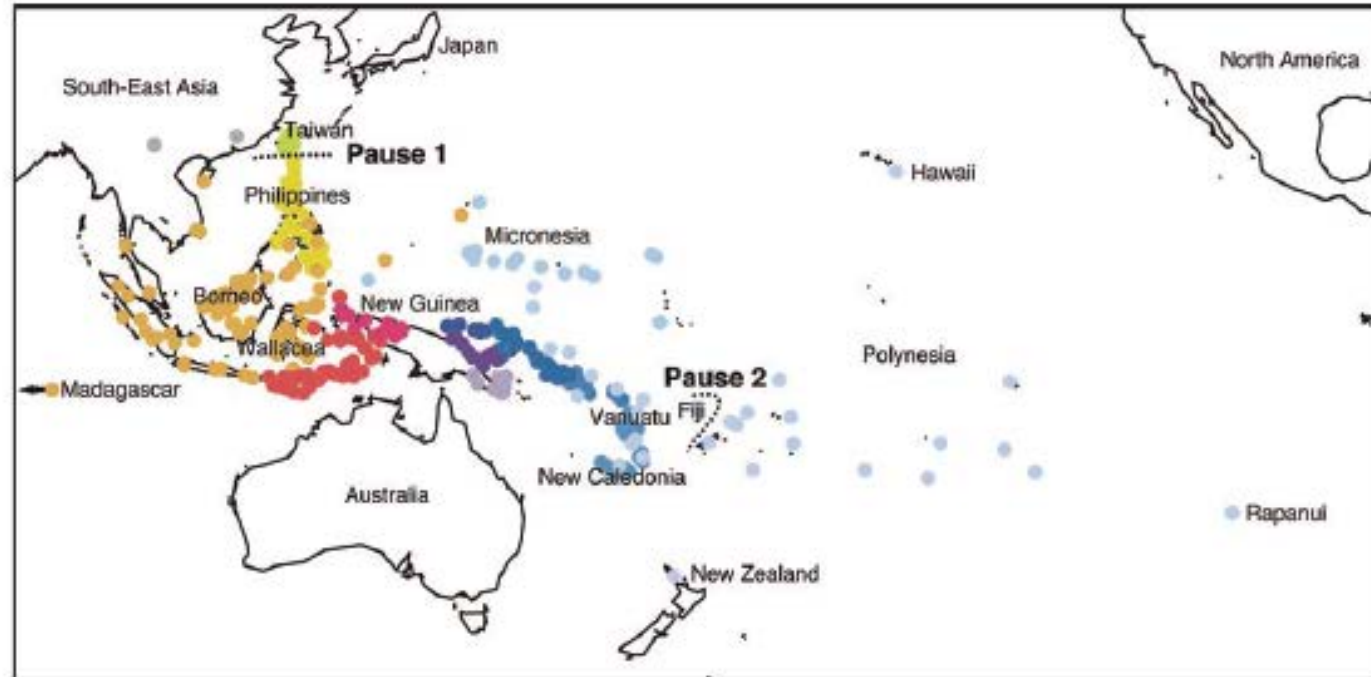
<http://www.mpcm-evolution.org/practice>

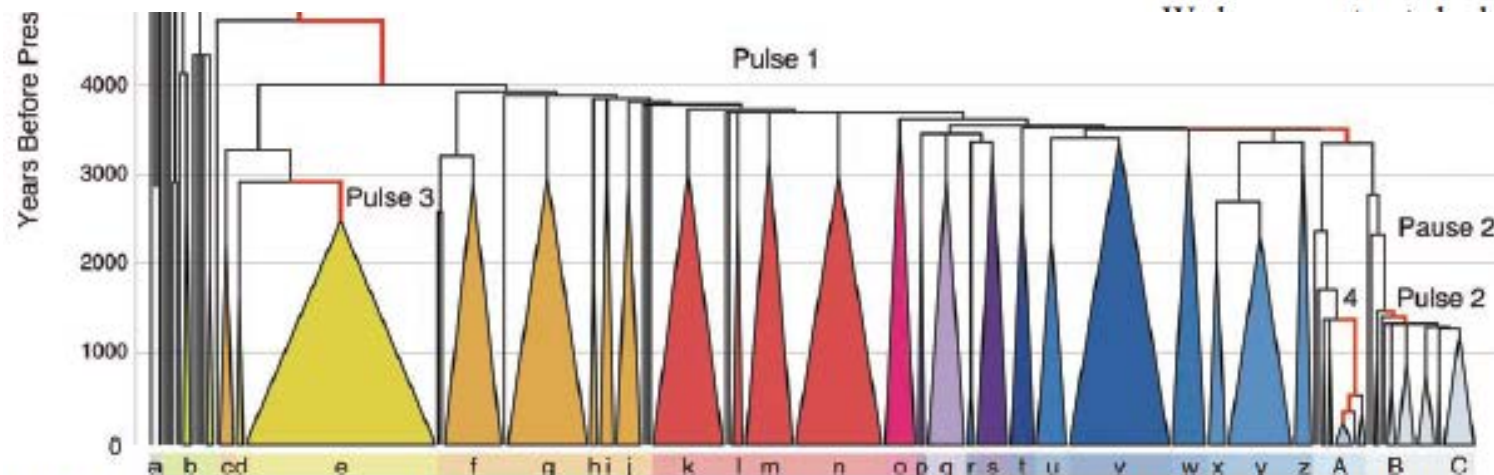
RESEARCH ARTICLE

Shared Cultural History as a Predictor of Political and Economic Changes among Nation States

Luke J. Matthews^{1,3*}, Sam Passmore², Paul M. Richard³, Russell D. Gray^{2,4}, Quentin D. Atkinson^{2,4}

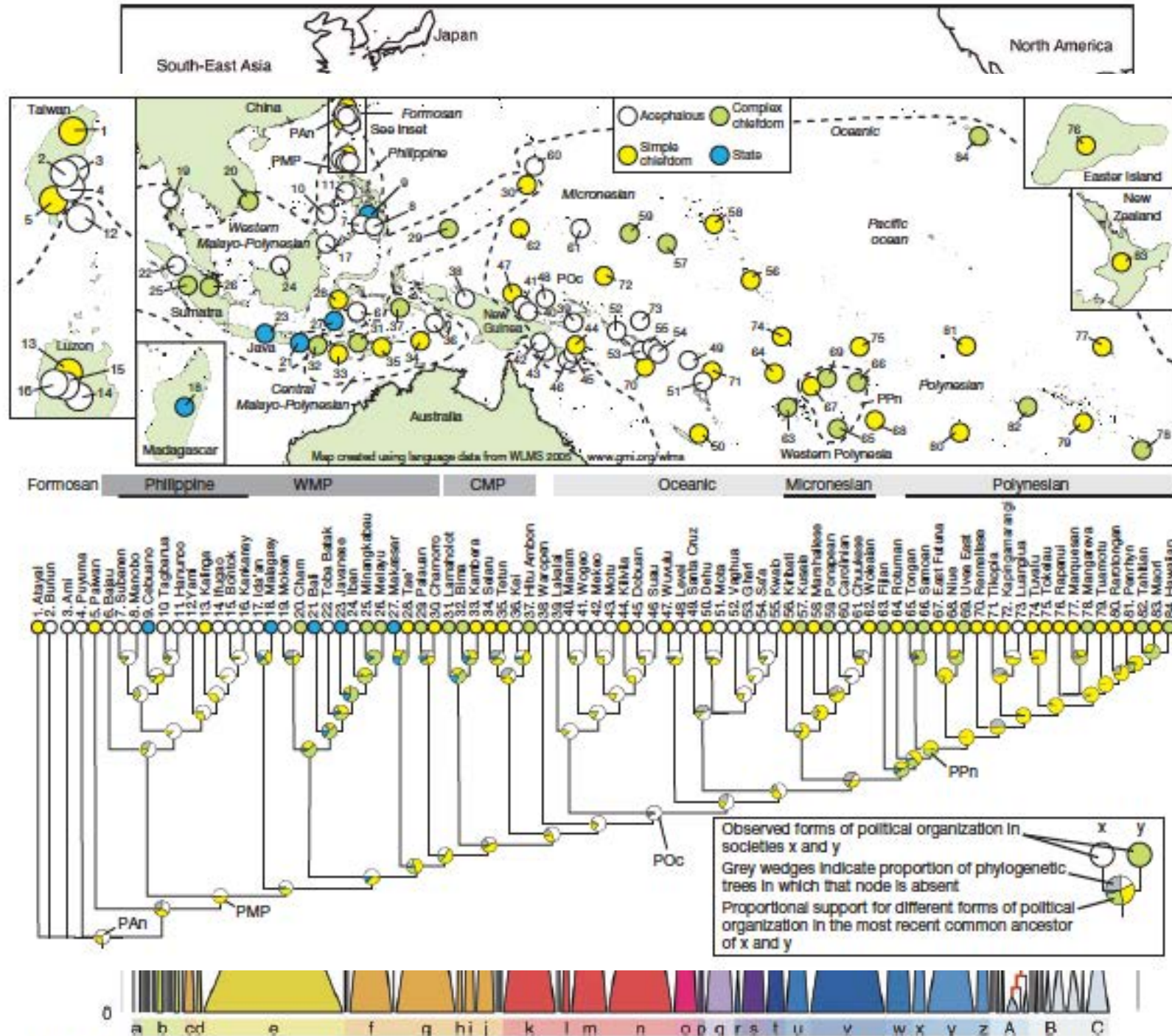






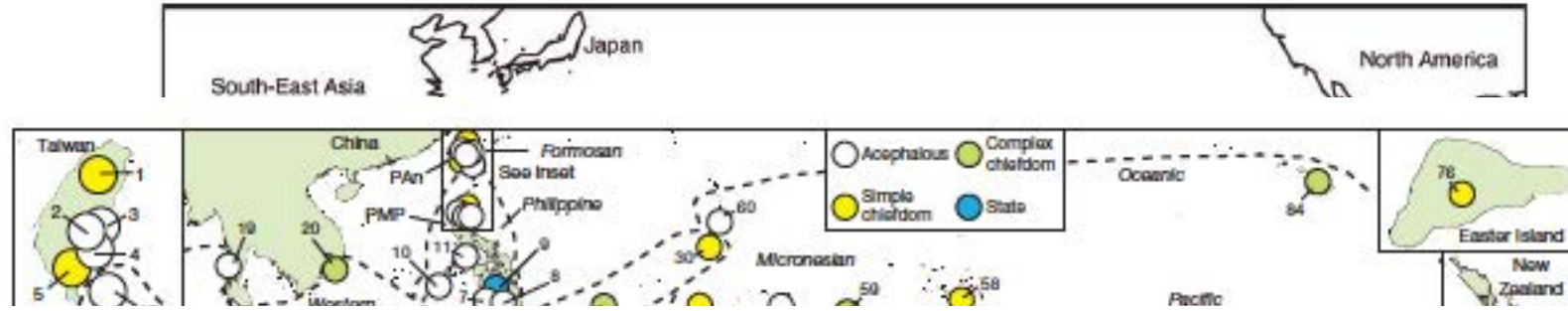
Language Expansion in the Pacific

R. D. Gray,¹ A.



TITLE

Gray and Jordan's analysis of Austronesian languages for the expansion of the seapause scenario used methods used to test the timing of the expansion of Austronesian language in the world, with evidence from Taiwan to Easter Island.



ARTICLE

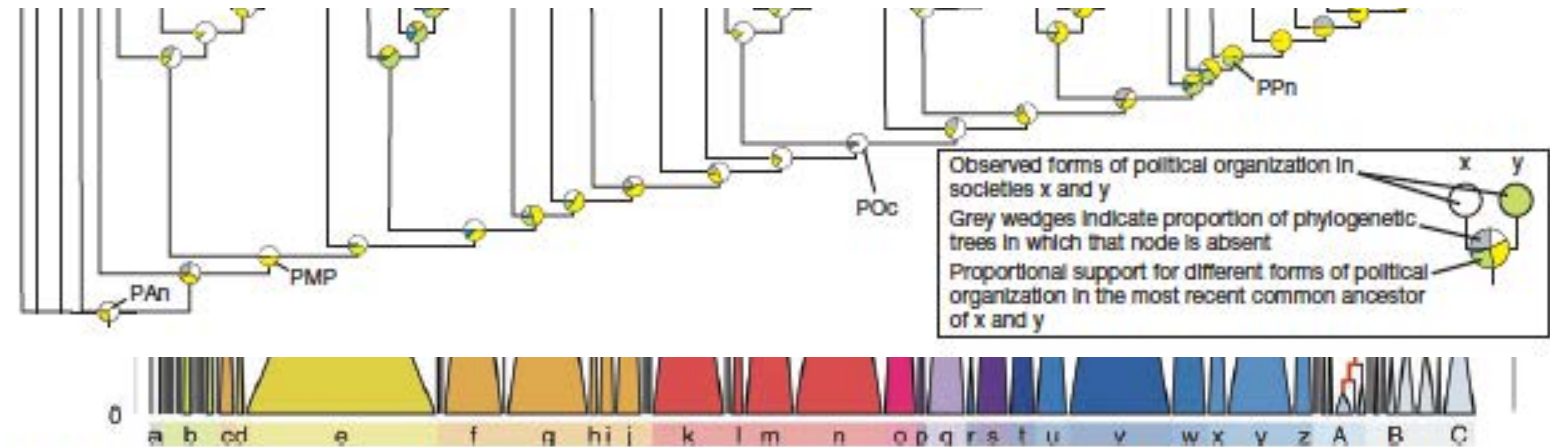
doi:10.1038/nature09461

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is of Austrone-
or the expansion
-pause scenario
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esian language
the world, with
from Taiwan to
to Easter Island.

ARTICLE

Rise and fall of political complexity in island South-East Asia and the Pacific

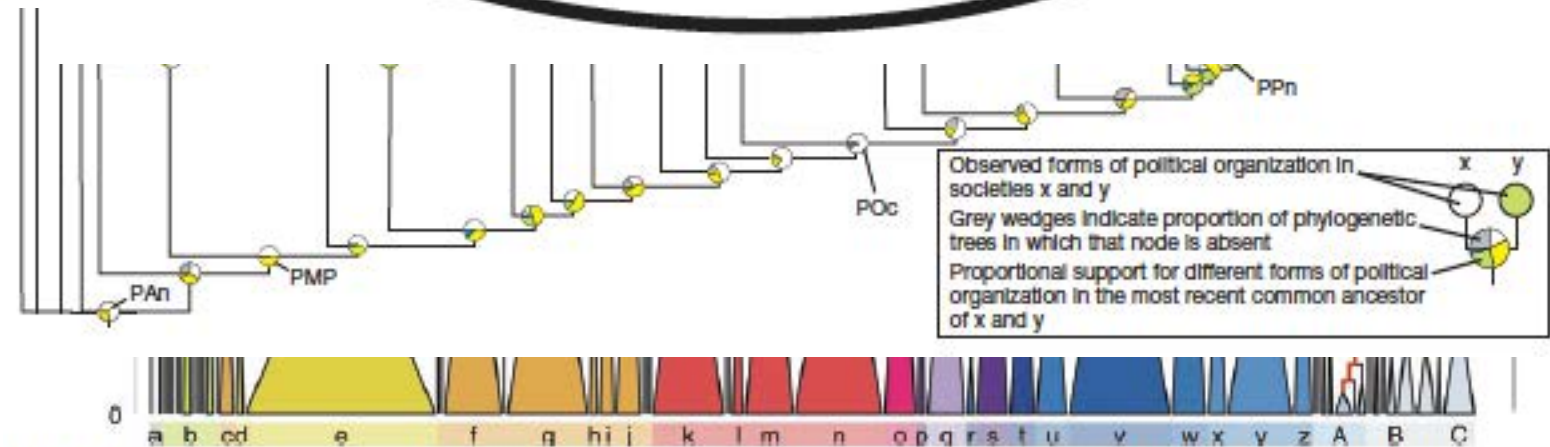
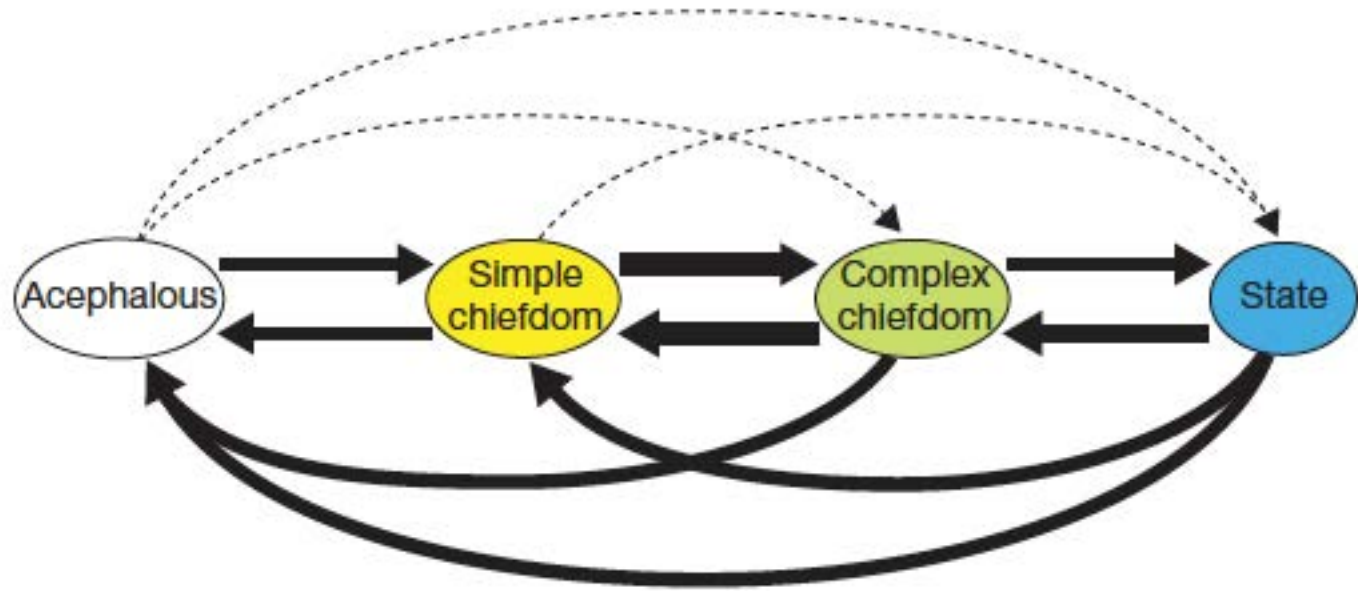
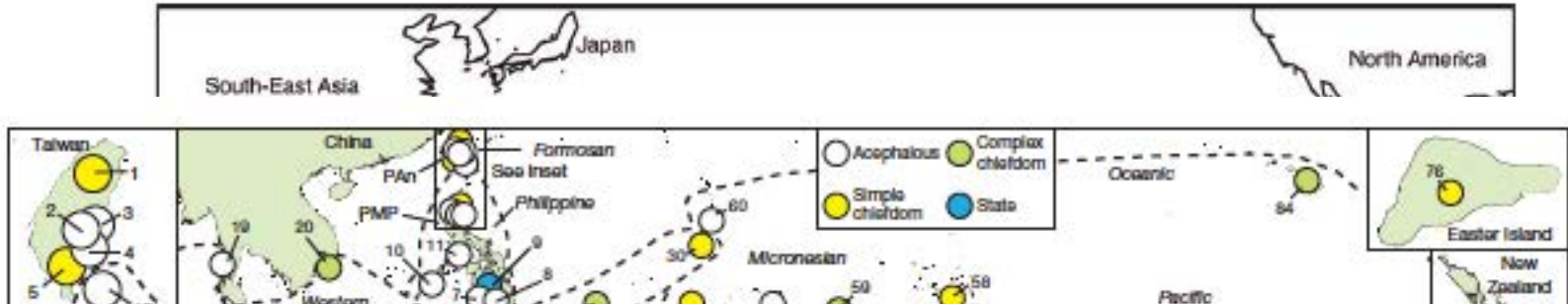
Thomas E. Currie^{1,2}, Simon J. Greenhill^{3,4}, Russell D. Gray³, Toshikazu Hasegawa¹ & Ruth Mace²



ARTICLE

Rise and fall of political organization in island Southeast Asia

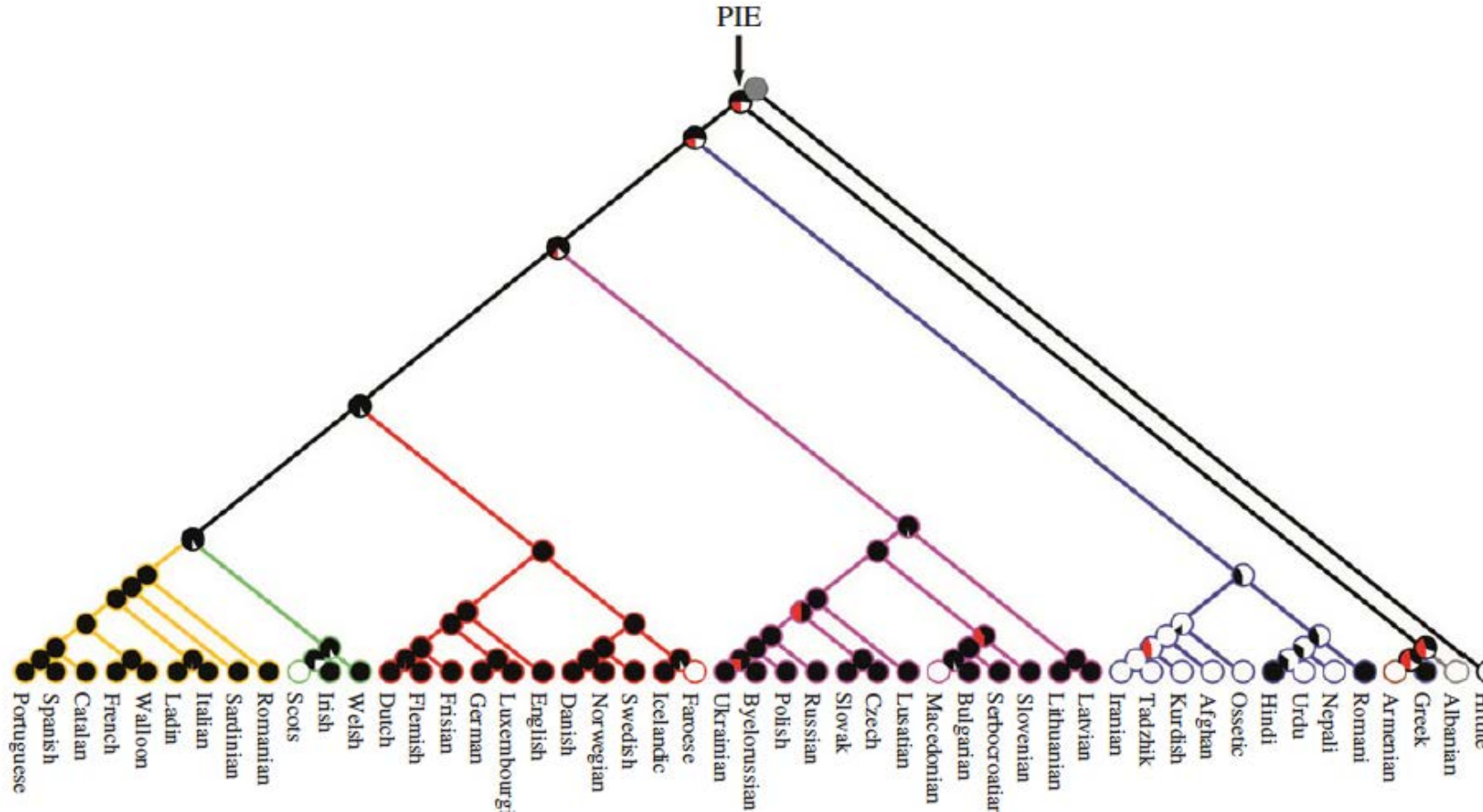
Thomas E. Currie^{1,2}, Simon J. Green¹



Currie and Jordan's analysis of Austronesian expansion or the expansion of the pause scenario. The methods used at the timing of the expansion. Austronesian language is the world, with Austronesian from Taiwan to Easter Island.

Comparative phylogenetic analyses uncover the ancient roots of Indo-European folktales

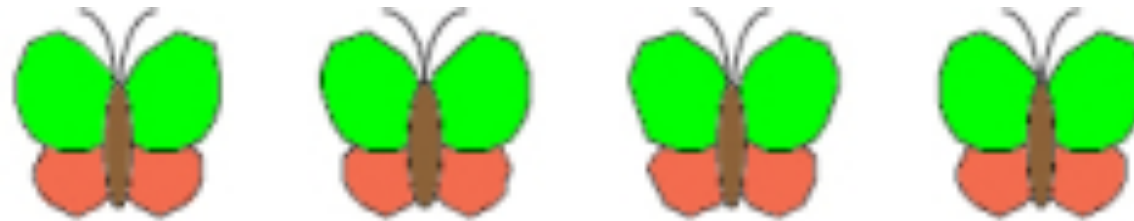
Sara Graça da Silva¹ and Jamshid J. Tehrani²



vertical processes of cultural inheritance. Moreover, we show that these oral traditions probably originated long before the emergence of the literary record, and find evidence that one tale ('The Smith and the Devil') can be traced back to the Bronze Age. On a broader level, the kinds of stories told in

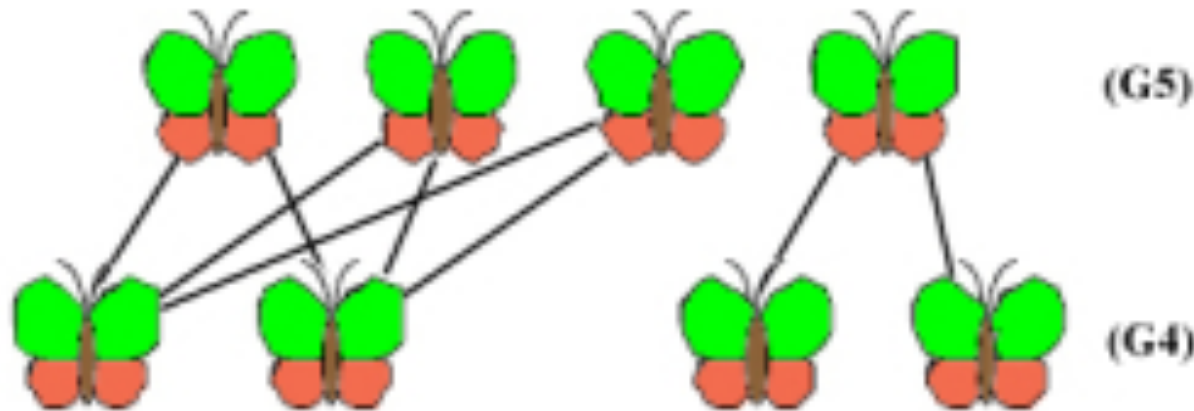
O que é uma filogenia? O que ela representa?

Indivíduos isolados



A

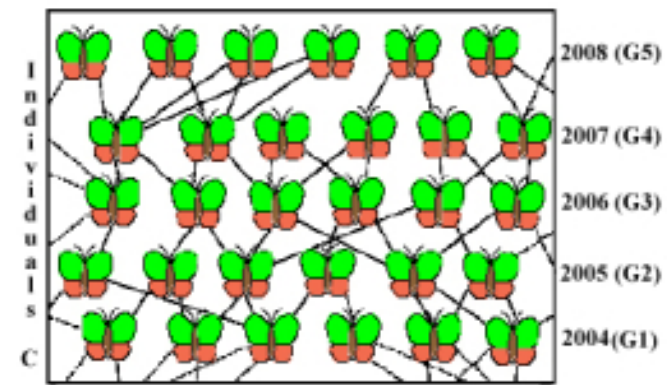
Four butterflies



B

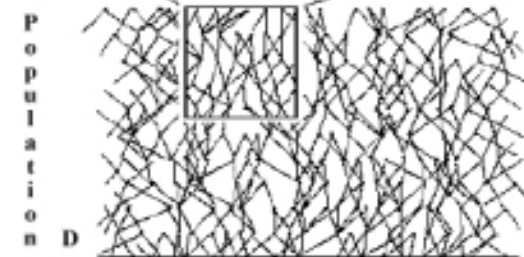
Four butterflies (top row),
connected to their parents (bottom row)

Família



Escala de Parentesco

Famílias



Populações



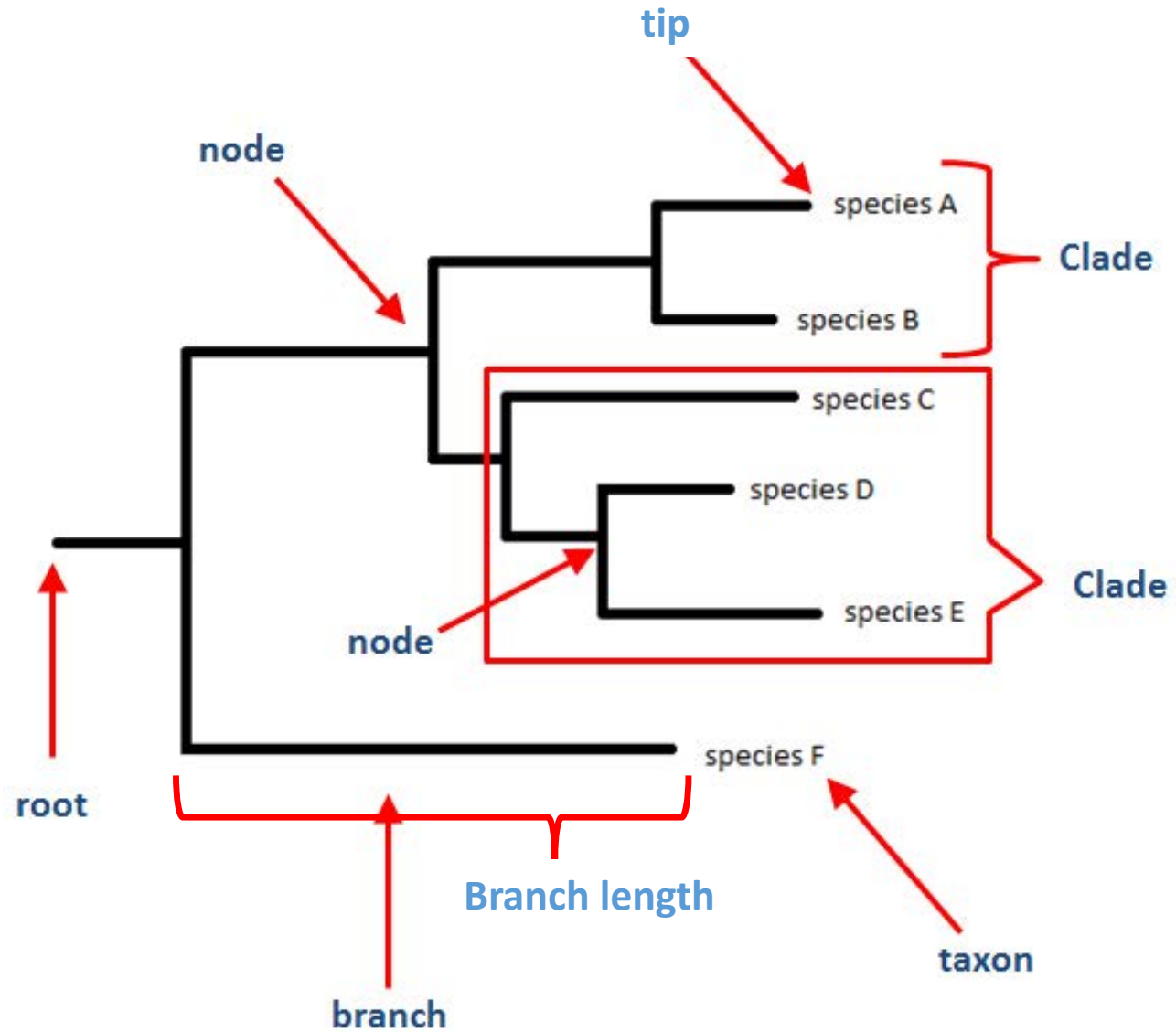
Espécies



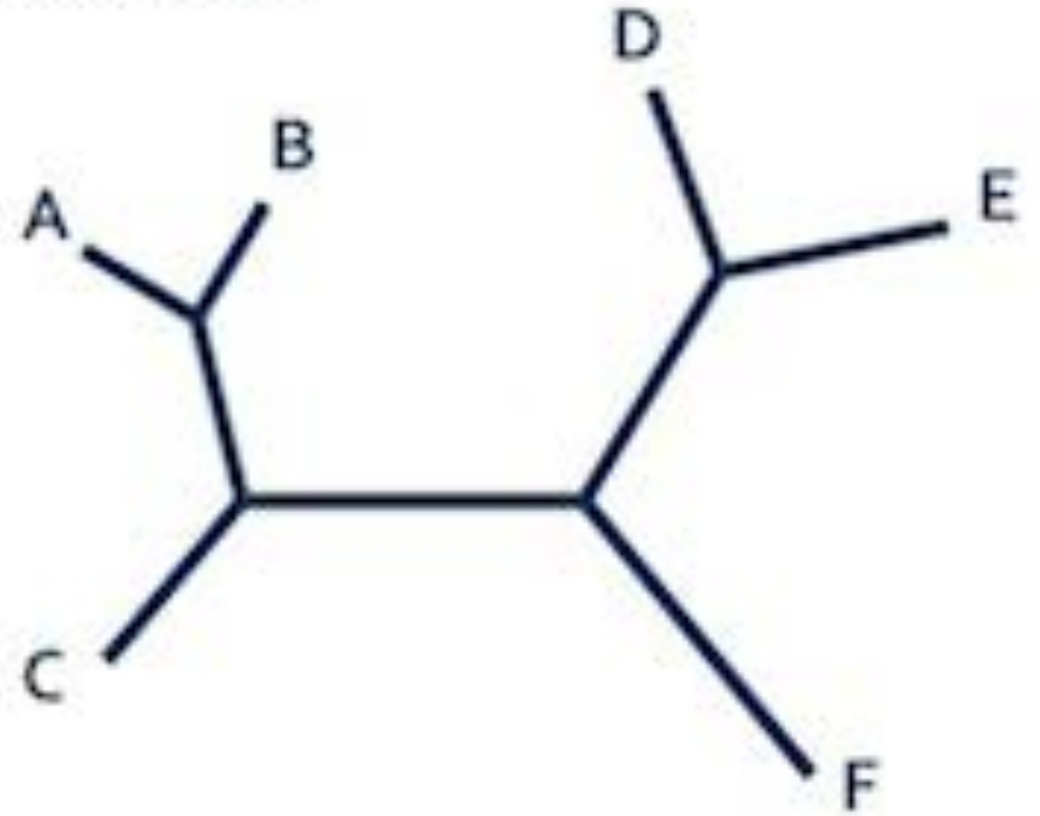
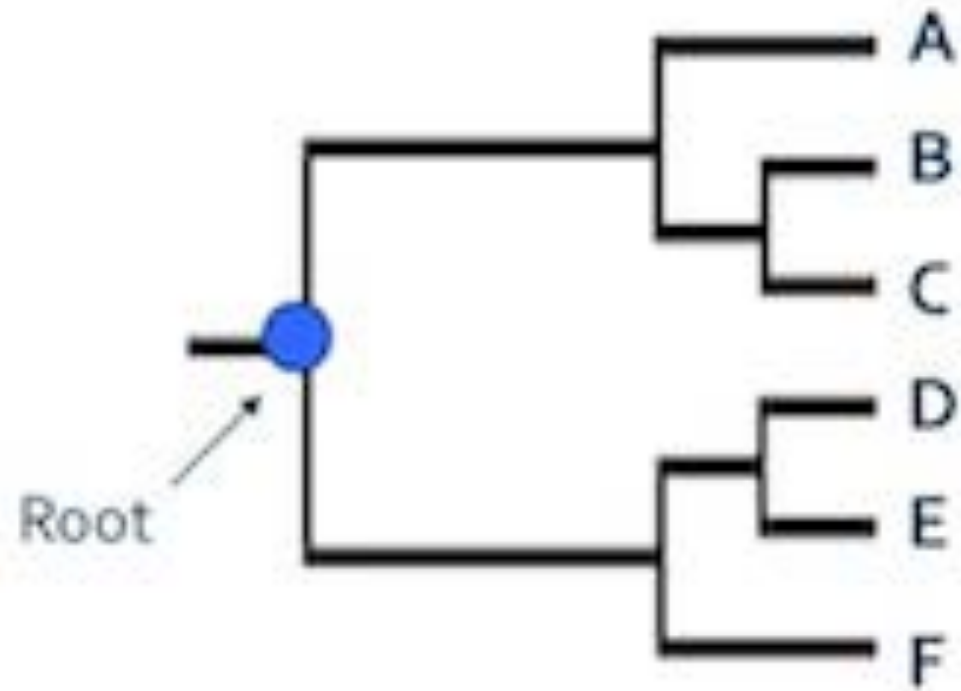
Filogenia (entre espécies)

Conceitos importantes

Parts of a phylogenetic tree

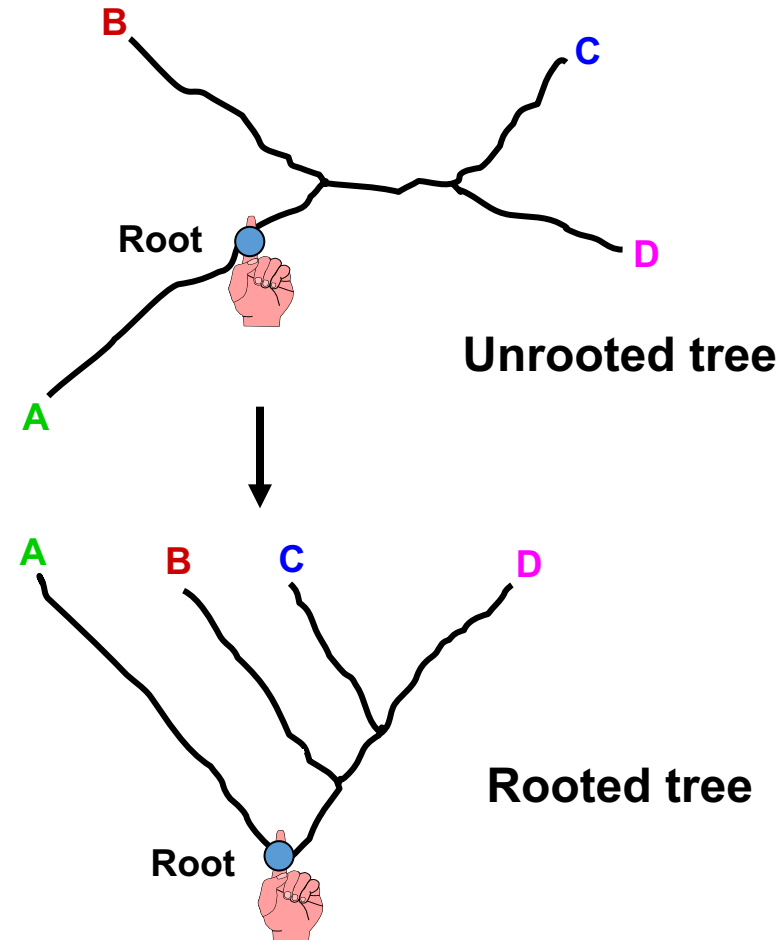


Rooted vs. unrooted trees

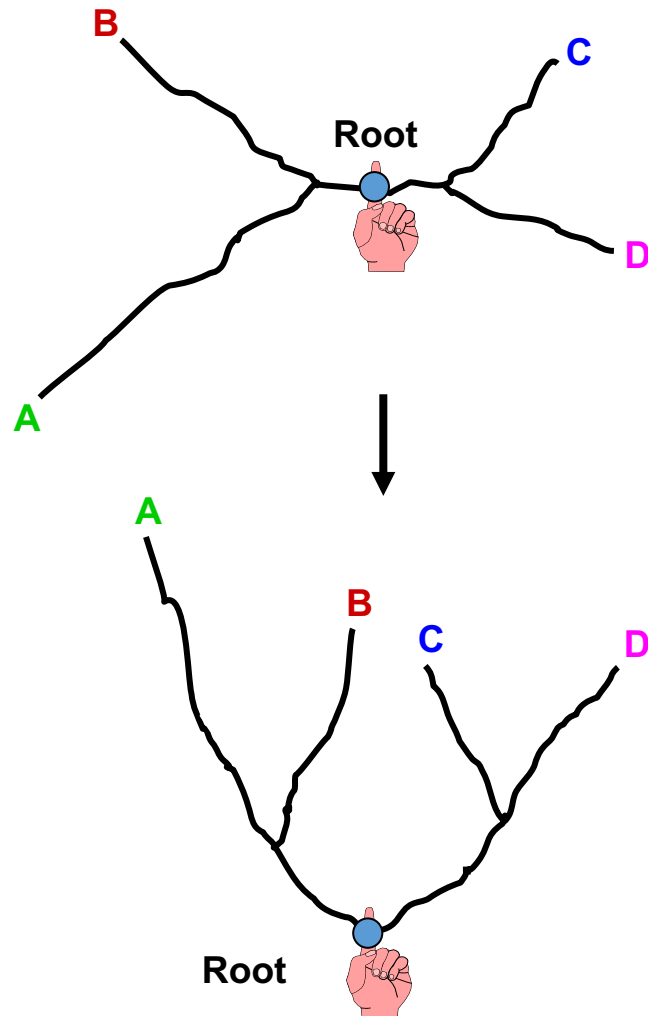


Enraizando a árvore

Taxon A não é mais aparentado com o B que C ou D.



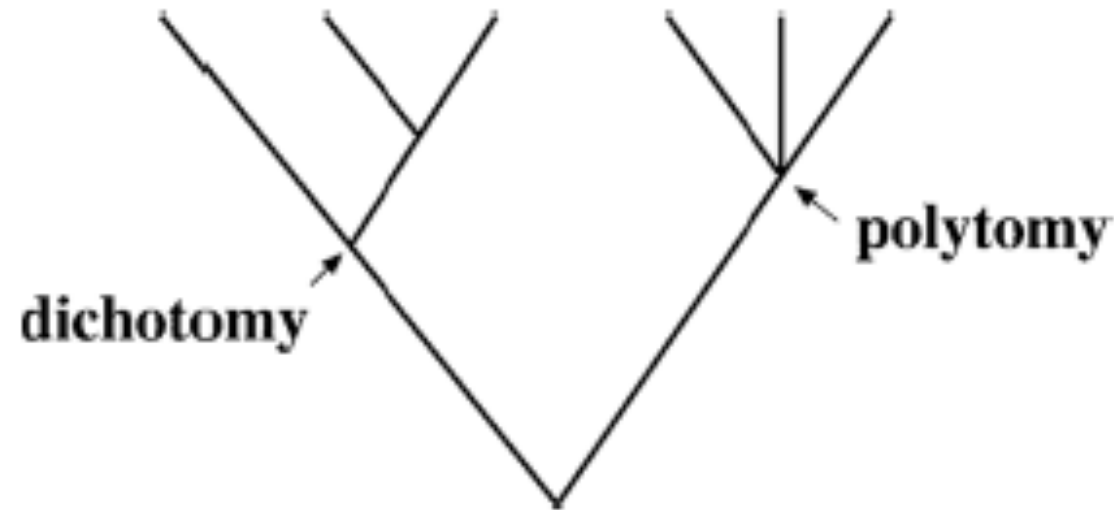
Enraizando em outra posição



Unrooted tree

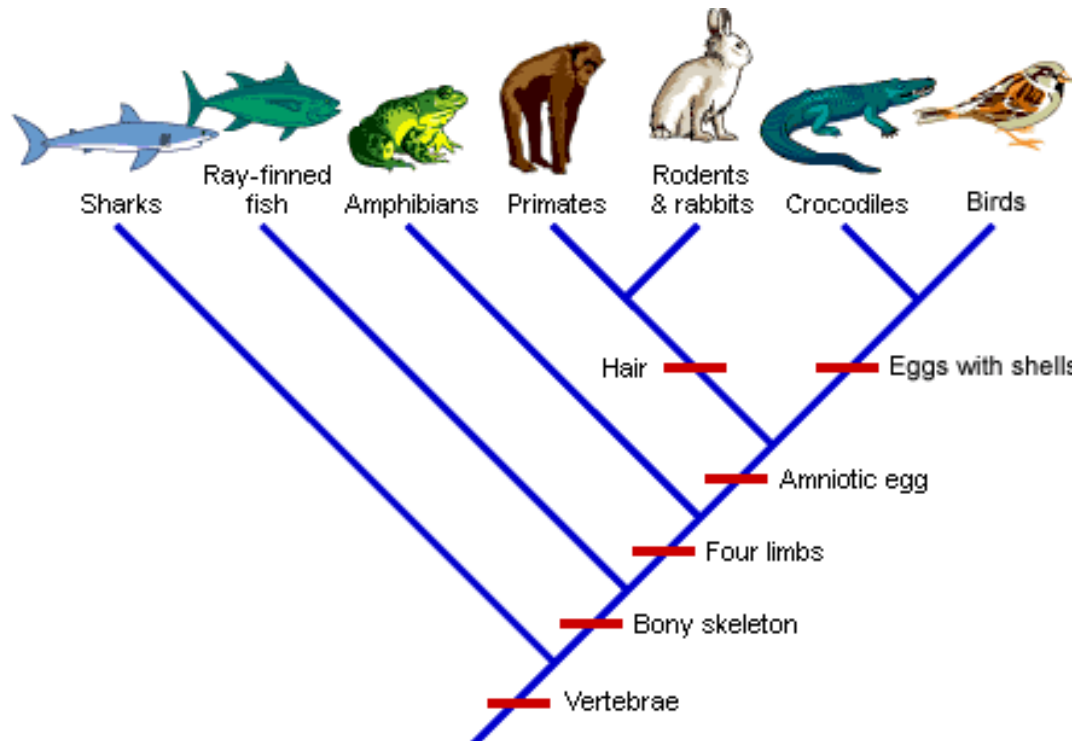
Rooted tree

Agora o taxon A é mais próximo de B, e ambos são grupo irmão de C e D.



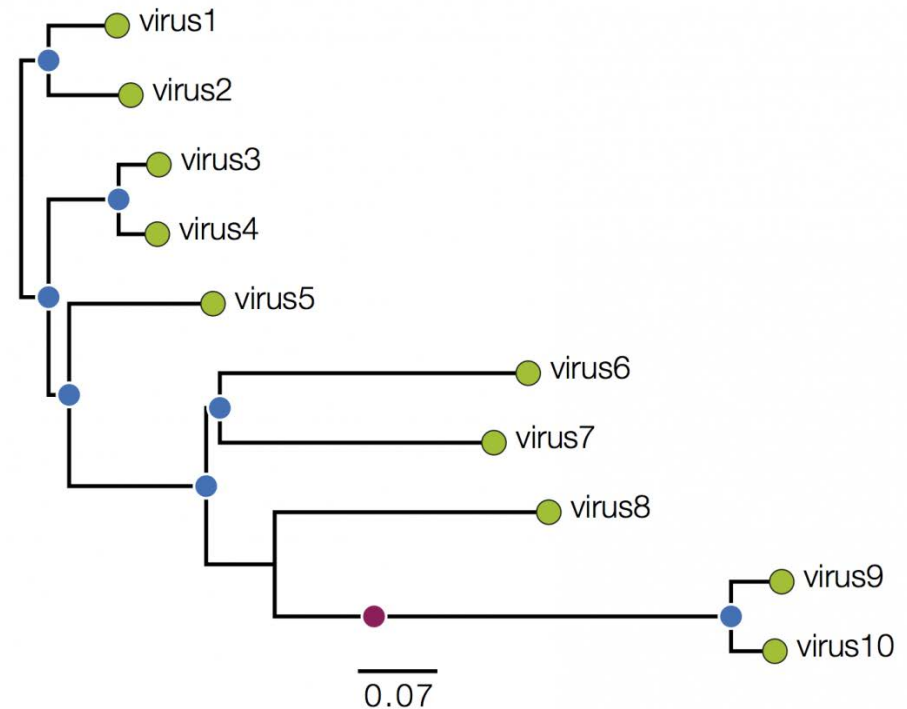
- **Politomia** – quando um nó (ancestral comum) dá origem a mais de 2 descendentes
 - Soft-polytomy: quando os dados disponíveis não conseguem discriminar a ordem de divergência, devido à rápida divergência
 - Hard-polytomy: quando de fato a politomia é real, usualmente representando uma "explosão" radiativa de espécies

Tipos de representação de árvores



Cladograma

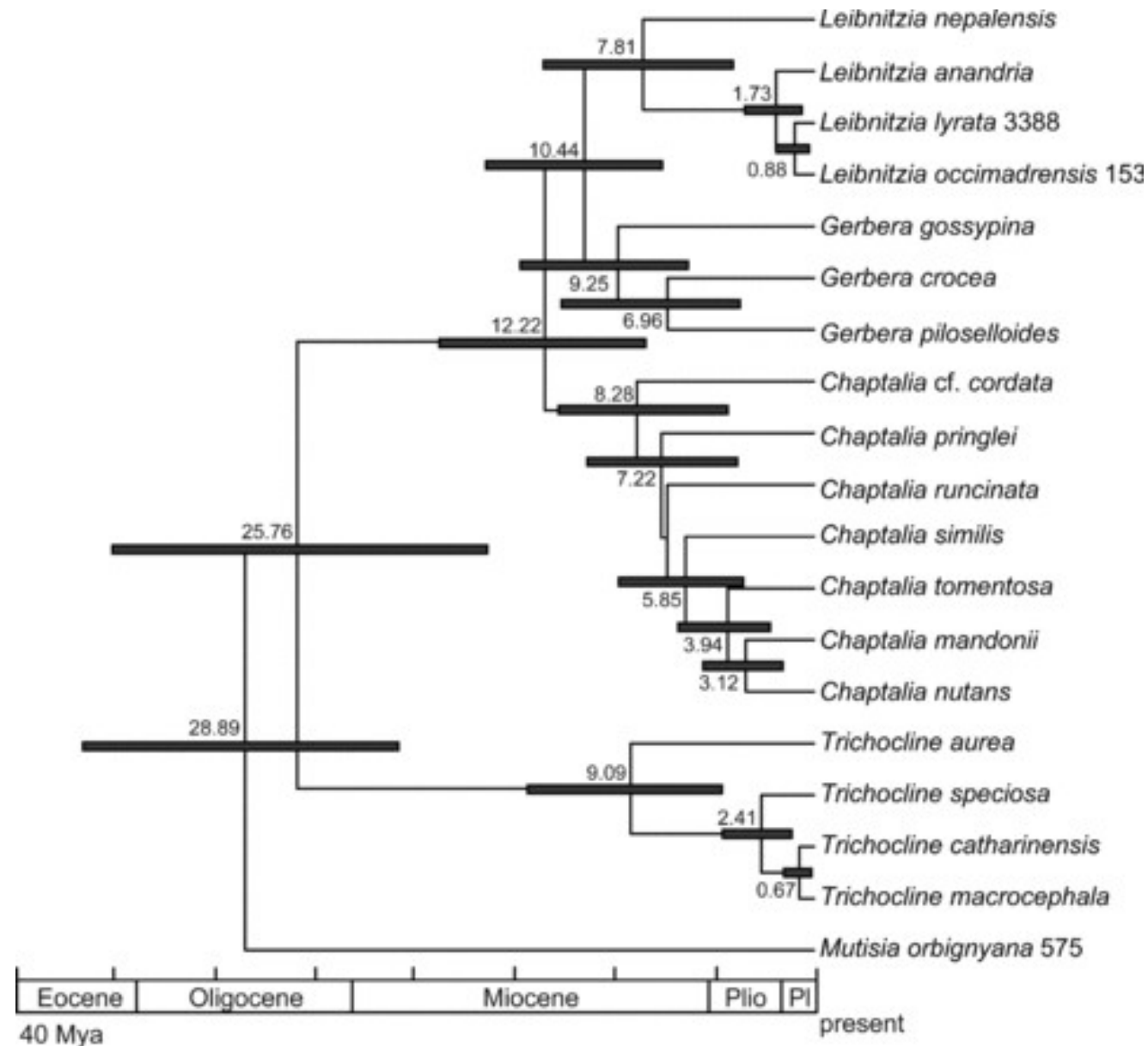
Enfatiza relações entre tips e
opcionalmente o compartilhamento de
caracteres derivados (sinapomorfias).
Comprimentos de ramo não importam



Filograma

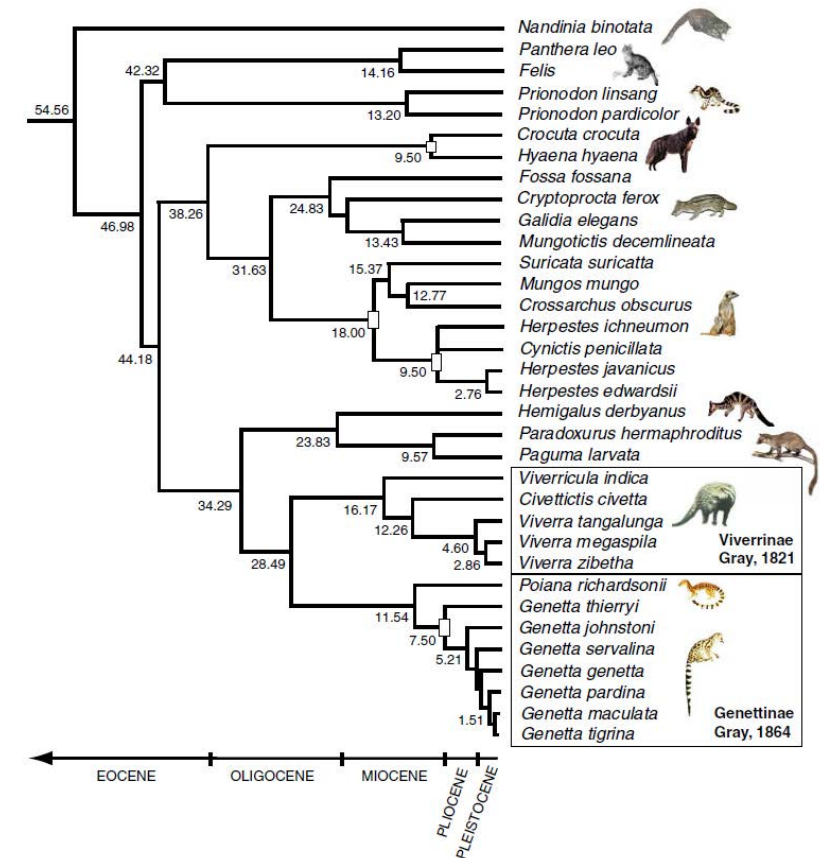
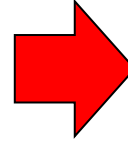
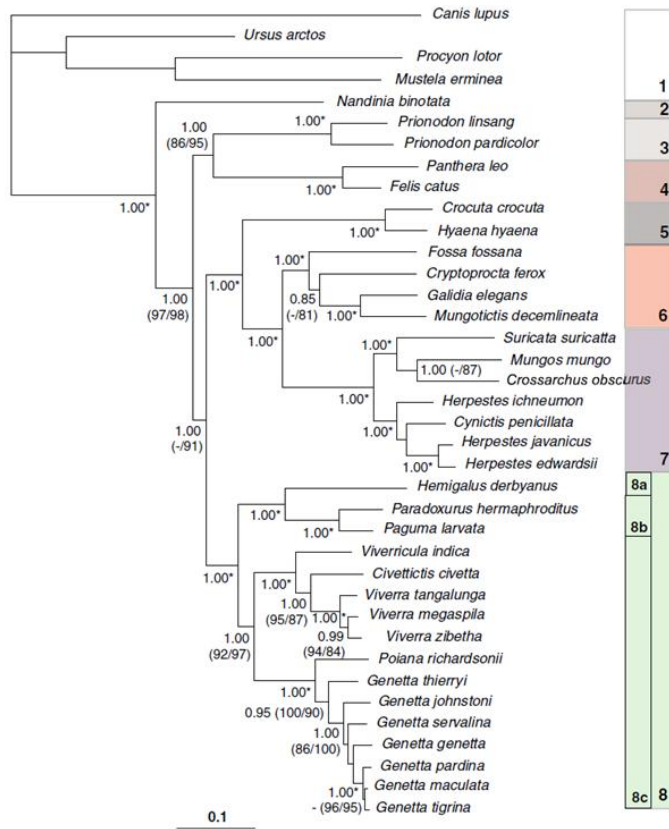
Comprimento de ramo refletem
divergência entre espécies

Tipos de representação de árvores



Cronograma

Comprimentos de ramo representam tempo de divergência em Milhões de anos



Filograma não ultramétrica

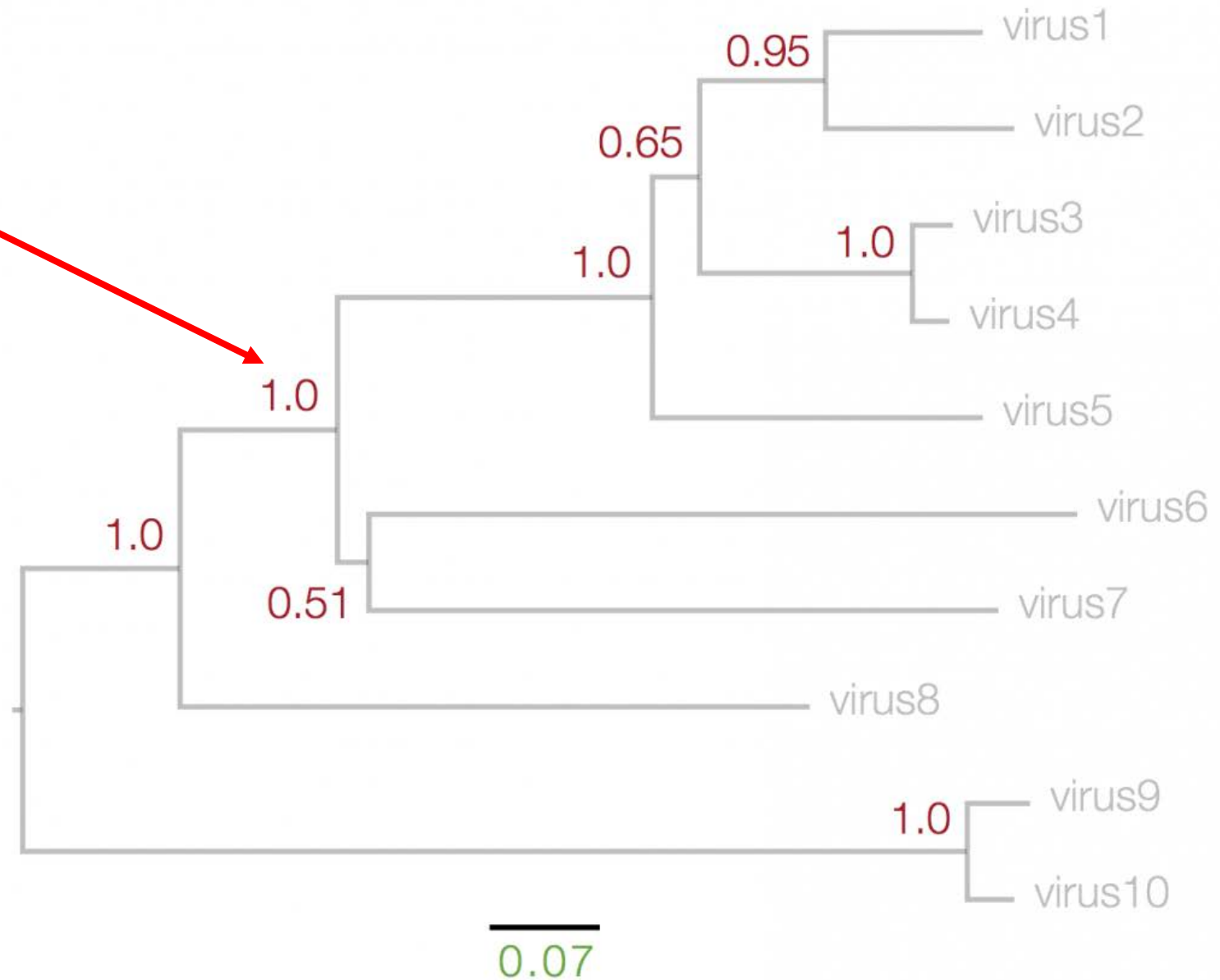
Árvore "ultramétrica"

Transformação de divergência (em termos de distâncias genéticas) em tempo utilizando um relógio molecular

Medida de suporte do nó

Existem várias formas, cada uma mais adequada pra um critério de otimização. P. ex. Bootstrap (usado em qualquer critério), Bremer (usada em MP), Probabilidade Posterior (BI)

Demonstram de maneira geral o grau de suporte para uma dada relação a partir do conjunto de dados usado pra reconstruir a filogenia

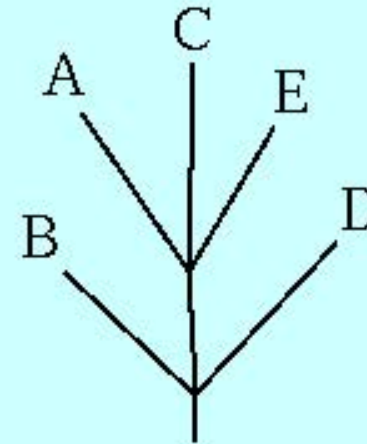


Formatos para descrever filogenias

The Newick tree format

Introduction

The Newick Standard for representing trees in computer-readable form makes use of the correspondence between trees and nested parentheses, a concept popularized by the mathematician [Arthur Cayley](#). If we have this rooted tree:



then in the tree file it is represented by the following sequence of printable characters:

```
(B, (A, C, E), D);
```



Examples

To help you understand this tree representation, here are some trees in the above form:

```
((raccoon:19.19959,bear:6.80041):0.84600,((sea_lion:11.99700, seal:12.00300):7.52973,((monkey:100.85930,cat:47.14069):20.59201,weasel:18.87953):2.09460):3.87382,dog:25.46154);  
  
(Bovine:0.69395,(Gibbon:0.36079,(Orang:0.33636,(Gorilla:0.17147,(Chimp:0.19268, Human:0.11927):0.08386):0.06124):0.15057):0.54939,Mouse:1.21460):0.10;  
  
(Bovine:0.69395,(Hylobates:0.36079,(Pongo:0.33636,(G._Gorilla:0.17147, (P._paniscus:0.19268,H._sapiens:0.11927):0.08386):0.06124):0.15057):0.54939,Rodent:1.21460);
```

Rooted and unrooted trees

In addition, the standard is representing a root when describing inferences in such cases. Here

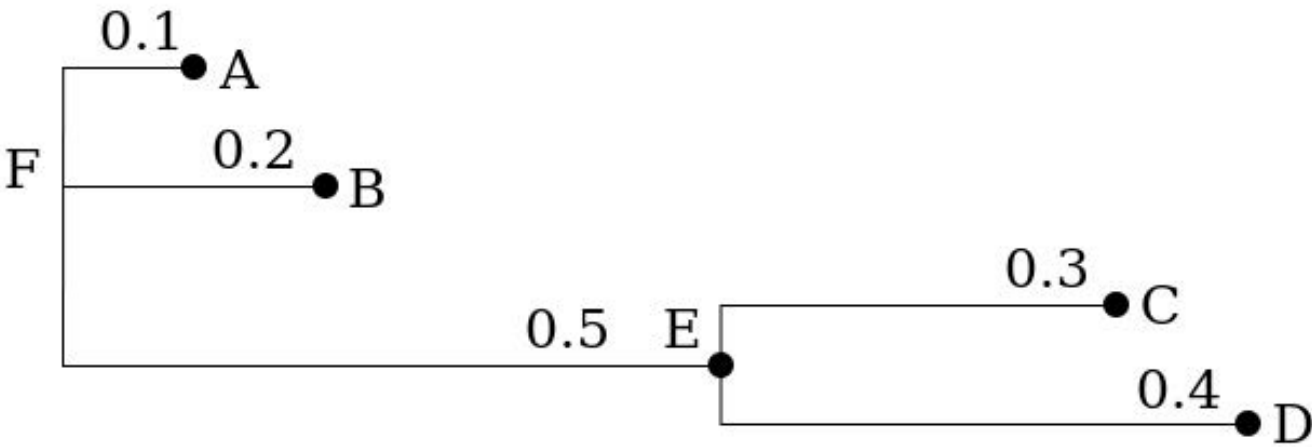
```
(B,(A,D),C);
```

would be the same unrooted tree as

```
(A,(B,C),D);
```

and as

```
((A,D),(C,B));
```

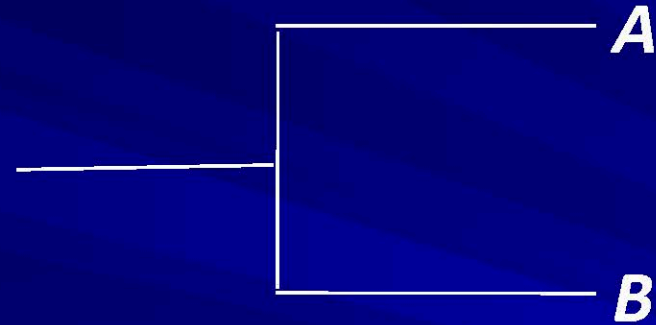


could be represented in Newick format in several ways

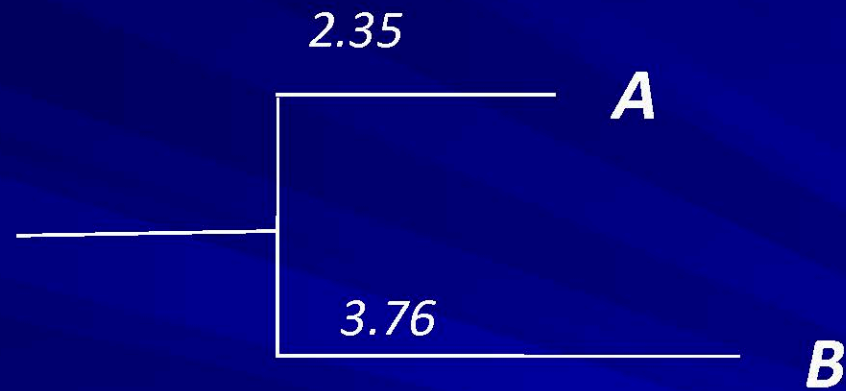
<code>(,,(,));</code>	<i>no nodes are named</i>
<code>(A,B,(C,D));</code>	<i>leaf nodes are named</i>
<code>(A,B,(C,D)E)F;</code>	<i>all nodes are named</i>
<code>(:0.1,:0.2,(:0.3,:0.4):0.5);</code>	<i>all but root node have a distance to parent</i>
<code>(:0.1,:0.2,(:0.3,:0.4):0.5):0.0;</code>	<i>all have a distance to parent</i>
<code>(A:0.1,B:0.2,(C:0.3,D:0.4):0.5);</code>	<i>distances and leaf names (popular)</i>
<code>(A:0.1,B:0.2,(C:0.3,D:0.4)E:0.5)F;</code>	<i>distances and all names</i>
<code>((B:0.2,(C:0.3,D:0.4)E:0.5)F:0.1)A;</code>	<i>a tree rooted on a leaf node (rare)</i>

The NEWICK (or New Hampshire) format

(A, B);



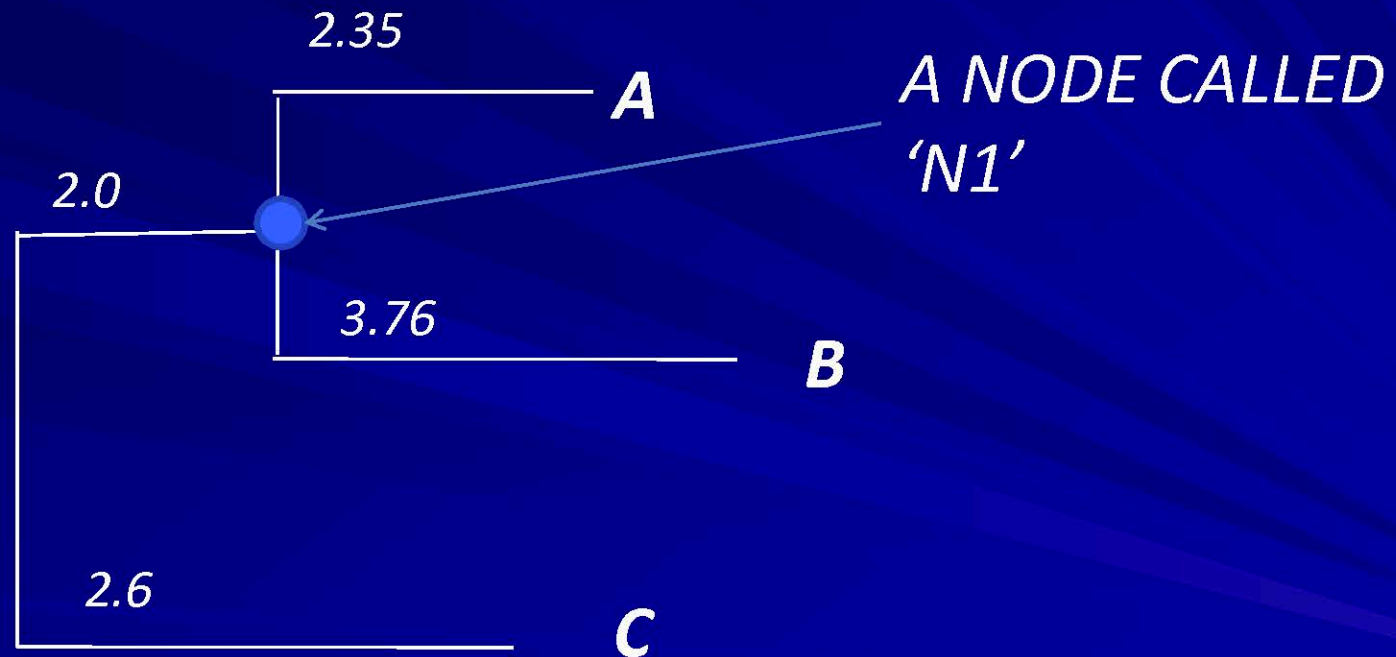
(A:2.35, B:3.76);



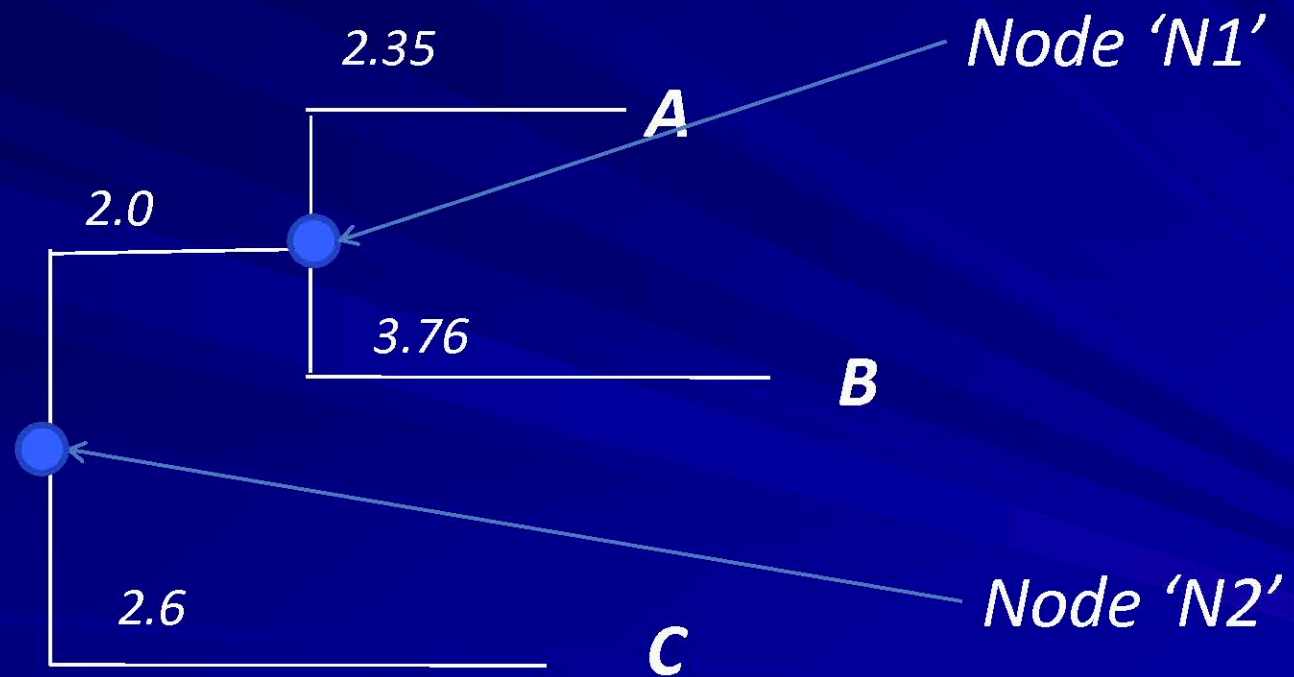
$((A:2.35, B:3.76): 2.0, C:2.6);$



((A:2.35, B:3.76)N1: 2.0,C:2.6);



$((A:2.35, B:3.76)N1:2.0, C:2.6)N2;$



NEXUS: AN EXTENSIBLE FILE FORMAT FOR SYSTEMATIC INFORMATION

DAVID R. MADDISON,¹ DAVID L. SWOFFORD,² AND WAYNE P. MADDISON³

¹*Department of Entomology, University of Arizona, Tucson, Arizona 85721, USA; E-mail: beetle@ag.arizona.edu*

²*Laboratory of Molecular Systematics, MRC 534, MSC, Smithsonian Institution, Washington, D.C. 20560, USA*

³*Department of Ecology and Evolutionary Biology, University of Arizona, Tucson, Arizona 85721, USA*



David R Maddison



Phylogenetics: NEXUS Format



EEB 5349: Phylogenetics

The goal here is to explain the most important features of the NEXUS file format commonly used in phylogenetics. There are no lab exercises here, just information. Use this as a reference.

Contents [\[hide\]](#)

1 The Nexus Data File Format

1.1 Nexus blocks

1.2 Nexus commands

1.3 Nexus comments

1.4 Commonly-used Nexus blocks

1.4.1 Taxa block

1.4.2 Data block

1.4.3 Trees block

1.4.4 Sets block

1.4.5 Assumptions block

1.4.6 Paup block

The Nexus Data File Format

Nexus blocks

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Tools

[What links here](#)[Related changes](#)[Special pages](#)[Printable version](#)[Permanent link](#)[Page information](#)

http://hydrodictyon.eeb.uconn.edu/eebedia/index.php/Phylogenetics:_NEXUS_Format



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Community portal
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Recent changes
Random page
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sitesupport

Tools

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Special pages
Printable version
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[Log in](#)

Page [Discussion](#)

Taxa block

[View history](#)



Phylogenetics: NEX



EEB 5349

The goal here is
here, just inform

Contents [\[hide\]](#)

- 1 The Nexus Data File Format
 - 1.1 Nexus blocks
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 - 1.4 Commonly-used Nexus blocks
 - 1.4.1 Taxa block
 - 1.4.2 Data block
 - 1.4.3 Trees block
 - 1.4.4 Sets block
 - 1.4.5 Assumptions block
 - 1.4.6 Paup block

The purpose of a taxa block is to provide names directly in the data block (see below). Here is an

```
#nexus
...
begin taxa;
  dimensions ntax=5;
  taxlabels
    Giardia
    Thermus
    Deinococcus
    Sulfolobus
    Haobacterium
;
end;
```

ed in phylogenetics. There are no lab exercises

The Nexus Data File Format

Nexus blocks

http://hydrodictyon.eeb.uconn.edu/eebedia/index.php/Phylogenetics:_NEXUS_Format

Data block

Phylo

The data block is the workhorse of Nexus blocks. This is where you place the actual sequence data, and names of your sequences. Here is an example of a data block:



```
#nexus
...
begin data;
  dimensions ntax=5 nchar=54;
  format datatype=dna missing=? gap=-;
  matrix
    Ephedra      TTAAGCCATGCATGTCTAAGTATGAACTAATTCCAAACGGTGAACTGCGGATG
    Gnetum       TTAAGCCATGCATGTCTATGTACGAACTAATC-AGAACGGTGAACTGCGGATG
    Welwitschia  TTAAGCCATGCACGTGTAAGTATGAACTAGTC-GAAACGGTGAACTGCGGATG
    Ginkgo       TTAAGCCATGCATGTGTAAGTATGAACTCTTTACAGACTGTGAACTGCGAATG
    Pinus        TTAAGCCATGCATGTCTAAGTATGAACTAATTGCAGACTGTGAACTGCGGATG
    [ ----+--10 | ----+--20 | ----+--30 | ----+--40 | ----+--50 | ---- ]
;
end;
```

The Nex

Nexus blocks

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1 The Nexu
 1.1 Ne
 1.2 Ne
 1.3 Ne
 1.4 Co
 1.4
 1.4
 1.4
 1.4
 1.4
 1.4

Phylo



- 1 The Nexus
- 1.1 Ne
- 1.2 Ne
- 1.3 Ne
- 1.4 Co
- 1.4
- 1.4
- 1.4
- 1.4
- 1.4
- 1.4

The Nex

Nexus blocks

Data block

The data block is the names of your sequence

Trees block

A trees block has the following structure:

```
#nexus
...
begin data;
  dimensions r
  format datat
  matrix
    Ephedra
    Gnetum
    Welwitschi
    Ginkgo
    Pinus
;
end;
```

```
#nexus
...
begin trees;
  translate
    1 Ephedra,
    2 Gnetum,
    3 Welwitschia,
    4 Ginkgo,
    5 Pinus
;
  tree one = [&U] (1,2,(3,(4,5)));
  tree two = [&U] (1,3,(5,(2,4)));
end;
```

the actual sequence data, an

```
AACGGTGAAACTGCGGATG
AACGGTGAAACTGCGGATG
AACGGTGAAACTGCGGATG
GACTGTGAAACTGCGAATG
GACTGTGAAACTGCGGATG
--40 | ----+--50 | ----]
```



The future data exchange standard is here!

NeXML is an exchange standard for representing phyloinformatic data — inspired by the commonly used NEXUS format, but more robust and easier to process.



nenhum arquivo selecionado

Overview

The [NEXUS flat file format](#) is a commonly used syntax for phylogenetic data. Unfortunately, over time, non-compliant NEXUS implementations have overloaded the standard - which has caused various [problems](#). Meanwhile, mature technologies around the [XML](#) standard have emerged. These technologies have the potential to greatly simplify and improve robustness in the processing of rich phylogenetic data. This website is the home for the community-driven NeXML project, which seeks to leverage XML technologies in the development of a data standard that translates NEXUS concepts into a syntax that is more easily validated and processed. This approach promises several advantages:

- ▶ **Syntax validation** — some of the issues hampering interoperability are caused by the fact that no formal specification exists for NEXUS and other flat files, and no unambiguous way to validate them. Using [XML Schema](#) we have defined a versioned grammar against which data files can be validated syntactically. In addition, this website has a validation service (the orange box in the center of every page) that also checks the semantics of uploaded NeXML files beyond ways that can be expressed in XSD schema language.

- ▶ **Semantic annotation** — an issue in current file formats is that their semantics are not well-defined. For example, what does it mean to use an ambiguity code in a matrix? Is it uncertainty or polymorphism? With the wider [EvoInfo working group](#) we are developing an

Quick links:

[Manual](#)
[Schema documentation](#)
[NeXML publication](#)
[Example files](#)
[Slide show](#)

Libraries:

[Java](#)[Python](#)[Perl](#)[C++](#)[JavaScript](#)[Ruby](#)

External links

- ▶ [nexml@github](#)
- ▶ [NeXML wiki](#)
- ▶ [CDAO](#)
- ▶ [PhyloWS](#)



The future data exchange standard is here!

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Syst. Biol. 61(4):675–689, 2012

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DOI:10.1093/sysbio/sys025

Advance Access publication on February 22, 2012

NeXML: Rich, Extensible, and Verifiable Representation of Comparative Data and Metadata

RUTGER A. VOS^{1,*}, JAMES P. BALHOFF^{2,3}, JASON A. CARAVAS⁴, MARK T. HOLDER⁵, HILMAR LAPP²,
WAYNE P. MADDISON⁶, PETER E. MIDFORD², ANURAG PRIYAM⁷, JEET SUKUMARAN⁵,
XUHUA XIA⁸, AND ARLIN STOLTZFUS⁹

translates NEXUS concepts into a syntax that is more easily validated and processed. This approach promises several advantages:

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- ▶ nexml@github
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- ▶ [CDAO](#)
- ▶ [PhyloWS](#)

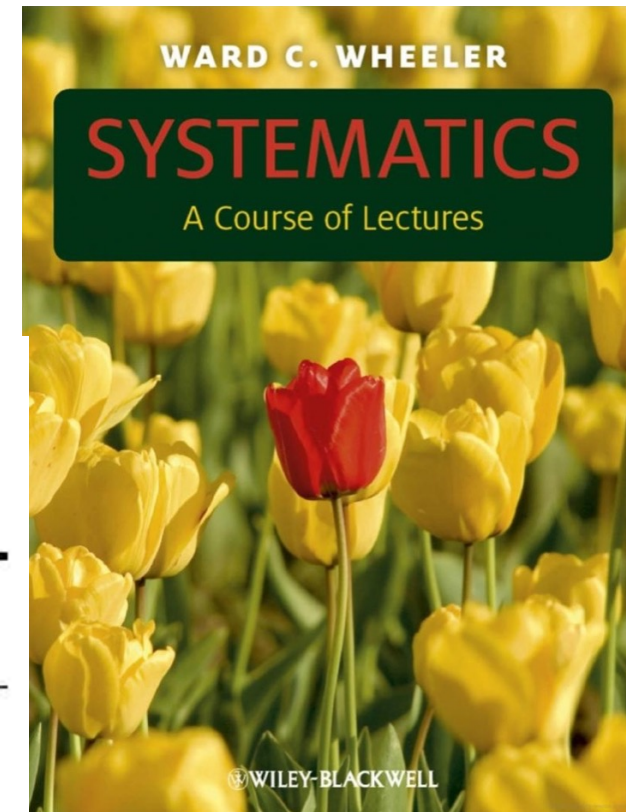
Features	Newick	NHX	NEXUS	PhyloXML	NeXML
Represents tree topology with branch lengths, support values	■	■	■	■	■
Represents labels for OTUs and internal nodes	■	■	■	■	■
Represents link from OTU to taxonomic concept		○		■	■
Represents molecular sequence data		○	■	■	■
Represents provenance of molecular data (e.g., accession)				■	■
Represents morphological and other non-molecular data			■		■
Represents provenance of morphological data (e.g., accession)					■
Supports annotation of data objects			○	○	■
Supports internal references to data objects			○		■
Supports user-defined extensions		○	■	■	■
Supports georeferences				■	■
Published format description			■	■	
Formally declared syntax				■	■
Actively developed				■	■

Breve revisão sobre métodos de inferência filogenética

Critérios de otimização

Table 1.5 Classification of phylogenetic analysis methods and their strategies

	Optimality search criterion	Clustering
Character state	Maximum parsimony (MP)	
	Maximum likelihood (ML)	
	Bayesian inference	
Distance matrix	Fitch–Margoliash	UPGMA
		Neighbor-joining (NJ)



Character-based methods

Non-character-based methods

Methods based on an explicit
model of evolution

Maximum likelihood methods

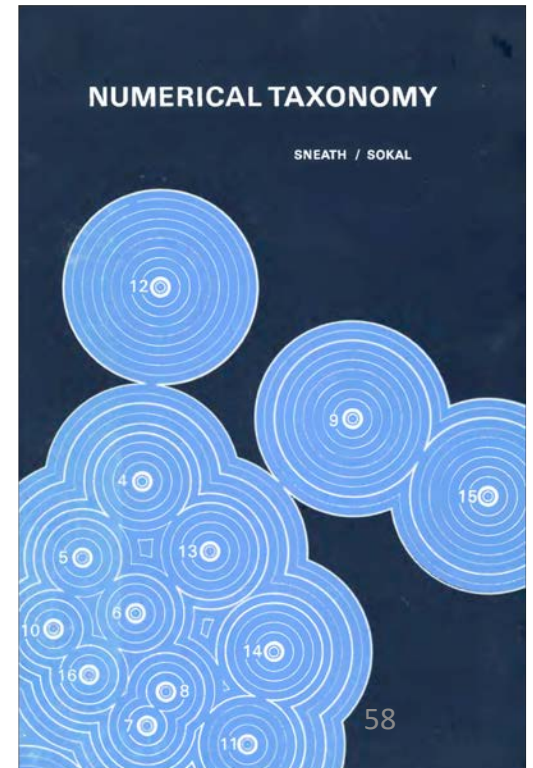
Pairwise distance methods

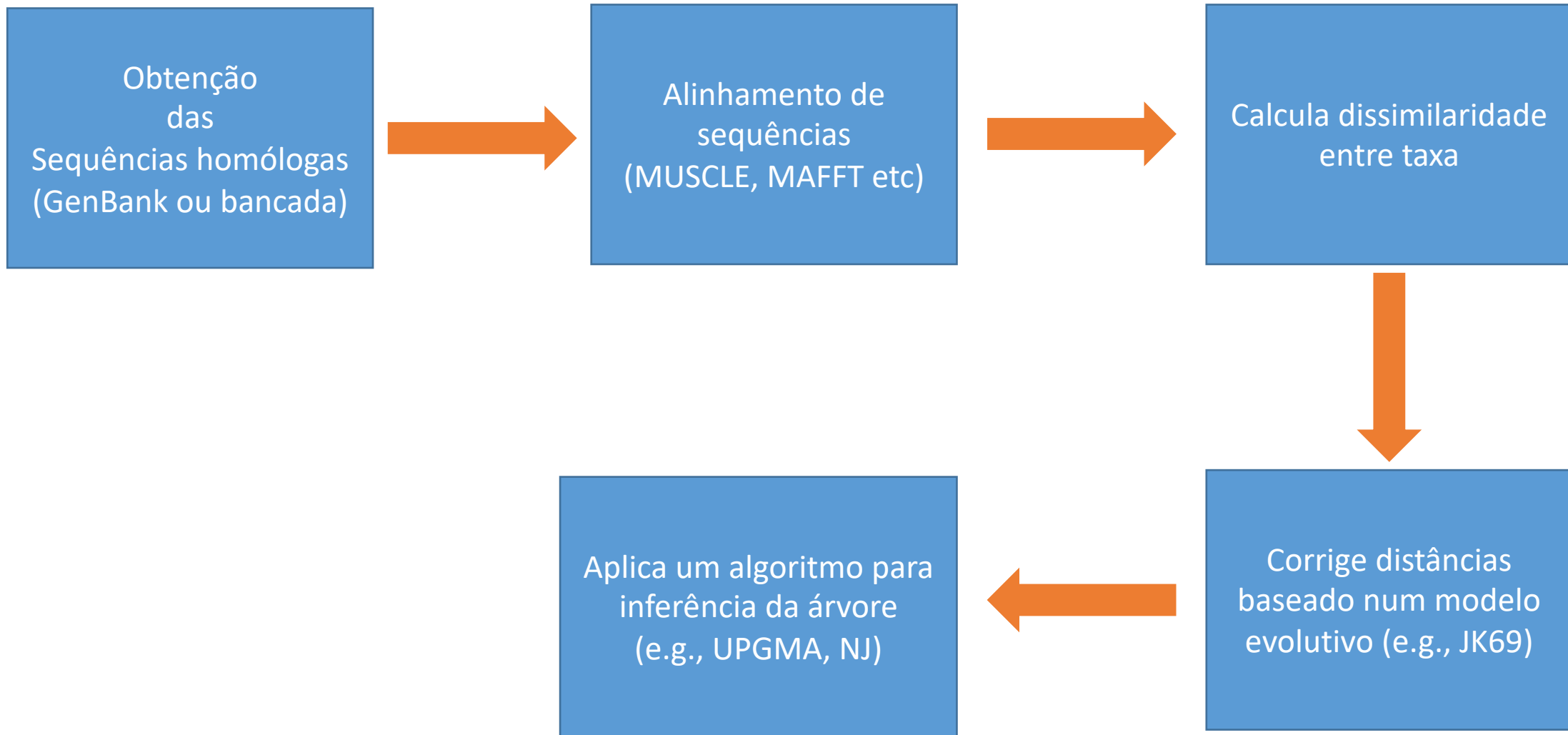
Methods not based on an
explicit model of evolution

Maximum parsimony methods

Métodos baseados em distância

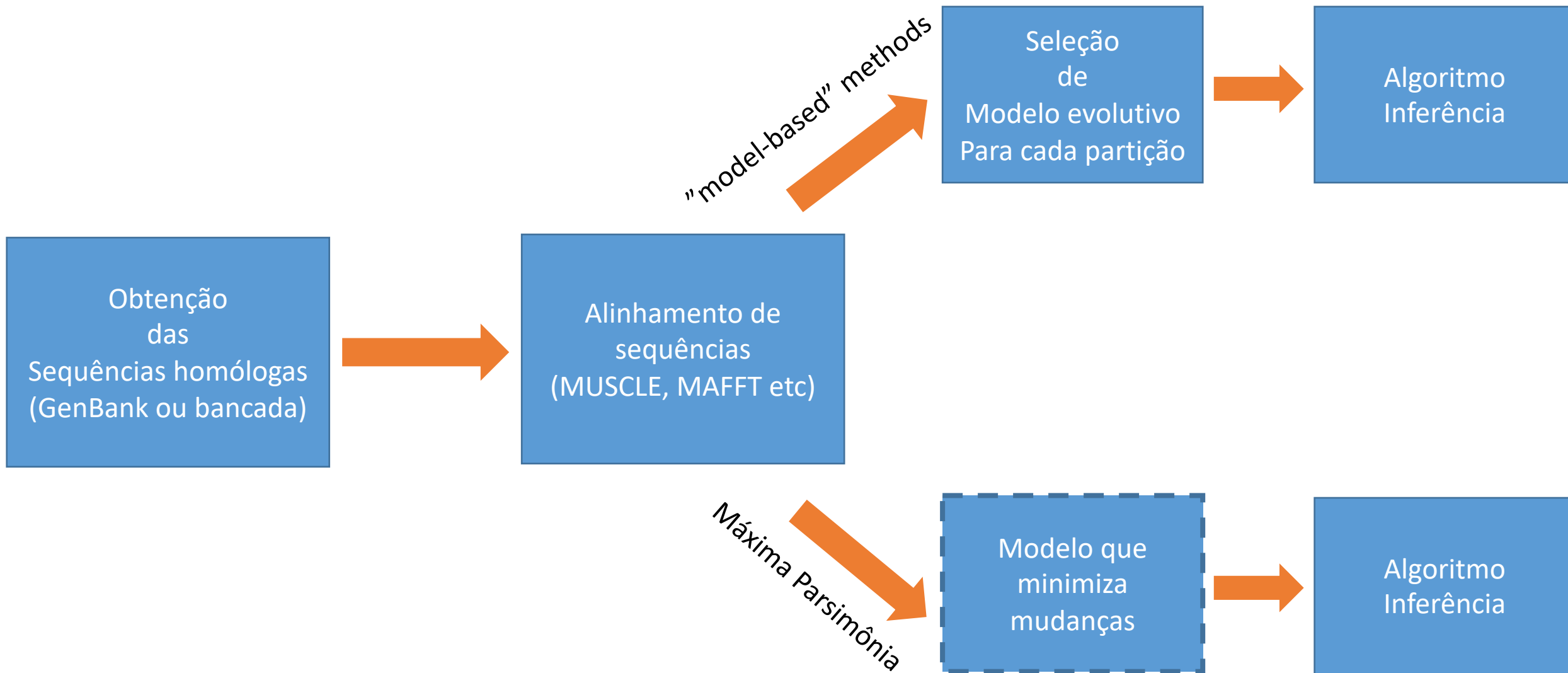
- Calculam uma medida de distância para medir a dissimilaridade entre pares de taxa (baseada nas suas sequências alinhadas) para produzir uma matriz de distância par a par, e então inferir as relações entre os taxa
- Posteriormente se corrige essas distâncias utilizando um modelo evolutivo
- Algoritmos mais usados => UPGMA e NJ
 - NJ assume modelo de evolução mínima





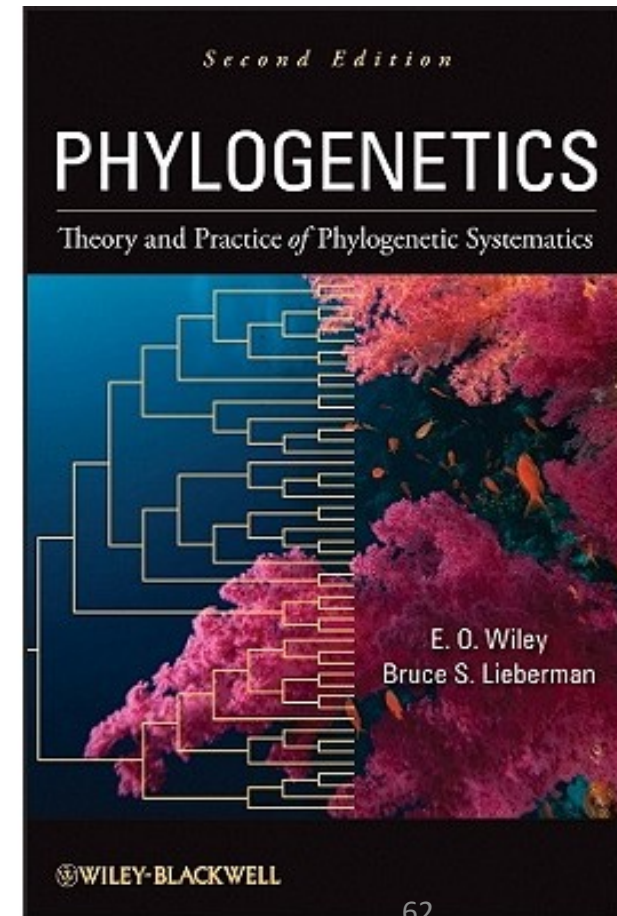
Métodos baseados em distância

- Vantagens
 - Rapidez, adequado para um grande conjunto de dados
- Desvantagens
 - Inconsistências na topologia e comprimento de ramo
 - Filosóficas (Fenética) => não está interessado em reconstruir a história dos taxa em termos de compartilhamento de caracteres derivados
 - Não é possível julgar qual árvore é melhor
 - Produzem uma única topologia
 - Não lida bem com heterogeneidade nas taxas de evolução
 - Toda a informação sobre as sequências é perdida, já que é transformada em distância
 - Não estima estados ancestral de caracteres



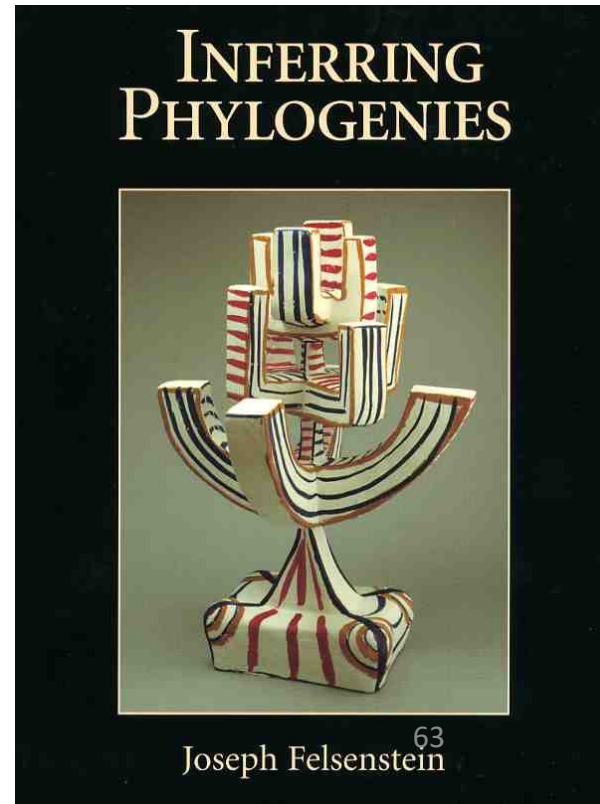
Máxima Parsimônia (MP)

- Procura a topologia que explica um conjunto de sequências alinhadas de forma a minimizar o número de mudanças de caracteres
- Vantagens
 - Simplicidade do critério de otimização
 - Trabalha tanto com dados morfológicos quanto moleculares
 - Rápida para muitos taxa
- Problemas
 - MP assume que um caracter comum é herdado diretamente de ancestral comum. Isso pode subestimar a divergência real entre taxons distantes (*long branch attraction*)
 - Não funciona bem quando a divergência entre sequências é alta
 - Estimativa enviesada de comprimento de ramo



Máxima Verossimilhança (ML)

- Examina diferentes topologias e avalia o suporte relativo somando todas as posições de sequências.
- Sustentada por critérios estatísticos ao invés de filosóficos
- Algoritmos de ML procuram pela topologia e os comprimentos de ramo que maximizam a probabilidade de se observar os estados de caracteres (verossimilhança), dada a topologia e um modelo de evolução
 - Precisa de uma topologia de início e um modelo de evolução explícito
- Vantagens
 - Menos problema com long branch attraction
 - Resultados entre MP e ML podem convergir ("non-common mechanism")
- Desvantagens
 - Pode fornecer a árvore errada se o modelo de evolução escolhido for o errado
 - Pode ser bem lenta



A Likelihood Approach to Estimating Phylogeny from Discrete Morphological Character Data

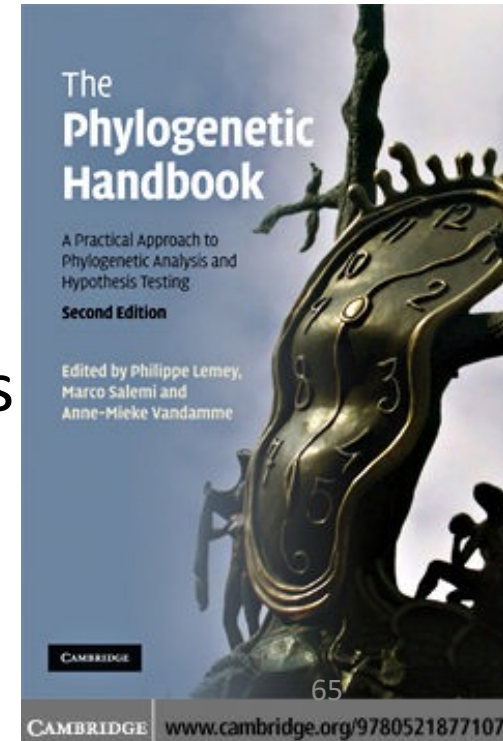
PAUL O. LEWIS

$$Q = \alpha \begin{bmatrix} 1 - k & 1 & \dots & 1 \\ 1 & 1 - k & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 - k \end{bmatrix}$$

Modelo de Markov
adaptado para
caracteres
morfológicos
discretos (binários)

Inferência Bayesiana (BI)

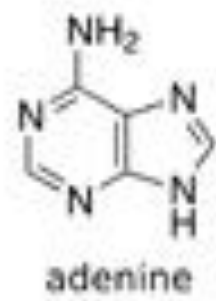
- Também usam o conceitos de verossimilhança (likelihood), mas não tentam buscar somente pela única melhor árvore. Ao invés disso buscam a distribuição de probabilidade de árvores. Esta probabilidade posterior possui uma estimativa de confiança.
- Distribuições posteriores de árvore são obtidas explorando o espaço de árvores usando um método chamado MCMC
- Normalmente se descarta ("burn-in") uma parte inicial destas árvores que foram estimadas antes do MCMC convergir
- Precisam de um *prior*, uma informação de crença prévia, que é formalizado pela distribuição estatística anterior dos parâmetros modelo



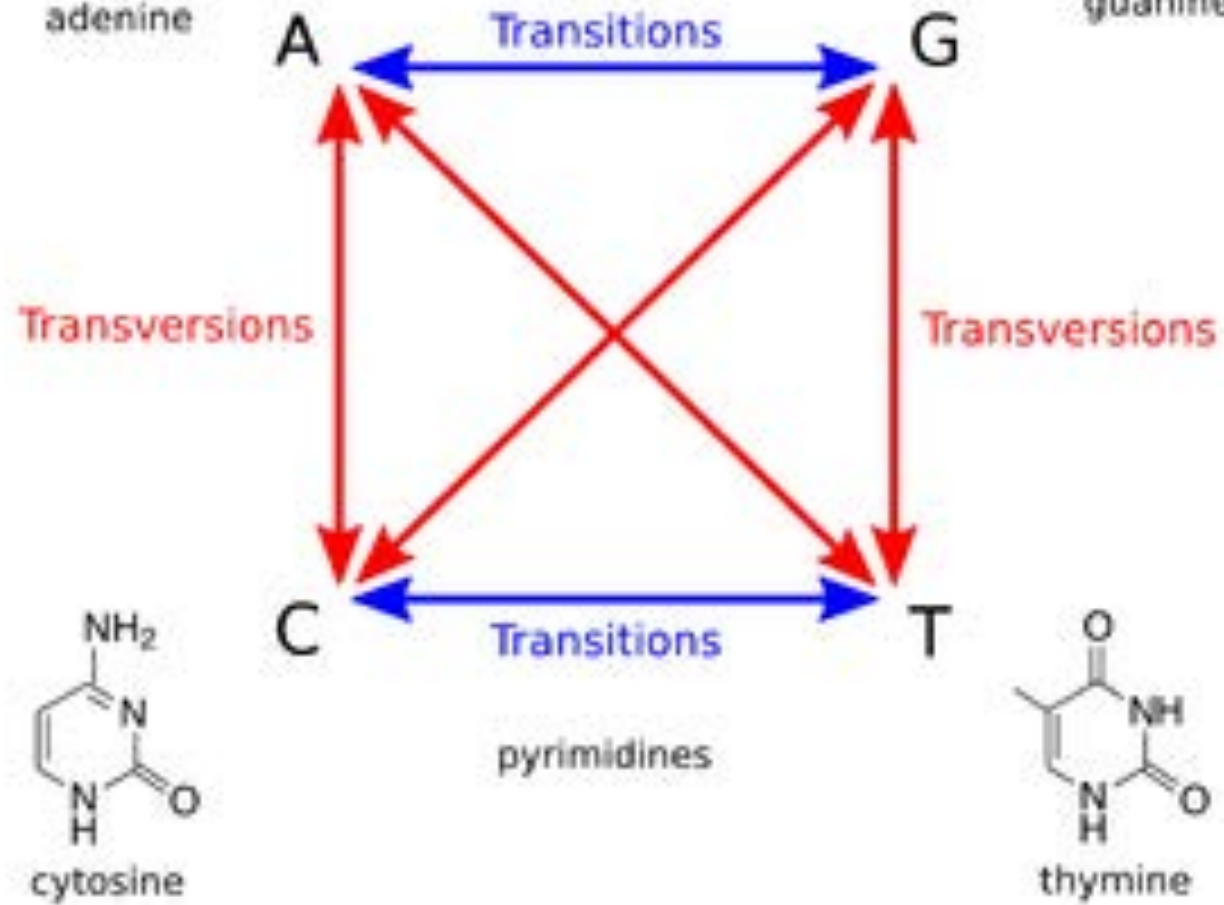
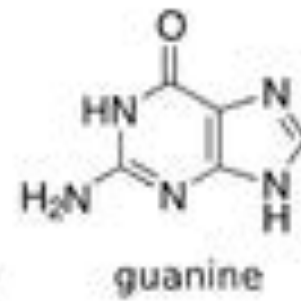
Modelos de evolução de sequências de nucleotídeos

- Processos de Markov homogêneos de tempo contínuo
- Processos de Markov são descritos em matrizes Q
 - Matrizes 4x4 na qual cada elemento descreve a taxa instantânea de mudança de uma base para a outra

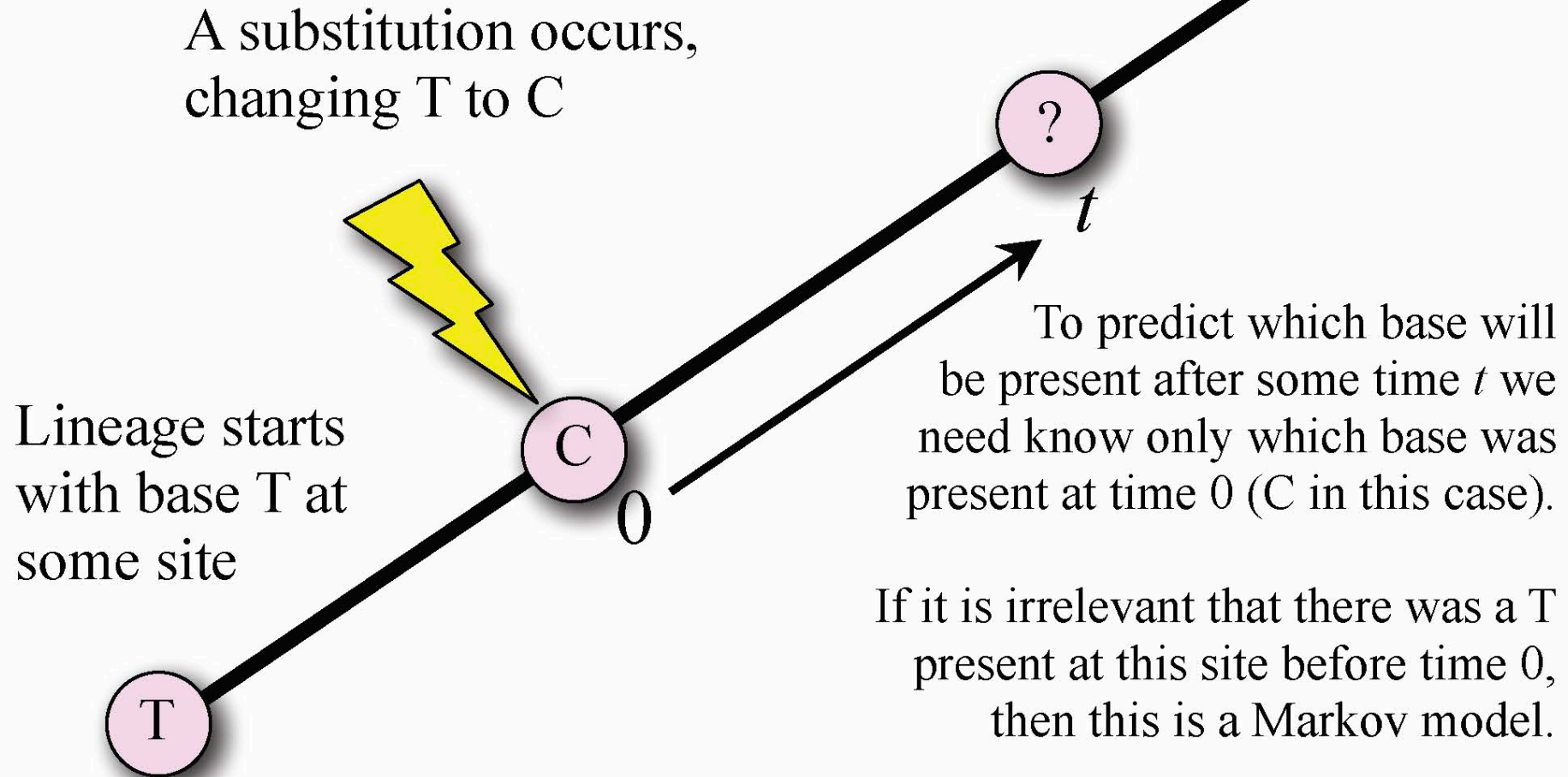
$$P(t) = \begin{pmatrix} p_{AA}(t) & p_{GA}(t) & p_{CA}(t) & p_{TA}(t) \\ p_{AG}(t) & p_{GG}(t) & p_{CG}(t) & p_{TG}(t) \\ p_{AC}(t) & p_{GC}(t) & p_{CC}(t) & p_{TC}(t) \\ p_{AT}(t) & p_{GT}(t) & p_{CT}(t) & p_{TT}(t) \end{pmatrix}$$



purines



What is a Markov process?



Modelos de Evolução de DNA

- Pressupostos

- Em qualquer instante, a taxa de mudança da base i para a j é independente da informação anterior a i . Dizemos que um processo de Markov tem "memória curta"
 - Tempo homogêneo/contínuo – taxas de substituição não mudam com o tempo
 - Estacionários – frequência relativa das bases está $(\pi_A, \pi_C, \pi_G, \pi_T)$ em equilíbrio
 - Muitos modelos são também time-reversible – a taxa de mudança de i para j é o mesmo de j para i .
- Necessários para modelar substituições como um processo estocástico

Modelos de Evolução de sequências

- **Jukes Cantor (JC69) – o modelo mais simples**
- Pressupostos:
 - Frequências de equilíbrio dos nucleotídeos são 25% ($\pi_A=\pi_C=\pi_G=\pi_T=1/4$)
 - Probabilidade iguais para qualquer substituição ($a=b=c=d=e=f=1$)

$$Q = \begin{pmatrix} -\frac{3}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & -\frac{3}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & -\frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & -\frac{3}{4} \end{pmatrix}$$

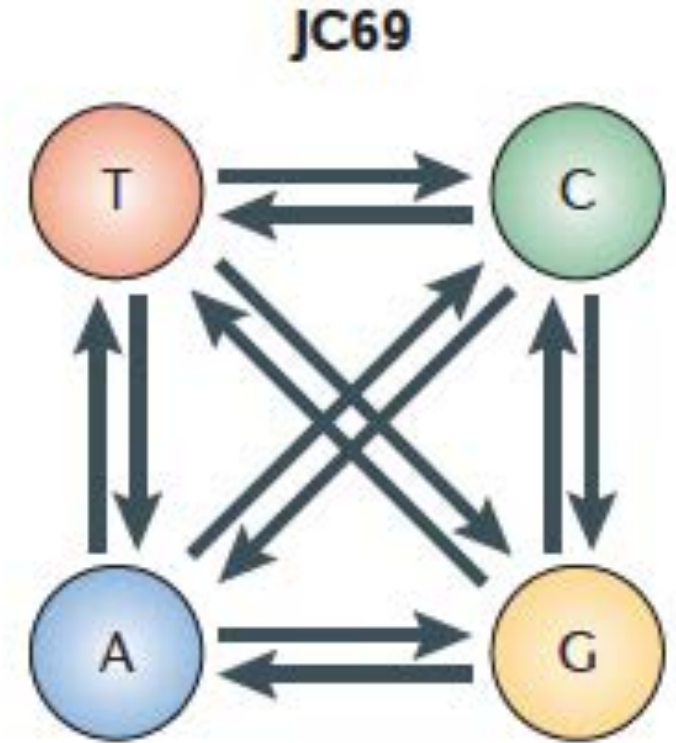
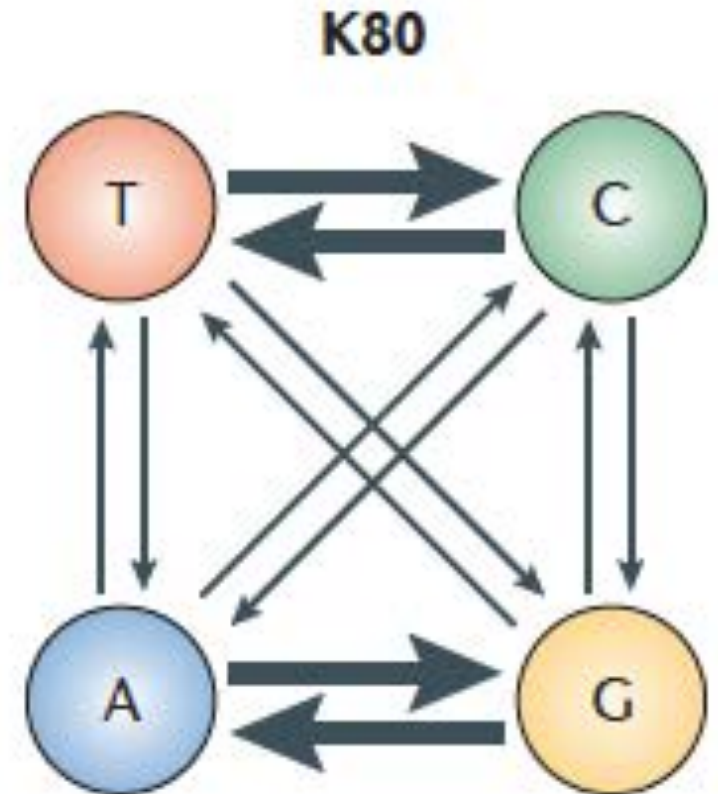
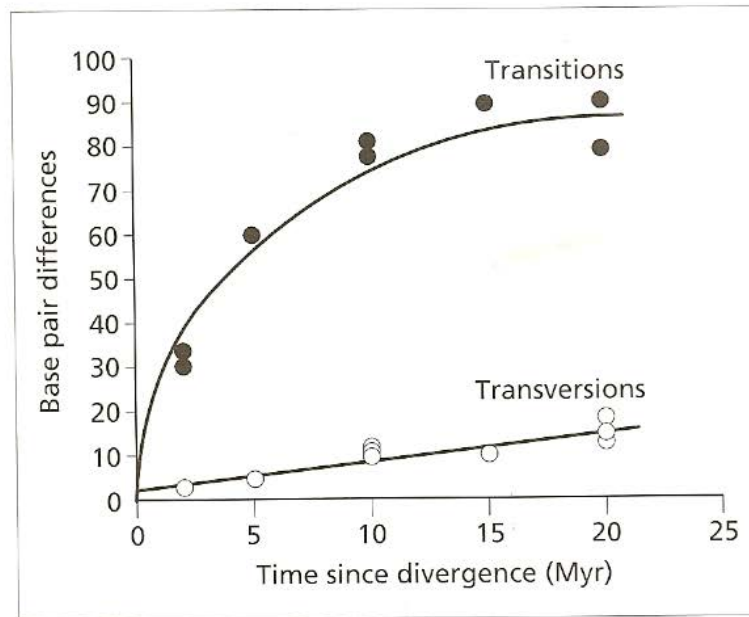


Figure 4.4 Instantaneous rate Q matrix for the Jukes and Cantor model (JC69).

Modelos de Evolução de sequências

- **Kimura 2-parameter (K2P)**
- Geralmente, *transições* ocorrem com taaxs maiores do que *transversões*
- Viola os pressupostos do JC69
- K2P adiciona um parâmetro a mais para modelar cada tipo de substituição
- Assume frequências iguais de nucleotídeos em equilíbrio

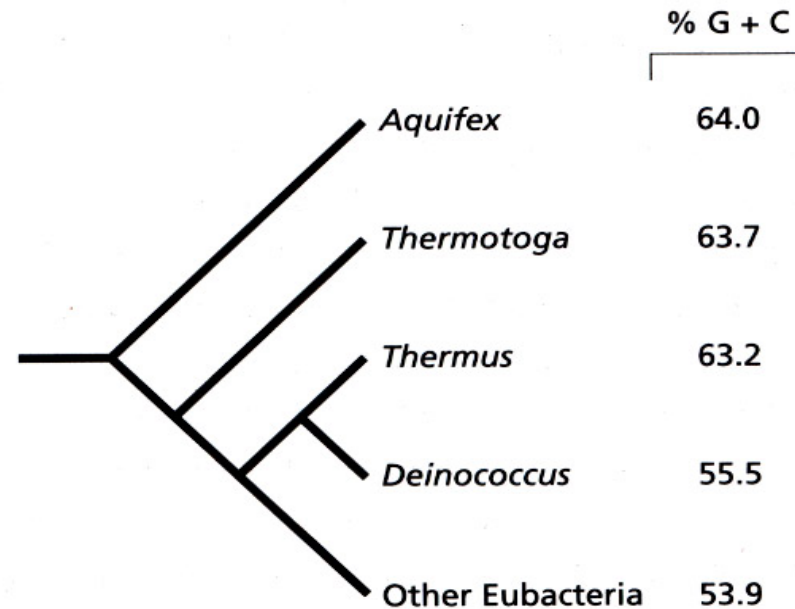
Fig. 5.13 The number of transitions and transversions between the same bovid mammal sequences used in Fig. 5.11. Transitions accumulate much more rapidly than transversions and become saturated, whereas transversions accumulate more slowly and show no evidence of saturation.



Modelos de Evolução de sequências

- **Felsenstein (1981) - F81**
- Na maioria das spp, A+T $\neq$ C+G
- Se houver pouco G's, a taxa de substituição de G para A será menor do que outras substituições

Organism	A	T	C	G
Human	30.9	29.4	19.9	19.8
Chicken	28.8	29.2	20.5	21.7
Grasshopper	29.3	29.3	20.5	20.7
Sea Urchin	32.8	32.1	17.7	17.3
Wheat	27.3	27.1	22.7	22.8
Yeast	31.3	32.9	18.7	17.1
E. Coli	24.7	23.6	26.0	25.7



$$\mathbf{P}_t = \begin{bmatrix} . & \pi_C \alpha & \pi_G \alpha & \pi_T \alpha \\ \pi_A \alpha & . & \pi_G \alpha & \pi_T \alpha \\ \pi_A \alpha & \pi_C \alpha & . & \pi_T \alpha \\ \pi_A \alpha & \pi_C \alpha & \pi_G \alpha & . \end{bmatrix}$$

Modelos de Evolução de sequências

- Hasegawa, Kishino and Yano (HKY85)

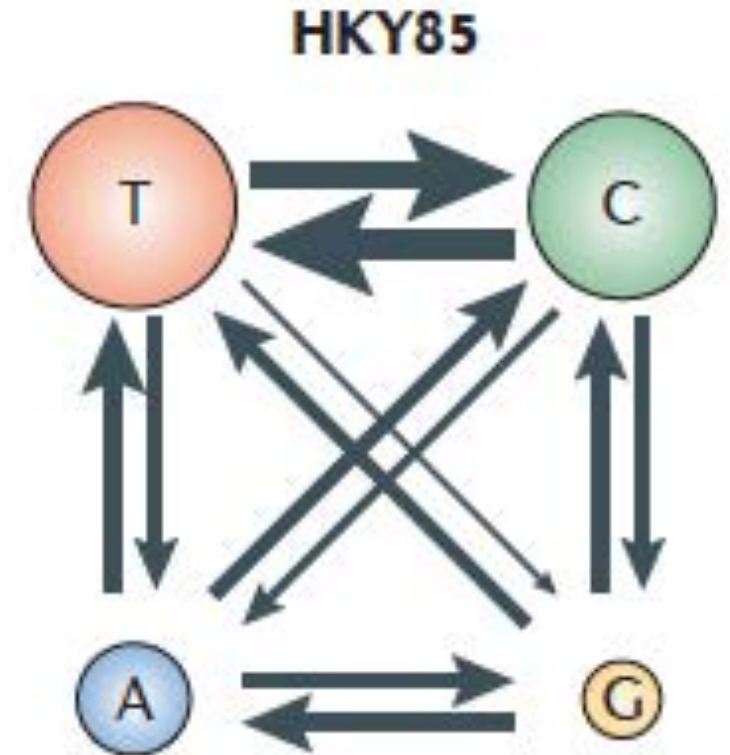
- Combina F81 e K2P

$$\mathbf{P}_t = \begin{bmatrix} . & \pi_C \beta & \pi_G \alpha & \pi_T \beta \\ \pi_A \beta & . & \pi_G \beta & \pi_T \alpha \\ \pi_A \alpha & \pi_C \beta & . & \pi_T \beta \\ \pi_A \beta & \pi_C \alpha & \pi_G \beta & . \end{bmatrix}$$

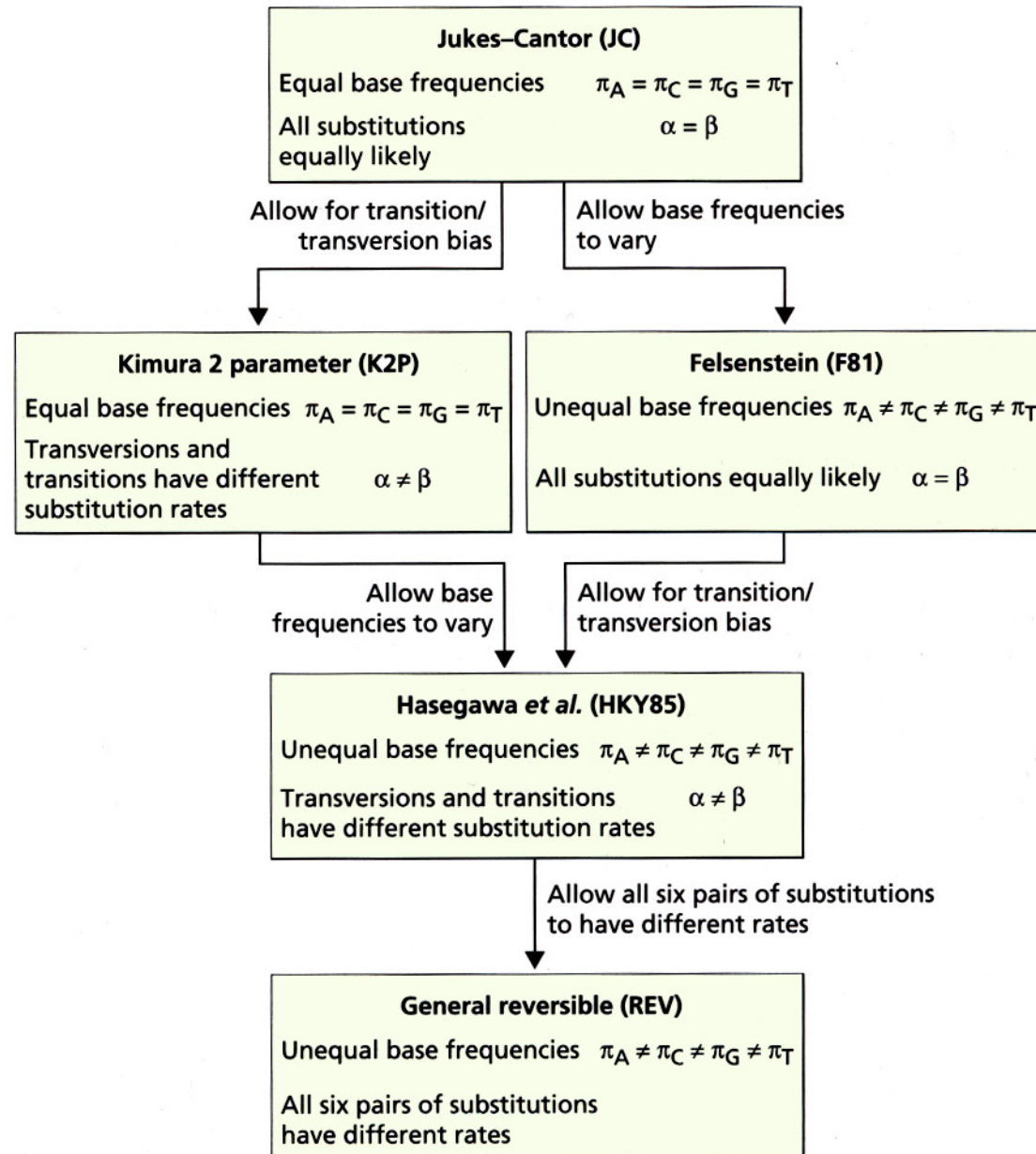
- General Time Reversible (GTR)

- Permite que todos os pares de substituição tenham taxas distintas
- Permite frequências de nucleotídeos diferentes

$$\mathbf{P}_t = \begin{bmatrix} . & \pi_C a & \pi_G b & \pi_T c \\ \pi_A a & . & \pi_G d & \pi_T e \\ \pi_A b & \pi_C d & . & \pi_T f \\ \pi_A c & \pi_C e & \pi_G f & . \end{bmatrix}$$



Modelos de Evolução de sequências



Mas porque isso é importante?

a GTR nucleotides

	A	G	C	T
A	-	r_{AG}	r_{AC}	r_{AT}
G	r_{AG}	-	r_{GC}	r_{GT}
C	r_{AC}	r_{GC}	-	r_{CT}
T	r_{AT}	r_{GT}	r_{CT}	-

b Binary correlation

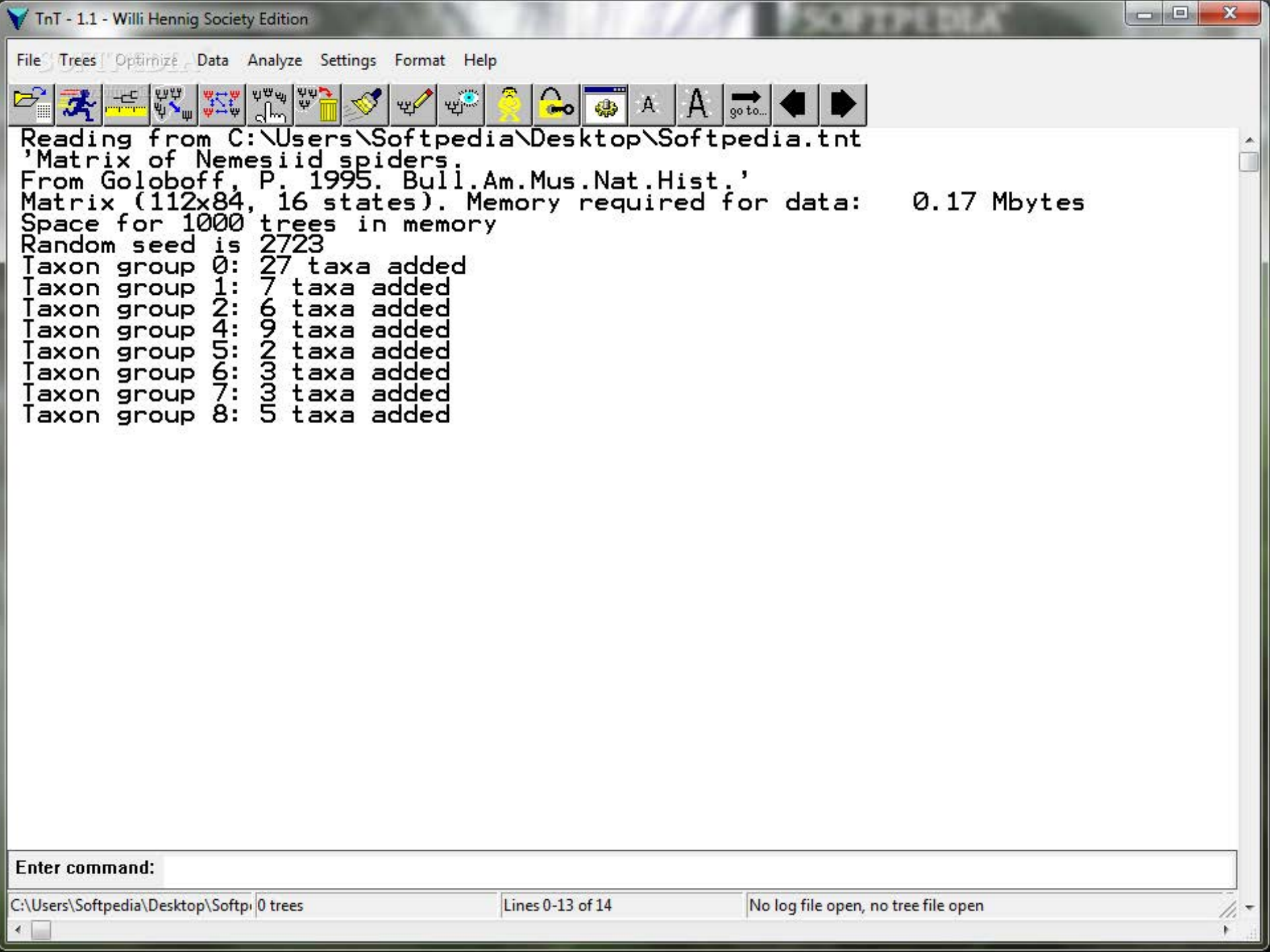
	00	01	11	10
00	-	r_A	0	r_B
01	r_C	-	r_D	0
11	0	r_E	-	r_F
10	r_G	0	r_H	-

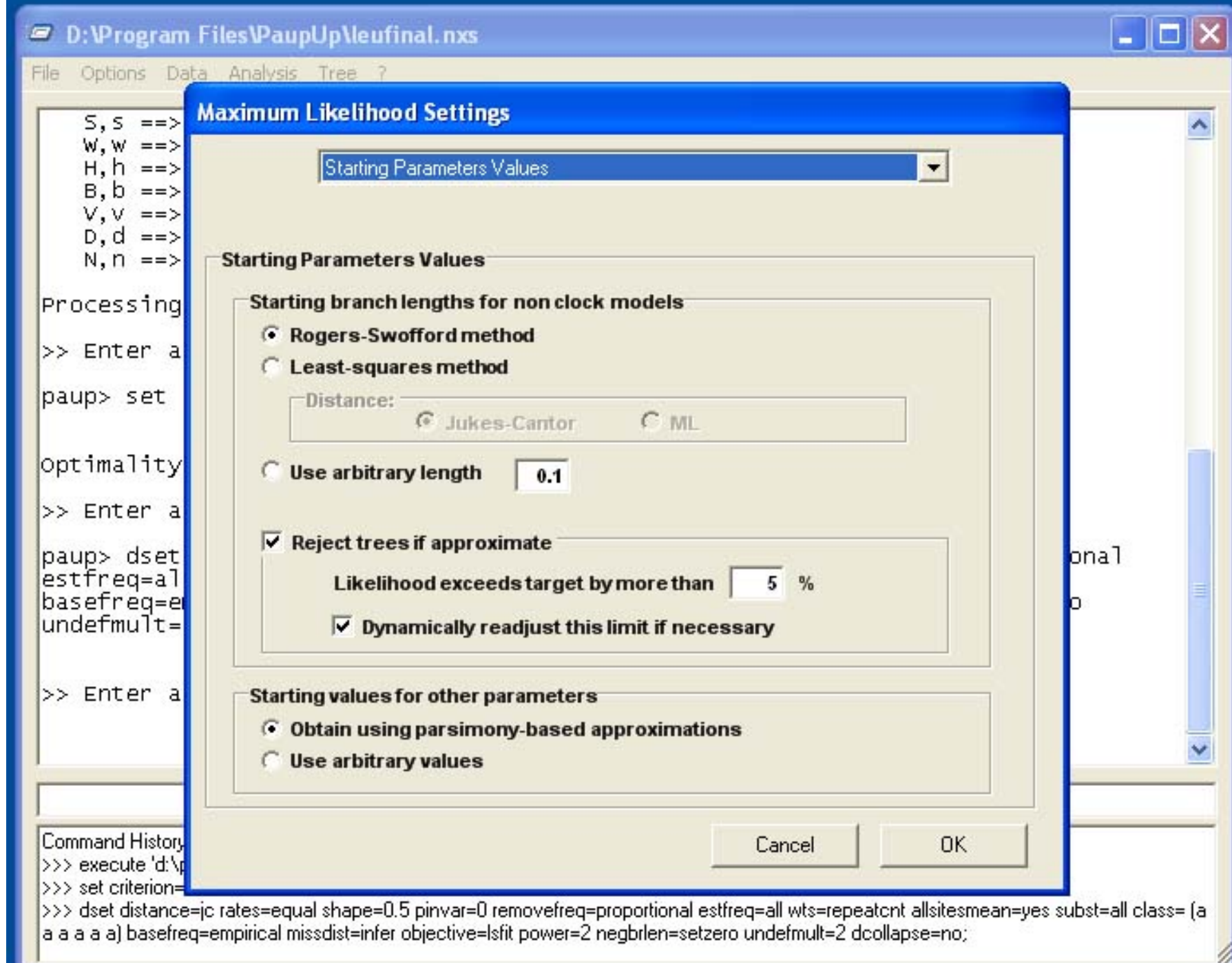
d Ordered transitions

	0	1	2	3
0	-	r_{01}	0	0
1	r_{10}	-	r_{12}	0
2	0	r_{21}	-	r_{23}
3	0	0	r_{32}	-

Além de serem necessários em métodos de inferência filogenética "model-based", esses modelos de substituição utilizam o mesmo arcabouço (processos de Markov) que métodos comparativos que iremos estudar, tais como correlação de caracteres binários e multiestado

Softwares para inferência filogenética





com.biomatters.iseek.application.ISeekMain File Edit View Tools Sequence Annotate & Predict Help

Geneious 7.0.0 Beta

Back Forward Sequence Search Agents Align/Assemble Tree Primers Cloning Workflows Back Up Support Help

Sources

- Local (1)
 - Constructs (14, 4 unread)
 - Genotyping (1346)
 - m037 (12)
 - Publications (15, 3 unread)
 - Sample Documents (1)
 - Alignments (18)
 - Cloning (15)
 - Contig Assembly (2)
 - Genomes (234)
 - PlasMapper Feature
 - Plasmids from NEB
 - Primers (12)
 - Protein Documents
 - Tree Documents (4)
 - Temp (100)
 - Deleted Items (711, 1)
 - Shared Databases
 - Operations
 - NCBI
 - Gene
 - Genome
 - Nucleotide
 - PopSet
 - Protein
 - PubMed
 - SNP
 - Structure
 - Taxonomy
 - Pfam (Not set up)
 - Domains
 - UniProt

Manage Workflows

Workflows

- Align then build tree
- Batch alignment with MUSCLE
- Filter sequence list
- Identify Organism
- Map reads then find variations/SNPs
- Map reads to each reference sequence
- Sequence Search, Align, and Build Tree

Enzymes that are in BioBrick RFC standards

- BioBrick
- pDEST15
- pDONR201
- pSB1C3
- TetR repressible GFP generator
- Type IIS

View Workflow

Workflow Name: Filter sequence list

Description: Takes a sequence list and creates a new sequence list consisting of only those sequences

Icon: Choose Custom Icon...

☐ Share (read-only) with other Shared Database users

+ Add Step - Delete Step View Options Help

For Each Sequence

Filter Documents Options: Sequence Length > '100'

Group Sequences

Rename Document(s) Options: Filtered + Name of document from 4 operations ago

OK

Using 184 / 493 MB memory
Using 45 / 64 MB permgen

Alt click on a sequence position or annotation, or select a region to zoom in. Alt-shift click to zoom out.



Beast2

Bayesian evolutionary analysis by sampling trees

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developers

February 4, 2016

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packages

December 13, 2015

ABOUT

BEAST 2 is a cross-platform program for Bayesian phylogenetic analysis of molecular sequences. It estimates rooted, time-measured phylogenies using strict or relaxed molecular clock models. It can be used as a method of reconstructing phylogenies but is also a framework for testing evolutionary hypotheses without conditioning on a single tree topology. BEAST 2 uses Markov chain Monte Carlo (MCMC) to average over tree space, so that each tree is weighted proportional to its posterior probability. BEAST 2 includes a graphical user-interface for setting up standard analyses and a suit of programs for analysing the results.

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The latest version of BEAST 2 is version 2.4.1:

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RevBayes

Bayesian graphical models

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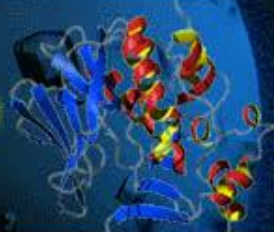


RevBayes

Bayesian phylogenetic inference using probabilistic graphical models and an interpreted language.

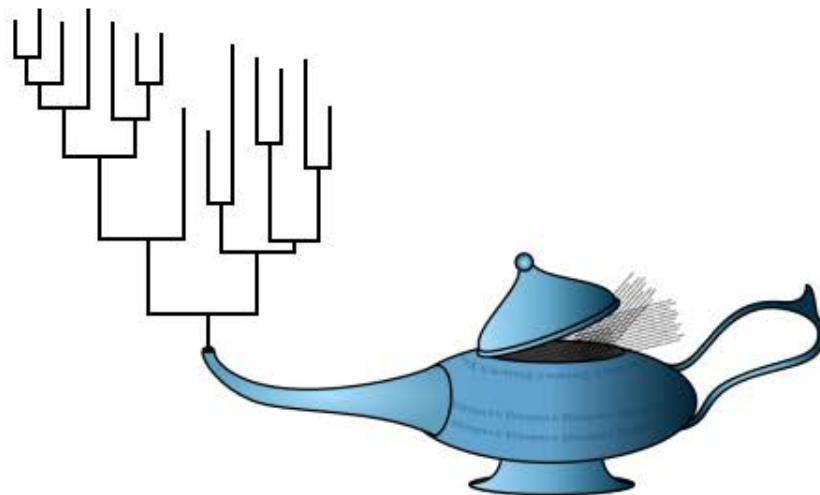
RevBayes was designed and developed by Sebastian Höhna, Fredrik Ronquist and John P. Huelsenbeck.
The core development team additionally includes Michael J. Landis, Bastien Boussau, Tracy A. Heath and Nicolas Lartillot.

RevBayes is free software released under the [GPL license v3](#). This site was generated using [Jekyll](#) and formatted using the [Herring Cove](#) theme.



Phylogeny.fr

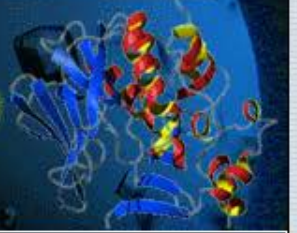
Robust Phylogenetic Analysis For The Non-Specialist



Phylogeny.fr is a free, simple to use web service dedicated to reconstructing and analysing phylogenetic relationships between molecular sequences.

Phylogeny.fr runs and connects various bioinformatics programs to reconstruct a **robust phylogenetic tree from a set of sequences**.

If you use this site, please cite:



Home

Phylogeny Analysis

Blast Explorer

Online Programs

Your Workspace

Documentation

Downloads

Contacts

Phylogeny.fr

Robust Phylogenetic Analysis For The Non-Specialist

Online phylogeny programs

Direct access to the individual tools available on this server.

Multiple Alignment:

MUSCLE
T-Coffee / 3D-Coffee
ClustalW
ProbCons

Phylogeny:

PhyML
TNT
BioNJ
MrBayes

Tree viewers:

TreeDyn
Drawgram
Drawtree
ATV

Utilities:

Gblocks
Jalview
Readseq
Format converter



Phylogeny.fr is a free, simple to use web service dedicated to reconstructing and analysing phylogenetic relationships between molecular sequences.

Phylogeny.fr runs and connects various bioinformatics programs to reconstruct a **robust phylogenetic tree from a set of sequences**.

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The CIPRES Science Gateway now offers BEAST2 and PhyloBayes MPI, along with RAxML, MrBayes and other codes.

First Time Users: Please review the [XSEDE Primer](#) and our [Fair Use Policy](#).

Status: All submissions are working normally.

CIPRES Gateway News

- CIPRES** [RAxML-HPC BlackBox](#) (8.2.8) ⓘ - Phylogenetic tree inference using maximum likelihood/rapid bootstrapping on XSEDE.
- CIPRES** [RAxML-HPC2 on XSEDE](#) (8.2.8) ⓘ - Phylogenetic tree inference using maximum likelihood/rapid bootstrapping run on XSEDE
- User** [RAxML-HPC2 Workflow on XSEDE](#) (8.2.8) ⓘ - Phylogenetic tree inference using maximum likelihood/rapid bootstrapping run on XSEDE
- User** [RAxML-HPC v.8 on XSEDE](#) (8.2.8) ⓘ - NEW Interface! Phylogenetic tree inference using maximum likelihood/rapid bootstrapping run on XSEDE
- Pass** [Readseq](#) (2.2) ⓘ - Readseq: biological sequence format interconversion utility
- [Seqboot](#) (Phylip 3.66) ⓘ - Bootstrap, Jackknife, or Permutation Resampling
- [TreeAnnotator on XSEDE](#) (1.8.3) ⓘ - Bayesian Evolutionary Analysis by Sampling Trees - run on XSEDE

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iPlant users login here:



iPlant Collaborative™

Empowering A New Plant Biology

[What is this?](#)

Status: All submissions are working
normally.

CIPRES Gateway News

CIPRES now provides
TreeAnnotator and LogCombiner

[New CIPRES Rest Tool
Configuration helper!](#)

[Error in Gotoh's alignment
algorithm](#)



CIPRES

Home

Phylogeny / Alignment Tools

If there is a tool or a feature you need, please [let us know](#).

XSEDE
Extreme Science and Engineering
Discovery Environment

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[BEAST2 on XSEDE](#) (2.3.2) ⓘ - Bayesian Evolutionary Analysis by Sampling Trees - run on XSEDE

[BEAST on XSEDE](#) (1.8.0; 1.8.1; 1.8.2; 1.8.3) ⓘ - Bayesian Evolutionary Analysis by Sampling Trees - run on XSEDE

[Clearcut](#) (1.0.9) ⓘ - Fast Implementation of Relaxed Neighbor Joining

[ClustalW](#) (1.82) ⓘ - Create Multiple Alignments from Sequences

[Consense](#) (Phylip 3.66) ⓘ - Find A Consensus Tree

[DPPDIV on XSEDE](#) (1.0) ⓘ - Estimating species divergence times and lineage-specific substitution rates on a fixed topology run on XSEDE

[FastTreeMP on XSEDE](#) (2.1.8) ⓘ - Fast (Approximate) Maximum Likelihood tree construction - run on XSEDE

[GARLI 2.01 on XSEDE](#) (2.01) ⓘ - Genetic Algorithm for Rapid Likelihood Inference - run on XSEDE

[RAxML-HPC BlackBox](#) (8.2.8) ⓘ - Phylogenetic tree inference using maximum likelihood/rapid bootstrapping on XSEDE.

[GARLI.conf Create](#)
[RAxML-HPC2 on XSEDE](#) (8.2.8) ⓘ - Phylogenetic tree inference using maximum likelihood/rapid bootstrapping run on XSEDE

[JModelTest2 on XSEDE](#)
[RAxML-HPC2 Workflow on XSEDE](#) (8.2.8) ⓘ - Phylogenetic tree inference using maximum likelihood/rapid bootstrapping run on XSEDE

[LogCombiner on XSEDE](#)
[RAxML-HPC v.8 on XSEDE](#) (8.2.8) ⓘ - NEW Interface! Phylogenetic tree inference using maximum likelihood/rapid bootstrapping run on XSEDE

[MAFFT on XSEDE](#)
[Readseq](#) (2.2) ⓘ - Readseq: biological sequence format interconversion utility

[Migrate-N on XSEDE](#)
[Seqboot](#) (Phylip 3.66) ⓘ - Bootstrap, Jackknife, or Permutation Resampling

[MrBayes Restart](#)
[TreeAnnotator on XSEDE](#) (1.8.3) ⓘ - Bayesian Evolutionary Analysis by Sampling Trees - run on XSEDE

[MrBayes on XSEDE](#) (2.2.6) ⓘ - Tree inference using Bayesian Analysis - run on XSEDE

[Muscle](#) (3.7) ⓘ - Create Multiple Alignments from Sequences or Profiles

[NCLconverter](#) (2.1) ⓘ - A file format transformation tool

[PAUPRat](#) (Not specified) ⓘ - Parsimony ratchet searches using PAUP*

[POY](#) (4.1.2) ⓘ - Phylogenetic tree inference using dynamic homologies.

[Probalign](#) (1.3) ⓘ - Multiple sequence alignment using partition function posterior probabilities.

[ProbCons](#) (1.12) ⓘ - Probabilistic Consistency-based Multiple Alignment of Amino/Nucleic Acid Sequences

ons are working

/ News

Base de dados para filogenias: como obter uma árvore

- Bases de dados de topologias e tempo de divergência
- Abordagens megaphylogenies
 - Supermatrix
 - Supertree
 - Next-generation sequencing (high-throughput sequencing data)
- Pipelines (semi)automatizadas "for dummies"

Bases de dados para topologias e tempo de
divergência

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[Submission Tutorial](#)

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Welcome to TreeBASE

TreeBASE is a repository of phylogenetic information, specifically user-submitted phylogenetic trees and the data used to generate them. TreeBASE accepts all kinds of phylogenetic data (e.g., trees of species, trees of populations, trees of genes) representing all biotic taxa. Data in TreeBASE are exposed to the public if they are used in a publication that is in press or published in a peer-reviewed scientific journal, book, conference proceedings, or thesis. Data used in publications that are in preparation or in review can be submitted to TreeBASE but will not be available to the public until they have passed peer review. Aside from the submitter, such data are only available to the publication editors or reviewers using a special access URL. TreeBASE is produced and governed by the **The Phyloinformatics Research Foundation, Inc.**

As of April 2014, TreeBASE contains data for 4,076 publications written by 8,777 different authors. These studies analyzed 8,233 matrices and resulted in 12,817 trees with 761,460 taxon labels that mapped to 104,593 distinct taxa.

The current release includes a host of new features and improvements over the previous TreeBASE prototype. New features include:

- Richer annotation of metadata (journal DOIs, specimen georeferences, Genbank accession numbers, etc)
- A mapping between taxon labels and taxonomic names in uBio and NCBI for improved normalization of names
- The ability to visualize and edit trees using Phylowidget
- The ability to search on tree topology
- Persistent and resolvable URIs for data objects in TreeBASE (i.e. studies, trees, matrices) serve as

iPhylo

Rants, raves (and occasionally considered opinions) on phyloinformatics, taxonomy, and biodiversity informatics. For more ranty and less considered opinions, see my Twitter feed.

ISSN 2051-8188 View this blog in [Magazine View](#).

has to be made. Way back in the mid 1990's when TreeBASE was first starting I was at Oxford University and Paul Harvey (coauthor of [The Comparative Method in Evolutionary Biology](#)) was sceptical of its merits. Given that the comparative method depends on phylogenies, and people like Andy Purvis were in the Harvey lab building supertrees (<http://dx.doi.org/10.1098/rstb.1995.0078>) this may seem odd (it certainly did to me) but Paul shared the view of many systematists. Phylogenies are labile, they change with increased data and taxon sampling, hence individual trees have a short life span.

Data, in contrast, is long-lived. You'd happily reuse GenBank sequences published a decade ago, you probably wouldn't use a decade-old phylogeny. I made this point in an earlier post

— Ross Mounce (@rmounce) [October 19, 2012](#)

In other words:

1. Databasing trees is the Right Thing™ to do
2. Few people are doing the Right Thing™
3. This is because those people are bad/misguided and must be made to see the light

popular posts



On having multiple DOI registration agencies for the same journal

On Friday I discovered that BHL has started issuing CrossRef DOIs for articles, starting with the journal *Revue Suisse de Zoologie*. The me...

Guest post: Response to the

APPLICATION

Treebase: an R package for discovery, access and manipulation of online phylogenies

Carl Boettiger^{1*} and Duncan Temple Lang²



HOME

BLOG

PACKAGES

COMMUNITY

DISCUSS

CONTACT

treebase tutorial

for v0.0-7.1

INSTALLATION

USING THE
PACKAGE

CITING

Neste post vamos ilustrar a funcionalidade do pacote com alguns exemplos introdutórios. Graças em parte às novas normas de periódicos como *Evolution*, *Am Nat*, e *Sys Bio* que requerem que os dados sejam depositados e planos de gerenciamento de dados requeridos pela NSF, esperamos que o pacote seja bastante útil para ensino na medida que replica os resultados ou para meta-análises que podem ser atualizadas automaticamente à medida que o repositório cresça.

Informações adicionais e reletórios de erro são bem-invos via [treebase page](#).

Search

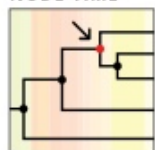
TimeTree is a public knowledge-base for information on the evolutionary timescale of life. Data from thousands of published studies are assembled into a searchable tree of life scaled to time. Three search modes are possible:

- **NODE TIME** - to find the divergence time of two species or higher taxa
- **TIMELINE** - to drill back through time and find evolutionary branches from the perspective of a single species
- **TIMETREE** - to build a timetree of a group of species or custom list

TIMEPANELS showing events in geological time and astronomical history are provided for comparison with timelines and timetrees. Results can be exported in different formats for additional analyses and publication.



NODE TIME



GET DIVERGENCE TIME FOR A PAIR OF TAXA

Specify 2 Taxon Names [?]

Taxon 1:

Taxon 2:

Clear

Search



Resolve Ambiguity [?]

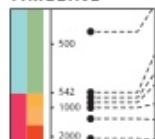
Taxon 1:

Taxon 2:

Show Time

Vamos fazer uma prática à tarde

TIMELINE



GET AN EVOLUTIONARY TIMELINE FOR A TAXON

Specify a Taxon Name [?]

Taxon:



Resolve Ambiguity [?]

Taxon:

Phylomatic

[[documentation](#)]

Version 3. This form sends data to the underlying web-service. To call the service directly, either GET or POST (for sending large trees) the parameters (named exactly as on this form, requirement as per *italics*) to <http://phylodiversity.net/phylomatic/pmws>. This is a new tool (2012-08-23), so *please let me know* if it is not working, or if you have comments. Thanks to the NESCent [PHYLOHACKATHON](#) for impetus and encouragement for upgrading.

tree =

Mandatory option. Paste in your megatree here (<100kB)

or

treeuri =

Mandatory option. URL of a tree online (<20MB)

or

storedtree =

Mandatory option. Locally stored [tree-of-trees megatrees](#), in newick format

informat =

Mandatory. Either a single newick tree: (...);, or NeXML, or a [CDAO](#) tree in XMLRDF (see [sample](#))

method =

☐ convert

Mandatory. Convert an input format to an output format

method =

☒ phylomatic

Mandatory. Graft your taxa into the megatree, and then prune the megatree to just your taxa

taxa =

Dependent mandatory option. Slash-delimited node paths, with returns between each taxon (see [phylomatic documentation](#), or [sample](#)). Please note: at the moment, using internal names (e.g. genera) in the megatree as terminal names in your input list will cause the system to hang; please use terminal 'dummy species,' even if your goal is a genus or family tree.

DateLife is currently being refactored to use OpenTree more thoroughly, as part of the Phylotastic project. The current functionality might break in the interim. Expect a relaunch in Summer 2016.

This will give the age of the most recent common ancestor of a set of taxa, or stretch an input tree, based on available time-calibrated trees. It is still actively in development, so it may break at times. Currently it mostly has mammals in the database, but we are actively adding taxa. See the [FAQ](#) page for more info on this project, and the [Phylotastic](#) page for more info. This was started at a NESCent hackathon, added to at others, and eventually NSF-funded as part of Phylotastic.

Use the syntax `Genus_species,Genus_species`: comma-delimited, no spaces, underscores in names, i.e.,

Rhinoceros_unicornis,Equus_caballus,Mus_musculus

for a list of species in the taxa box, **OR** a Newick tree string, ending with a semicolon:

((Rhinoceros_unicornis,Equus_caballus),Mus_musculus);

The tree can have branch lengths. Note that stretching a tree is much slower than just getting ages for a list of taxa. It uses a method by [Eastman et al. 2013](#)

Input:

Return format:

Note that to return a tree you'll need to provide at least three taxon names. The "best guesses" are just the medians of the results by study.

Partial match:

Use embargoed data:

How to deal with single point estimates (which have no uncertainty):



cellular organisms

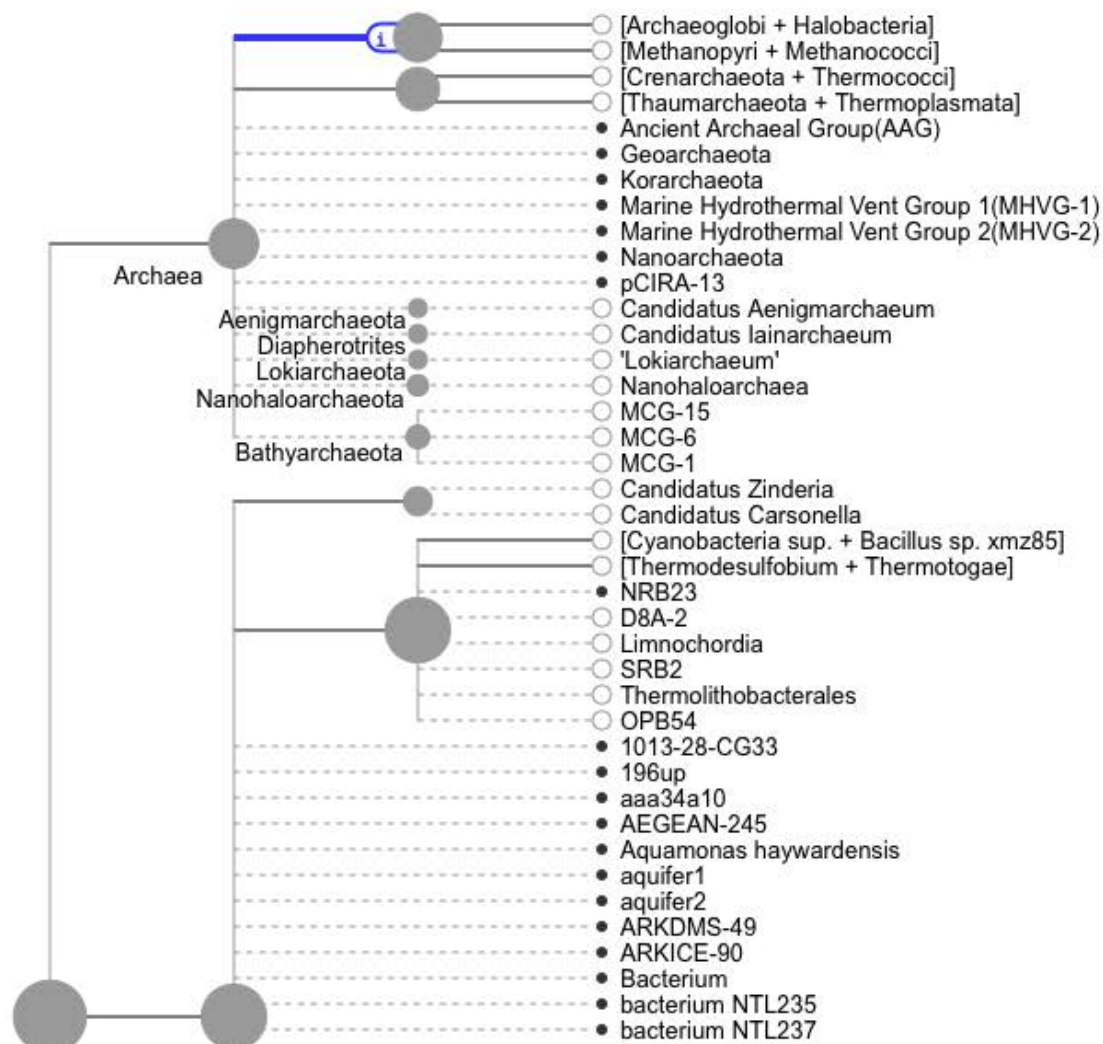
1 Show comments

Legend

Zoom tree view + -

Exploring the current synthetic tree

Hide properties

<https://tree.opentreeoflife.org>



APPLICATION

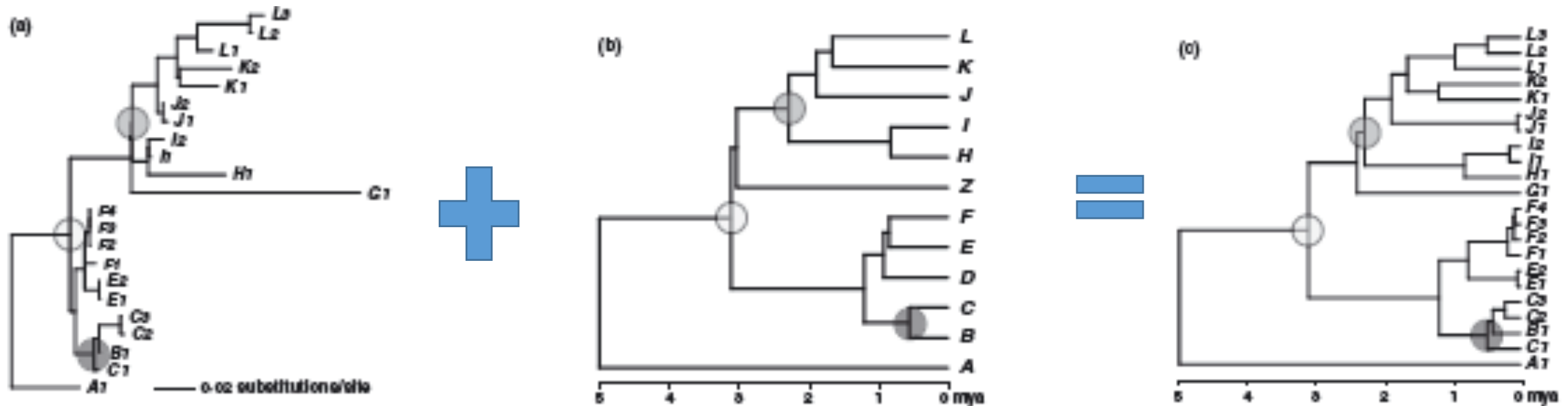
`rotl`: an R package to interact with the Open Tree of Life data

François Michonneau^{1,2*}, Joseph W. Brown³ and David J. Winter⁴

APPLICATION

Congruification: support for time scaling large phylogenetic trees

Jonathan M. Eastman^{1,2*}, Luke J. Harmon^{1,2} and David C. Tank^{2,3}



Target tree

Reference tree

Abordagens de megaphylogeny

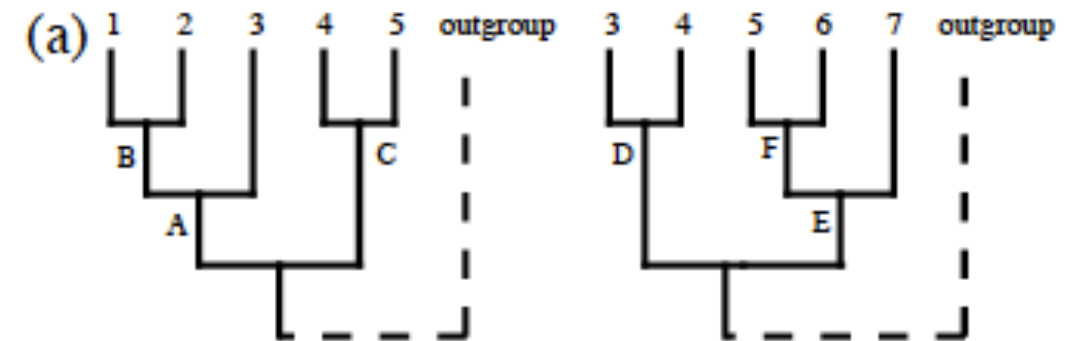
Estimativa de relações
filogenéticas para um
grande número de
espécies



Estudos ecológicos ou
comparativos em larga escala,
sem preocupação com taxonomia

Supertree

- Filogenias grande escala reconstruídas pela junção de outras menores, que não necessariamente devem ter o mesmo conjunto de espécies, conquanto que haja alguma sobreposição.
- Vantagens
 - Permitem combinar espécies fósseis com atuais
 - São necessários somente as topologias como dados de entrada.
 - Permitem combinar topologias obtidas com diferentes fontes de dado



(b)

	Component					
	A	B	C	D	E	F
outgroup	0	0	0	0	0	0
1	1	1	0	?	?	?
2	1	1	0	?	?	?
3	1	0	0	1	0	0
4	0	0	1	1	0	0
5	0	0	1	0	1	1
6	?	?	?	0	1	1
7	?	?	?	0	1	0

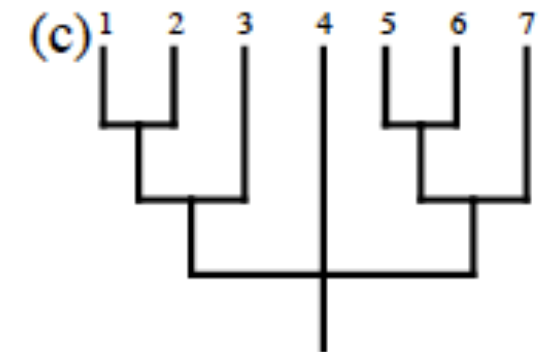


TABLE 1. Supertree methods examined in this work. +, – indicate objective functions that are maximized or minimized, respectively.

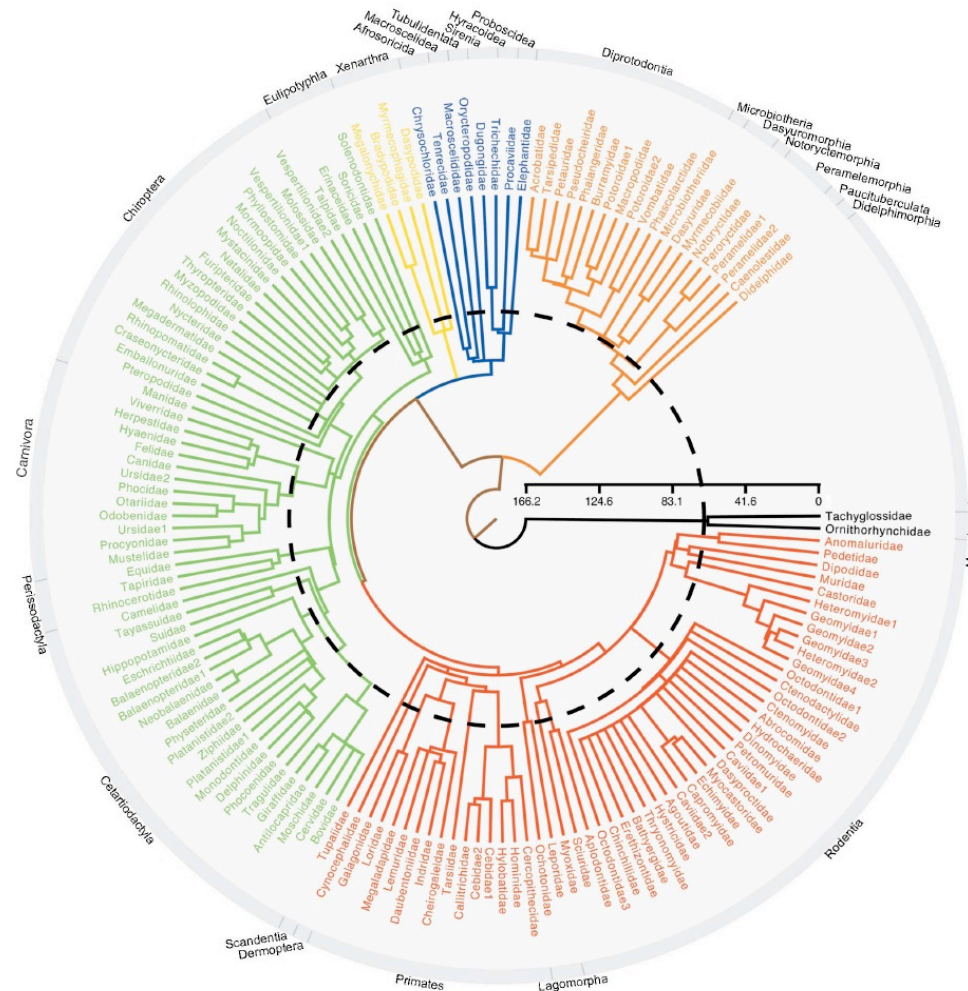
Method	Representation	Objective function
1. Standard MRP	Components	Fitch parsimony fit of composite matrix to supertree (–)
2. Irreversible MRP	Components	Camin-Sokal parsimony fit of composite matrix to supertree (–)
3. Purvis MRP	Sister groups	Fitch parsimony fit of composite matrix to supertree (–)
5. MinFlip	Components	Number of flips needed to render all input trees compatible with the supertree (–)
4. Split fit	Components	Number of matrix elements entailed by the supertree (+)
6. Triplet fit	Resolved triplets	Number of matrix elements entailed by the supertree (+)
7. Quartet fit	Resolved quartets	Number of matrix elements entailed by the supertree (+)
8. Average consensus	EBL distances (EBL)	Least squares fit of average matrix to supertree (–)
9. Average dendrogram	Ultrametric distances	Least squares fit of average matrix to ultrametric supertree (–)
10. MSS	EBL distances	Weighted sum of absolute differences between each tree and corresponding pruned supertree
11. MinCut	—	—
12. Modified MinCut	—	—
13. GTP (DL)	—	Number of duplications and losses needed to reconcile all input trees with the supertree
14. GTP (D)	—	Number of gene duplications to needed reconcile all input trees with the supertree

The delayed rise of present-day mammals

Olaf R. P. Bininda-Emonds^{1†}, Marcel Cardillo^{2†}, Kate E. Jones⁴, Ross D. E. MacPhee⁵, Robin M. D. Beck⁶, Richard Grenyer⁷, Samantha A. Price⁸, Rutger A. Vos⁹, John L. Gittleman¹⁰ & Andy Purvis^{2,3}

Vol 446 | 29 March 2007 | doi:10.1038/nature05634

nature



COMMENTARY

Open Access

Do we still need supertrees?

Arndt von Haeseler*

See research article <http://www.biomedcentral.com/1741-7007/10/12>

Críticas

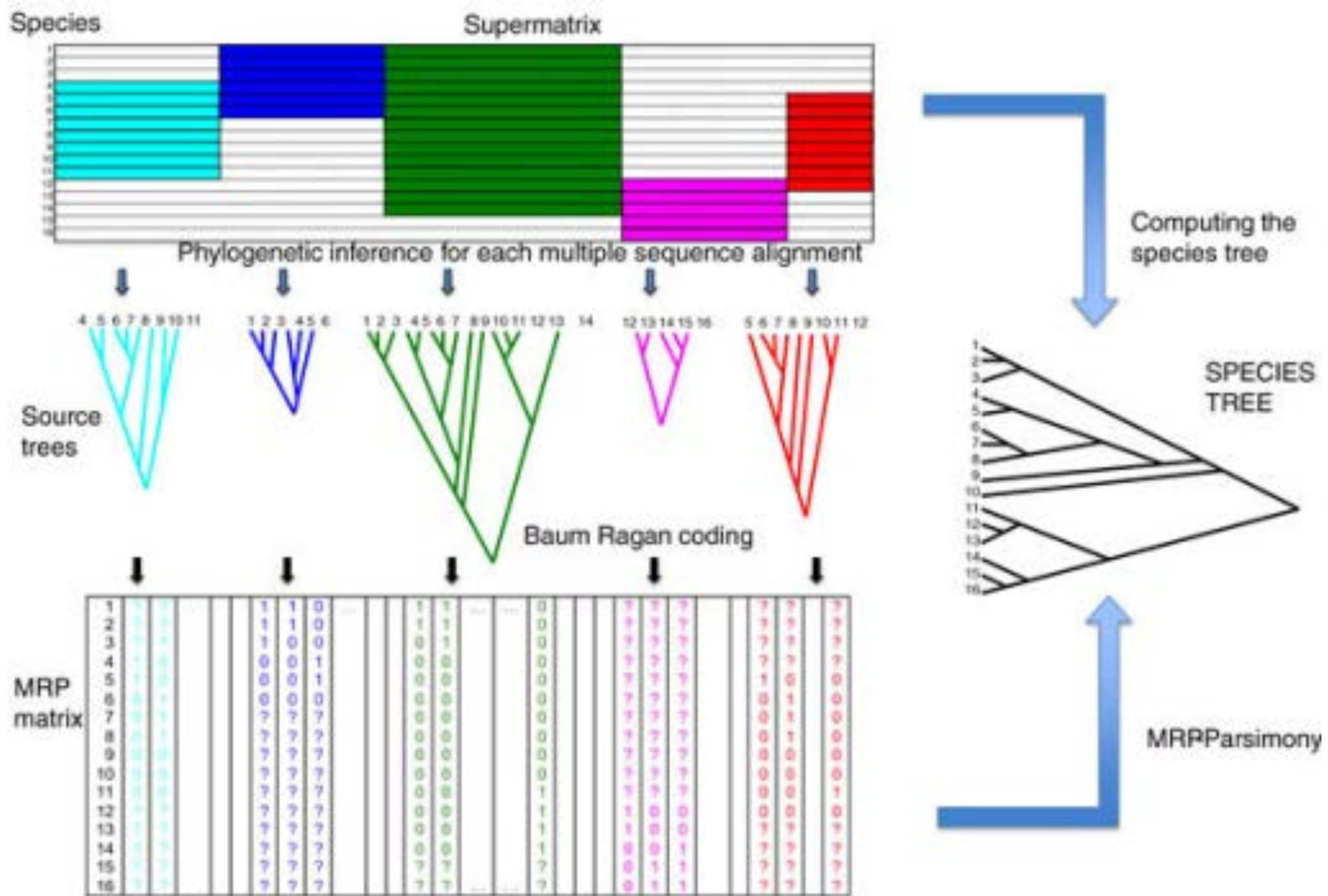
- Lidar com incongruências entre árvores construídas com diferentes métodos e dados não é simples
- Resolução e suporte é limitada às filogenias base menores, já que não distingue entre topologias com alto e baixo suporte. Incongruências acabam como politomias (que são muito comuns)
- Comprimento de ramo não tem significado (problema para análises comparativas)
- Resolução é bem baixa e não é confiável para representar relações muito finas (gênero e espécie)

Supermatrix

- Utiliza sequências de nucleotídeos depositadas no GenBank por estudos anteriores
 - Seleção de genes (marcadores) para minimizar missing data
 - Alinhamento
 - Estimativa de topologia utilizando algoritmos para dados em larga escala
- Limitações
 - GenBank não é padronizado, nomenclatura destoante, muitos dados ruins
 - Garbage in, garbage out
 - Dependendo do grupo existem poucos genes sequenciados
 - Alinhamento difícil ou de baixa qualidade
 - Gasta-se muito tempo filtrando e escolhendo sequências

Supermatrix

- Algoritmos para inferência de árvores que comportam grande número de terminais
 - RAxML
 - ExaML
 - ExaBayes
- Algoritmos para alinhamento
 - MAFFT
 - MUSCLE
 - ClustalW
 - T-Coffee
- Estimativa de divergência com muitos terminais
 - treePL



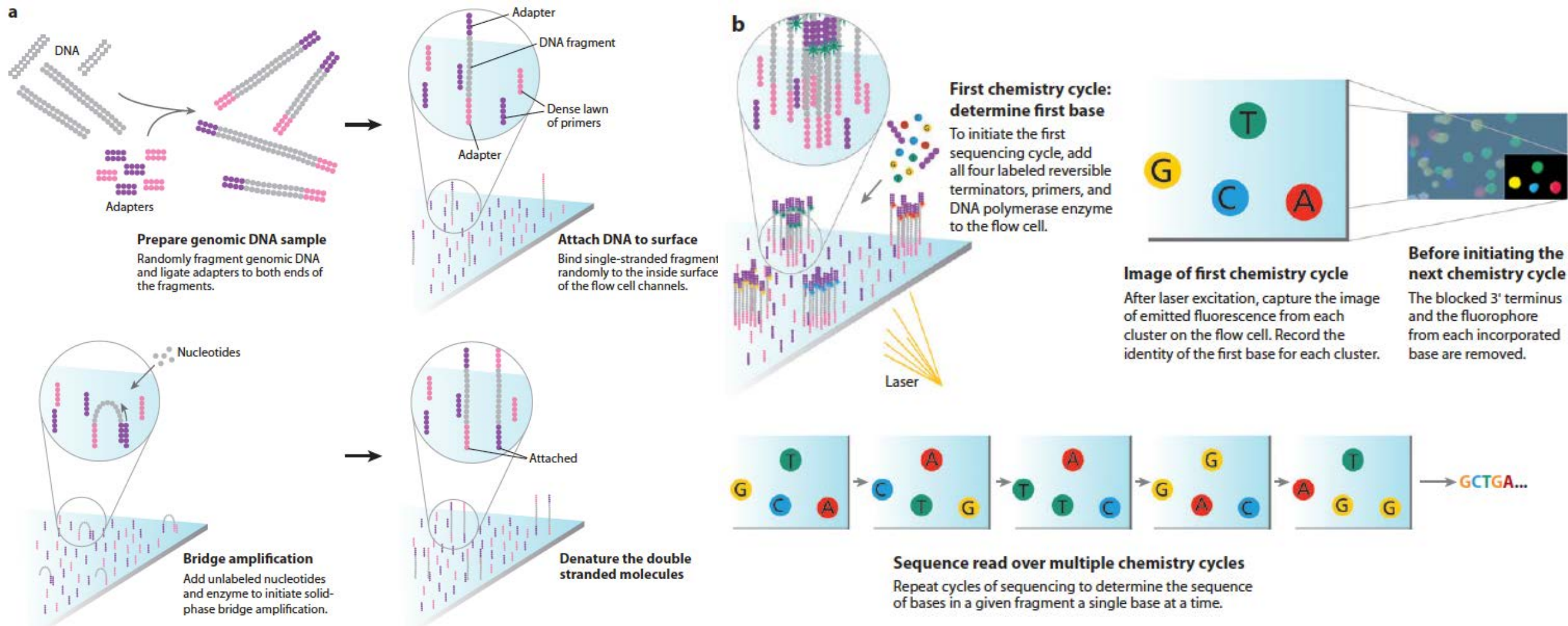
Next-generation sequencing e Phylogenomics

- Permitem sequenciar uma grande quantidade DNA e RNA de forma muito mais rápida e barata que o método Sanger comumente usado
- Geram uma quantidade enorme de dados para vários genes (marcadores) ao mesmo tempo (em paralelo)
- Possibilitam aumentar exponencialmente a quantidade de dados utilizados em reconstrução filogenética e filogeográfica
- Atualmente existem 4 plataformas comerciais
 - **Illumina (Solexa) sequencing**
 - Roche 454 sequencing
 - Ion torrent: Proton / PGM sequencing
 - SOLiD sequencing



© 2014 Illumina, Inc. All rights reserved.

Workflow Illumina NGS

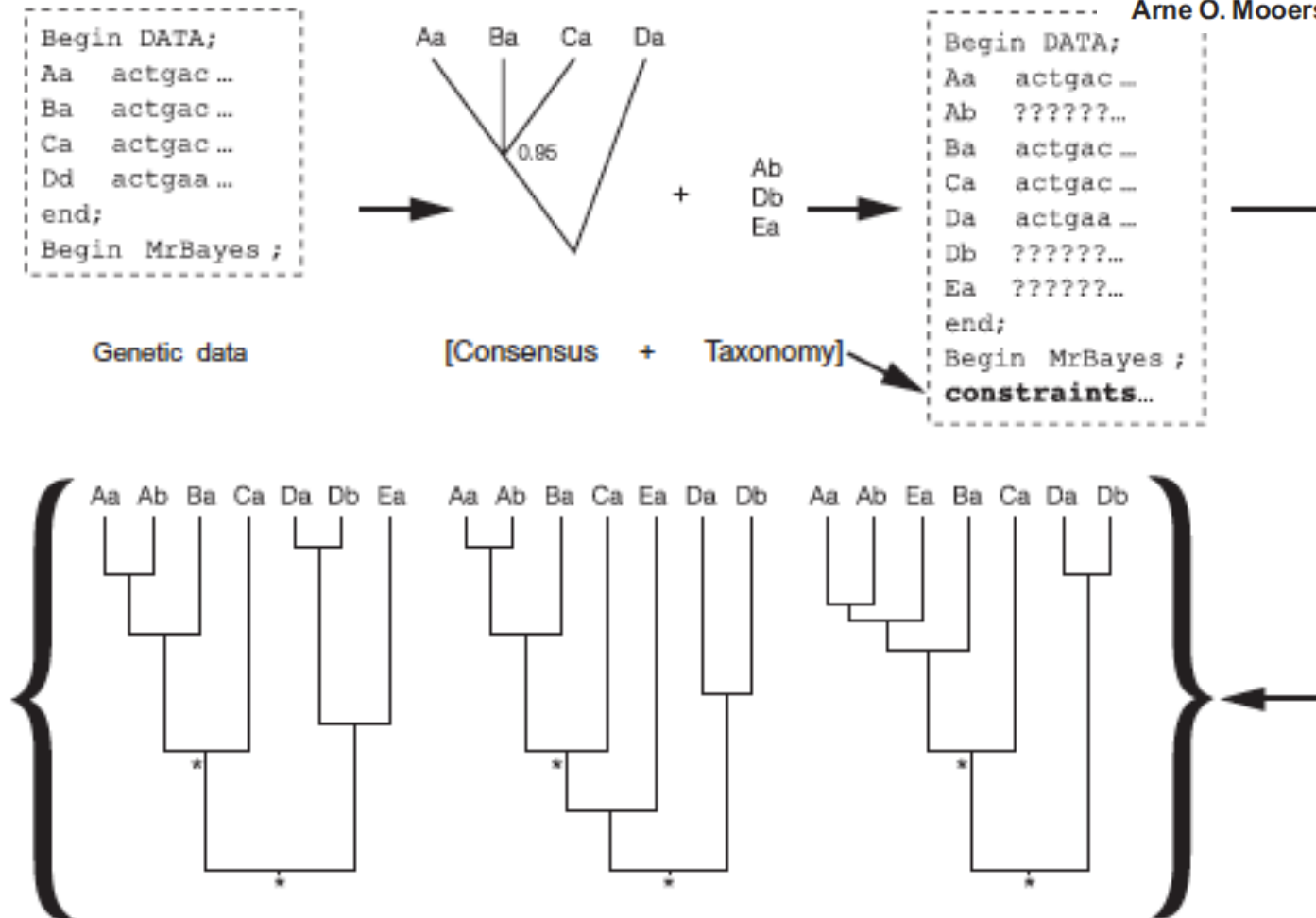


Pipelines (semi)automatizadas “for dummies”

APPLICATION

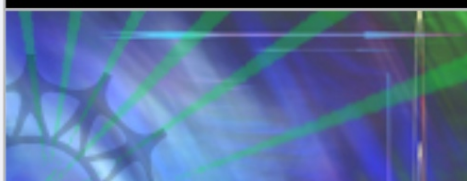
PASTIS: an R package to facilitate phylogenetic assembly with soft taxonomic inferences

Gavin H. Thomas^{1*†}, Klaas Hartmann^{2†}, Walter Jetz³, Jeffrey B. Joy⁴, Aki Mimoto⁴ and Arne O. Mooers⁴





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Award Abstract #1441737

Collaborative Research: VertLife Terrestrial: A complete, global assembly of phylogenetic, trait, spatial and environment characteristics for a model clade

NSF Org:

[DEB](#)

[Division Of Environmental Biology](#)

Initial Amendment Date: September 8, 2014

Latest Amendment Date: December 4, 2015

Award Number: 1441737

Award Instrument: Standard Grant

Program Manager: Simon Malcomber
DEB Division Of Environmental Biology
BIO Direct For Biological Sciences

Start Date: October 1, 2014

[illegible]

http://www.onezoom.org/EDGE_birds.htm

Federal Demonstration Partnership

October 1, 2014

A GLOBAL DIVISION OF



Contents lists available at [ScienceDirect](#)

Biological Conservation

journal homepage: www.elsevier.com/locate/bioc



Fully-sampled phylogenies of squamates reveal evolutionary patterns in threat status

João Filipe Riva Tonini^{a,*}, Karen H. Beard^b, Rodrigo Barbosa Ferreira^{b,c}, Walter Jetz^d, R. Alexander Pyron^a

^a Department of Biological Sciences, The George Washington University, 2029 G St NW, Washington, DC 20052, USA
^b Department of Wildland Resources and the Ecology Center, Utah State University, Logan, UT 84322-5230, USA
^c Laboratório de Ecologia de Populações e Conservação, Universidade Vila Velha, Rua Comissário José Dantas de Melo 21, Boa Vista, Vila Velha, ES 29102-920, Brazil
^d Department of Ecology and Evolutionary Biology, Yale University, 165 Prospect Street, New Haven, CT 06520, USA

<http://species.mol.org/info/>
Explore results on OneZoom tree:
http://www.onezoom.org/EDGE_birds.htm

Cooperative Agreement Conditions Special Conditions Federal Demonstration Partnership	Program Manager:	Simon Malcomber DEB Division Of Environmental Biology BIO Direct For Biological Sciences
	Start Date:	October 1, 2014



Ecography 35: 001–014, 2012

doi: 10.1111/j.1600-0587.2012.07773.x

© 2012 The Authors. Ecography © 2012 Nordic Society Oikos

Subject Editor: Ken Kozak. Accepted 23 October 2012

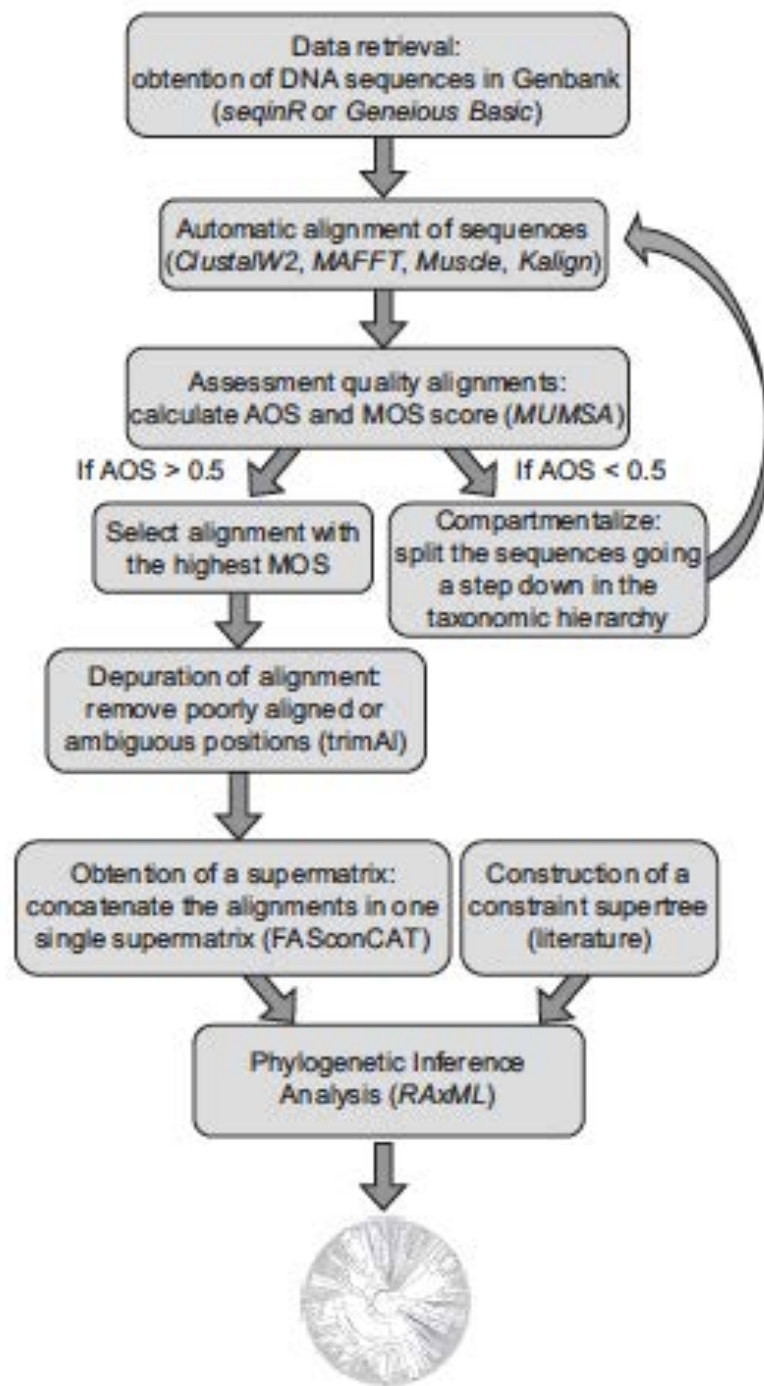
Building megaphylogenies for macroecology: taking up the challenge

Cristina Roquet, Wilfried Thuiller and Sébastien Lavergne



Building megaphylogeny challenge

Cristina Roquet, Wilfried Thuiller



Ecography 35: 001–014, 2012
doi: 10.1111/j.1600-0587.2012.07773.x
Authors. Ecography © 2012 Nordic Society Oikos
Act Editor: Ken Kozak. Accepted 23 October 2012

y: taking up the

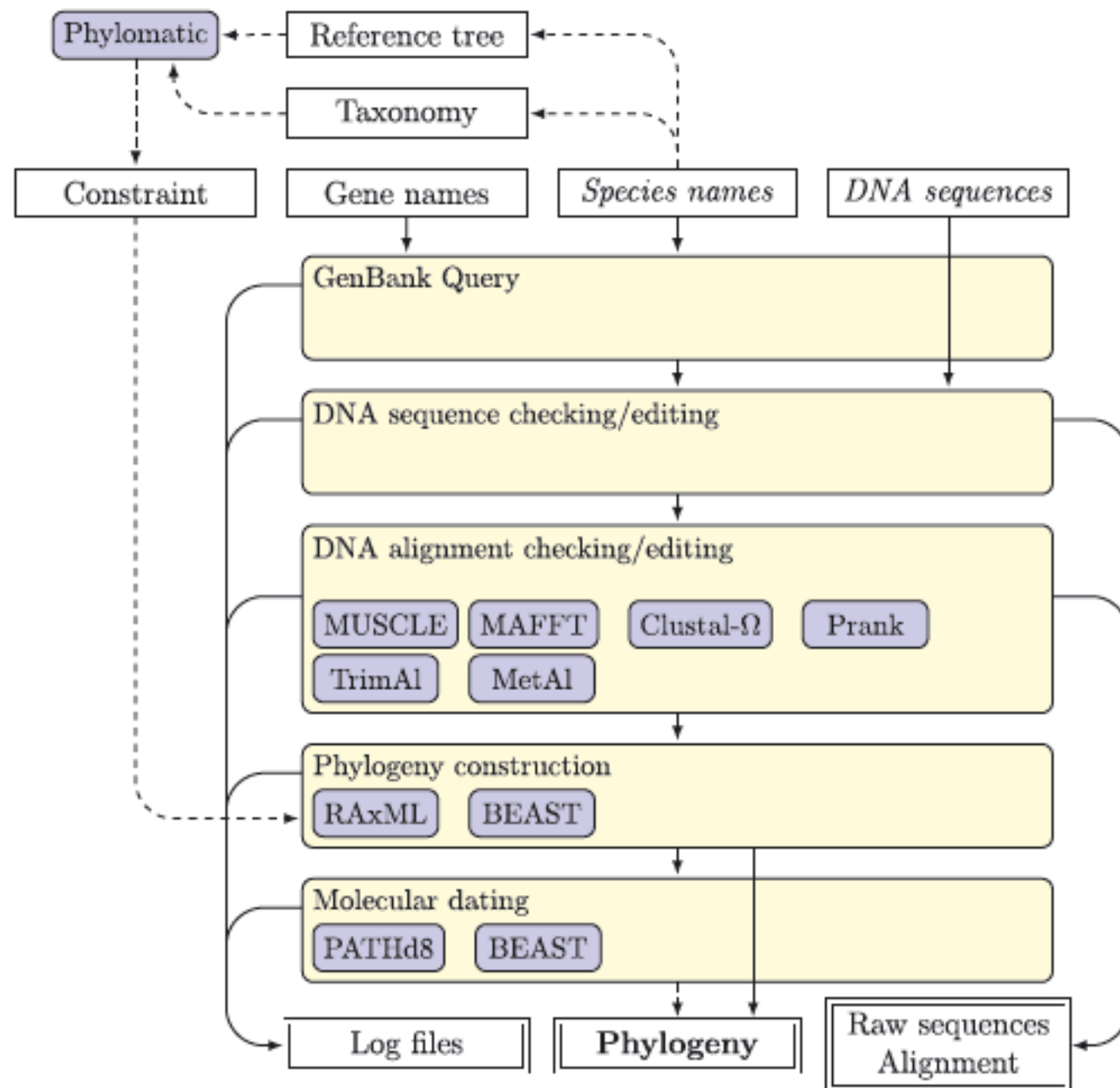
APPLICATION

phyloGenerator: an automated phylogeny generation tool for ecologists

William D. Pearse^{1,2*} and Andy Purvis¹

Table 3. Programs with features similar to phyloGenerator. In order from left to right, each column describes whether a program: downloads DNA data from the Internet, aligns DNA data, heuristically searches for an acceptable phylogeny, doesn't require the user to manually customise or run its subcomponents, conducts analyses on the user's computer and attempts to check the user's data or output for obvious sources of error. In each column, ✓ and × indicate whether a program does or does not have a feature, respectively; Phylomatic does not attempt to build a novel phylogeny and so is listed as NA under some columns

Program	DNA download	DNA alignment	Tree search	One-click	Local	Sanity checks
Phylomatic (Webb <i>et al.</i> 2008)	NA	NA	NA	✓	✓	×
Peters <i>et al.</i> (2011)	✓	✓	✓	×	✓	×
rPlant (Banbury <i>et al.</i> 2012)	×	✓	✓	×	×	×
GeneFinder (Lanfear & Bromham 2011)	✓	×	×	×	✓	×
SATé-II (Liu <i>et al.</i> 2012)	×	✓	✓	✓	✓	×
phyloGenerator	✓	✓	✓	✓	✓	✓



PUmPER: phylogenies updated perpetually

Fernando Izquierdo-Carrasco^{1,*}, John Cazes², Stephen A. Smith³ and
Alexandros Stamatakis^{1,4}

- Bem complicado de instalar e usar
- Não é atualizado muito frequentemente

Manual

Installation

GenBank database management

Quick start run

Run options

Guide trees

Outlier detection

Sequence similarity

Advanced searches using SQL

Updating runs

User sequences

Issues

Databases

Publications

NEWS AND UPDATES

New methods

New feature: advanced queries with SQL

outlier detection

no more phyutility dependency

New division codes

Welcome

Here you will find...

- [instructions](#)
- [news](#)
- [prebuilt databases](#)
- [publications](#)

[link](#)

New methods

We are currently developing new methods for homology detection to continue to improve results in PHLAWD. In particular, to reduce the interaction and need for corrections. Developments will be pushed to github ASAP.

[link](#)

New feature: advanced queries with SQL

PHLAWD now has the ability to accept raw SQL query strings to use when searching the source database for potential sequences of interest. This provides a great deal more power than the default search functionality that simply matches against a set of terms. Now one can use exclusion, be explicit about the treatment of flanking whitespace, use all available SQL wildcards, and use boolean operators such as AND and OR, among other things.

Of course, with great power comes great responsibility...

For more information see the [manual page for SQL queries](#)

Fork me on GitHub

Syst. Biol. 66(2):152–166, 2017

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DOI:10.1093/sysbio/syw066

Advance Access publication August 19, 2016

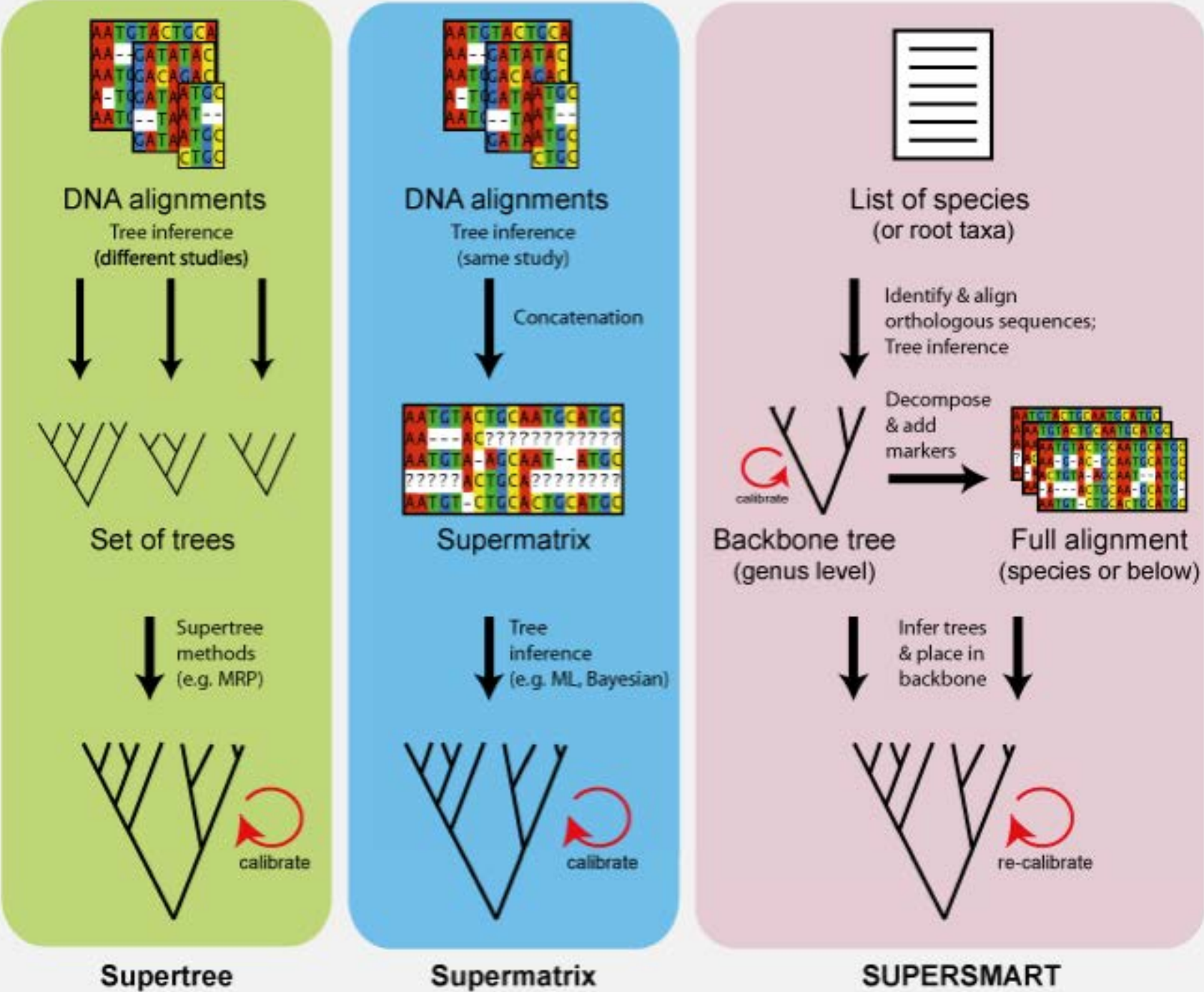
Toward a Self-Updating Platform for Estimating Rates of Speciation and Migration, Ages, and Relationships of Taxa

ALEXANDRE ANTONELLI^{1,2,*}, HANNES HETTLING³, FABIEN L. CONDAMINE^{1,4}, KARIN VOS¹, R. HENRIK NILSSON¹,
MICHAEL J. SANDERSON⁵, HERVÉ SAUQUET⁶, RUUD SCHARN¹, DANIELE SILVESTRO^{1,7}, MATS TÖPEL^{8,9},
CHRISTINE D. BACON¹, BENGT OXELMAN¹, AND RUTGER A. VOS³

Syst. Biol. 66(2):152–166, 2017.
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DOI:10.1093/sysbio/syw001
Advance Access publication: February 1, 2017

Toward a S

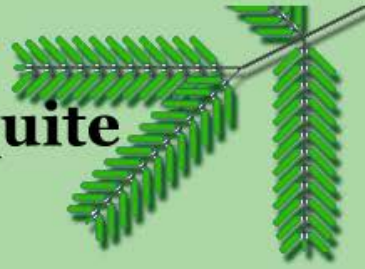
ALEXANDRI
MICH



on, Ages,

ILSSON¹,
,9,
,

Mesquite


 Search

Mesquite Home Page

Overview of Mesquite

- What Mesquite Does
- List of Features
- Development Team

Getting Started

- Download and Installation
- Additional Packages
- Beginning Mesquite
- Example Files

Basics

- Files
- Menus
- Windows
- Charts
- Selection
- Modules
- Simplifying Mesquite

mesquiteproject.wikispaces.com

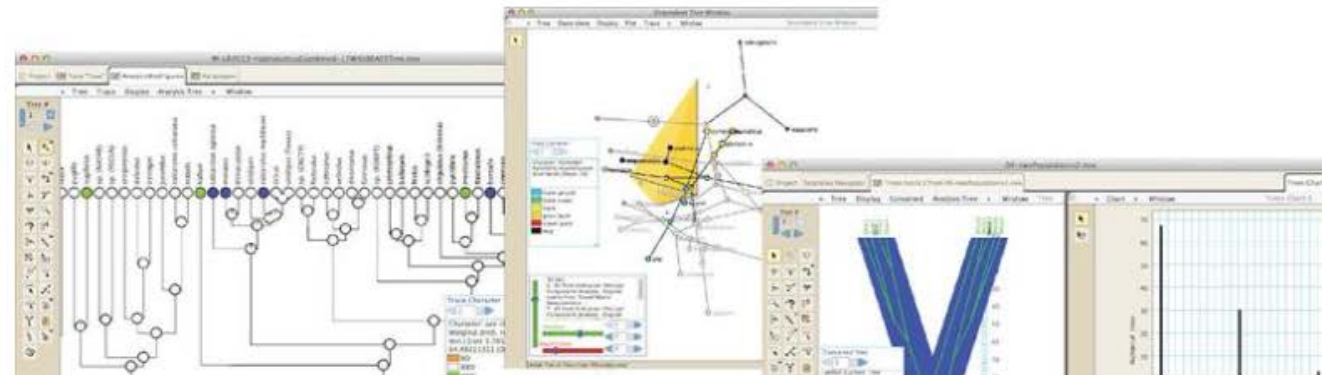
If you're looking for Mesquite Software, Inc. or its CSIM toolkit for building simulation models, go [here](#)

Mesquite: A modular system for evolutionary analysis

Current release version: 3.04 [[download](#)]

If you have a previous version, please [update](#) because of [bug fixes and new features](#).

Mesquite is modular, extendible software for evolutionary biology, designed to help biologists organize and analyze comparative data about organisms. Its emphasis is on phylogenetic analysis, but some of its modules concern population genetics, while others do non-phylogenetic multivariate analysis. Because it is modular, the analyses available depend on the modules installed.



Última opção: fazer uma topologia sem comprimento de ramo na mão!



YouTube BR

Search: phylocom

Enviar

Safari Arquivo Editar Visualizar Histórico Favoritos Janelas Ajuda

TreeBASE Web

treebase.org/treebase-web/home.html

TreeBASE A Database of Phylogenetic Knowledge

Search TreeBASE
Submission Tutorial
Submit
About
Overview
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People
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References
NSF Data Management
Data Access
Journals
Contact

Welcome to TreeBASE

TreeBASE is a repository of phylogenetic information, specifically user-submitted phylogenetic trees and the data used to generate them. TreeBASE accepts all types of phylogenetic data (e.g., trees of species, trees of populations, trees of genes) representing all biotic taxa. Data in TreeBASE are exposed to the public if they are used in a publication that is in press or published in a peer-reviewed scientific journal, book, conference proceedings, or thesis. Data used in publications that are in preparation or in review can be submitted to TreeBASE but are only available to the authors, publication editors, or reviewers using a special access code. TreeBASE is produced and governed by the The Phyloinformatics Research Foundation, Inc.

Some recent additions:

- Historical biogeography and diversification of truffles in the Tuberales and their newly identified Southern hemisphere sister lineage
November 02, 2012
- The analysis of rRNA sequence-structure in phylogenetics: An application to the family Pectinidae (Mollusca: Divalvia)
November 02, 2012
- Floral paedomorphy leads to secondary specialization of Malagasy Dalechampia (Euphorbiaceae)
November 02, 2012
- A dated phylogeny of the papaya family (Caricaceae) reveals the crop's closest relatives and the family's biogeographic history
November 02, 2012

0:03 / 26:43

Analytics Gerenciador de vídeos

Reprodução automática

Como datar uma filogenia usando o Phylocom 4.2

Diogo Borges Provete

Configurações do canal

Próximo

Introduction to phytools and phangorn: Phylogenetics tools for R
phyloseminar.org
2.256 visualizações

59:38

O maior bate-papo sobre GUERRA CIVIL I

Datação dessa filogenia utilizando o BLADJ a partir de tempos de divergência obtidos em bases de dados (e.g., TimeTree)

Bases de dados de atributos

Bases de datos online de atributos

Ecological Data Wiki

Welcome back, data wiki users

The Ecological Data Wiki has been updated to a newer version of Drupal. If you are a returning user, you may need to [request a new password](#) to log in.

[HOME](#) [FIND DATA](#) [GETTING STARTED](#) [HELP](#) [LOGIN](#)

Welcome to the Ecological Data Wiki

[View](#) [Revisions](#)

Ecological data is everywhere, but how do you find what you need and learn how to use it?

The Data Wiki helps scientists collaborate on gathering information about data and how to use it.

The site is a source for finding ecological datasets and quickly figuring out the best ways to use them.



Recent comments

- Can someone point me in the 8 months 1 week ago
- Thanks Jessica. I just 9 months 1 week ago
- Thank you so much for this 9 months 1 week ago

Bases de datos online de atributos

Ecological Data Wiki

[Home](#)[About TRY](#)[Data Portal](#)[Feedback](#)[Registration](#)

Quantifying and scaling global plant trait diversity

TRY is a network of vegetation scientists headed by [Future Earth](#) and the [Max Planck Institute for Biogeochemistry](#), providing a global archive of curated plant traits. The TRY database is a research platform of [iDiv](#).

5.6 million trait records
100,000 plant species
largely open access

[Data Portal](#)

News

Paper published (2016-06-15)
Hébert F. et al.: Recovery of plant community functional traits following severe soil perturbation in plantations: a case-study, *International Journal of Biodiversity Science, Ecosystem Services & Management*. ([link](#))

Paper published (2016-06-13)
Negolta L. et al.: Isolation-driven functional assembly of plant communities on islands. *Ecography*. ([link](#))

Activity Report (2016-06-01)
In May 2016, TRY received 133 requests and released 9.7 million trait records for 92 requests; 1 new publication was reported. This brings the totals to 2105 received requests, 179 million trait records released for 1594 requests, and 76

ou may need to [request a new password](#) to log in.



Recent comments

- [Can someone point me in the](#) 8 months 1 week ago
- [Thanks Jessica. I just](#) 9 months 1 week ago
- [Thank you so much for this](#) 9 months 1 week ago

them.



navegação

- [Página Inicial](#)
- [About](#)
- [Trait research](#)
 - [Trait research – Protocols](#)
 - [Trait research – Core references](#)
 - [Trait research – Datasets](#)
 - [Trait research – List of Datasets](#)
- [TraitNet Participants](#)
- [Ecoinformatics](#)
- [Using the Site](#)

Trait research – List of Datasets

List of databases that contain plant trait data:

- [TRY](#) – 65 datasets from more than 90 contributors. It contains about 2.400.000 trait entries for more than 64.000 plant species and about 1000 different traits, covering a variety of biomes and geographic areas
- [GLOPNET](#) – A database of plant traits from different biomes
- [LEDA](#) – Database on the life history traits of Northwest European flora
- [BROT](#) – Plant trait database for Mediterranean Basin species
- [USDA Plants](#) – Database provides standardized information about the vascular plants, mosses, liverworts, hornworts, and lichens of the U.S. and its territories.
- [CalFlor](#) – Database of the flora of California, USA
- [NY-Flora](#) – Database of the flora of New York, USA
- [UConn](#) – Database of flora of Connecticut, USA
- [NZ](#) – Database of ecological traits of New Zealand

[acesse para adicionar comentários](#)

**100,000 plant species
largely open access**

Data Portal

news

Registry of plant trait data – COMING SOON!

word to log in.

Plant Functional Traits Ontology – COMING SOON!



nts

point me in the 8 months 1 week ago

. I just 9 months 1 week ago

nuch for this 9 months 1 week ago

Negoita L. et al.: Isolation-driven functional assembly of plant communities on islands. *Ecography*. [\(link\)](#)

Activity Report (2016-06-01)

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TraitNet

você está aqui: página inicial → trait research → trait research – list of datasets

navegação

- [Página Inicial](#)
- [About](#)
- [Trait research](#)
 - [Trait research – Protocols](#)
 - [Trait research – Core references](#)
 - [Trait research – Datasets](#)
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Trait research – List of Datasets

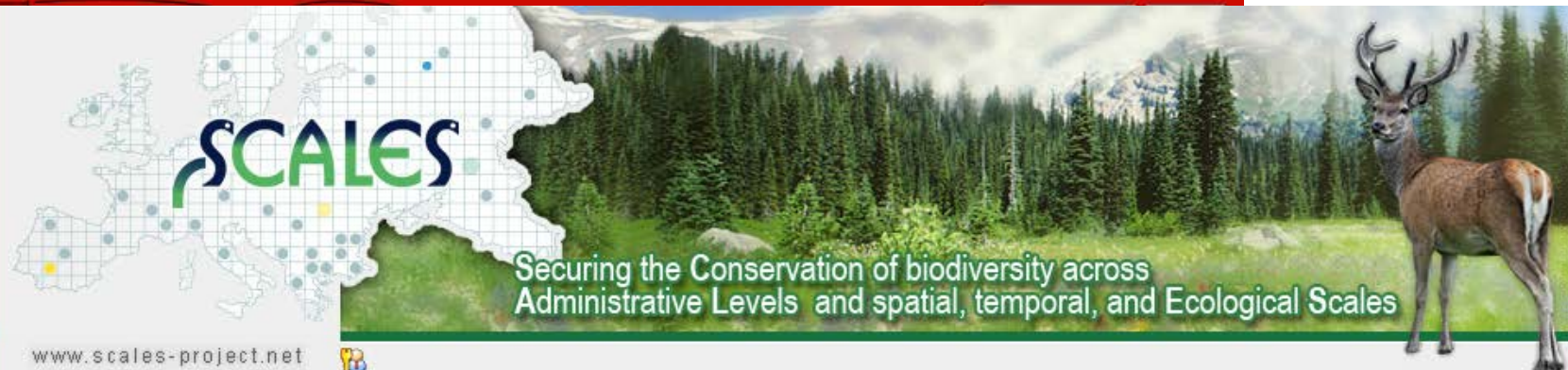
List of databases that contain plant trait data

- [TRY](#) – 65 datasets from more than 1000 different traits, covering a wide range of plant functional traits
- [GLOPNET](#) – A database of plant traits from 1000 different species
- [LEDA](#) – Database on the life history traits of plants
- [BROT](#) – Plant trait database for the British Isles
- [USDA Plants](#) – Database providing information on the distribution and ecology of plants in the United States
- [CalFlor](#) – Database of the flora of California
- [NY-Flora](#) – Database of the flora of New York
- [UConn](#) – Database of flora of Connecticut
- [NZ](#) – Database of ecological traits of New Zealand plants

[acesse para adicionar comentários](#)

100,000 plant
largely open

Data Portal



www.scales-project.net

SCALETOOL

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Models and tools

Species traits databases

[Reptile Trait Database](#)[Minimum Area Requirements of species](#)[Plant Trait Database](#)[Wasps & Bees Database](#)[BioMat](#)

[Home](#) [Species traits databases](#)

Species traits databases

Species traits describe characteristics of species and are relevant to population dynamics across space and time. They affect ecosystem processes indirectly through abiotic control and directly through changes in biotic control (e.g. predators, pathogens). On the other hand, they are also affected by other species traits and environmental conditions which are highly sensitive to global change pressures, such as land use changes and climate change. Since species traits seem to play one of the most important roles in ecosystem processes and are thus highly relevant for conservation planning, it is inevitable to have overviews and compilations for different species available.

For that purpose, existing databases and review literature on species traits and, in the following, dispersal related traits, were compiled. This compilation also includes derived traits like functional traits, area requirements, and habitat selectivity to dispersal and functional connectivity. These data provide an excellent overview on species traits, which can be used for parameterizing model inputs, and will help to assess and visualize scaling effects across taxa and spatial scales. Databases for plants, insects, reptiles, and birds across Europe are either available online or upon author request.

[Ecography](#) ([link](#))

Activity Report (2016-06-01)

In May 2016, TRY received 133 requests and released 9.7 million trait records for 92 requests; 1 new publication was reported. This brings the totals to 2105 received requests, 179 million trait records released for 1594 requests, and 76



[Home](#)



Building the Tree of Life with phenotypes

FOR SCIENTISTS
Use the Tools

FOR SCIENTISTS & THE PUBLIC
See Published Research

Comparative biologists at work with these tools now....

SEE TOTAL ACTIVITY

89
SCIENTISTS WORKING

2864
SITE VISITORS

69444
CELLS SCORED

1632
IMAGES UPLOADED

21026/1389
PROJECT
VIEWS/DOWNLOADS

8751/117
MATRIX
VIEWS/DOWNLOADS

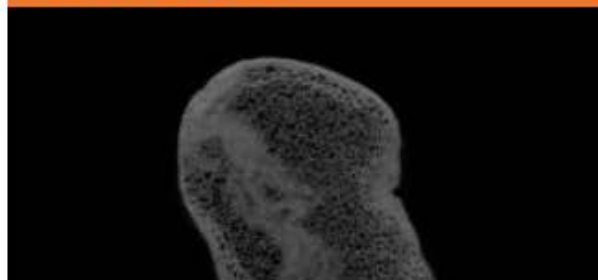
79170/14
MEDIA
VIEWS/DOWNLOADS

Stats for Last 30 Days

LATEST VISITORS



Featured Project



Citation Information

**How to
cite**



Comparative biology

89
SCIENTISTS WORKING

2864
SITE VISITORS

Stats for Last 30 Days

LATEST VISITORS



TraitBank

TraitBank® is a [searchable](#), comprehensive, open digital repository for organism traits, measurements, interactions and other facts for all taxa across the tree of life.

TraitBank currently features over [11 million records](#) related to more than [330 attributes](#) for [1.7 million taxa](#) obtained from over [50 data sources](#).

[Search TraitBank](#)

Popular attribute types





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English | [Español](#) | [Português \(Br, Pt\)](#) | [Français](#) | [Deutsch](#) | [Italiano](#) | [Nederlands](#) | [简体中文](#) | [繁體中文](#) | [日本語](#) | [More...](#)



(33200 Species, 317200 Common names, 56900 Pictures,
52600 References, 2230 Collaborators, 700000
Visits/Month)



FishBase consortium



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[FishWatcher](#) | [Ichthyology Course](#) | [LarvalBase](#) | [Team](#) | [Collaborators](#) | [Quick Identification](#) | [Services](#)

Common Name

is (e.g. rainbow trout)

[A](#) [B](#) [C](#) [D](#) [E](#) [F](#) [G](#) [H](#) [I](#) [J](#) [K](#) [L](#) [M](#) [N](#) [O](#) [P](#) [Q](#) [R](#) [S](#) [T](#) [U](#) [V](#) [W](#) [X](#) [Y](#) [Z](#)

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Scientific Name

☐ [Advanced Match](#)

Genus is (e.g. Rhincodon)

Species is (e.g. typus) ☐ Random Species

Genus + Species Sp. ID

[A](#) [B](#) [C](#) [D](#) [E](#) [F](#) [G](#) [H](#) [I](#) [J](#) [K](#) [L](#) [M](#) [N](#) [O](#) [P](#) [Q](#) [R](#) [S](#) [T](#) [U](#) [V](#) [W](#) [X](#) [Y](#) [Z](#)

[Why name assessments may be different](#) between FishBase and the independent [Catalog of Fishes \(Eschmeyer, 2014\)](#)

organism traits, measurements, interactions and other

[Search TraitBank](#)

n [330 attributes](#) for [1.7 million taxa](#) obtained from





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[DRYAD](#)

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***** Please note that all manuscripts submitted after 1 January 2016 follow separate guidelines for supporting information!**

***** Ecological Archives through the end of 2015 will be hosted on FigShare once the transition to publishing with Wiley is completed. Thereafter, supplemental material may be hosted on Wiley Online, and/or data deposited with FigShare, Dryad, and other repositories.**

What is Ecological Archives?

Ecological Archives publishes materials that are supplemental to articles that appear in the ESA journals (*Ecology*, *Ecological Applications*, *Ecological Monographs*, *Ecosphere*, *Ecosystem Health and Sustainability* and *Bulletin of the Ecological Society of America*), as well as peer-reviewed data papers with abstracts published in the printed journals. *Ecological Archives* is published in digital, Internet-accessible form.

Three kinds of publications appear in *Ecological Archives*: [appendices](#), [supplements](#), and [data papers](#). However, these types vary among the journals.

TABLE 1. *Ecological Archives* categories published by journal.

	Appendices	Supplements	Data Papers
<i>Ecosphere</i>	Yes†	Yes	No
<i>Ecology</i>	Yes	Yes	Yes

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SCIENTIFIC DATA



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Structural biology applications of X-ray lasers

Featured Collection

van der Schot et al. (2016) - CC-BY 4.0

Data Descriptor | 01 August 2016 | OPEN

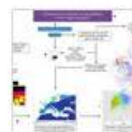
Global positioning system survey data for active seismic and volcanic areas of eastern Sicily, 1994 to 2013



Alessandro Bonforte, Sonia Fagone [...] Benedetto Saraceno

Data Descriptor | 01 August 2016 | OPEN

Species interactions in occurrence data for a community of tick-transmitted pathogens



Agustín Estrada-Peña & José de la Fuente

Announcement

The 10 principles of open research data

Aug 4 | by Iain Hrynaszkiewicz

scientificdata
updates

Q

All Author Title

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Resolving the publishing bottleneck for biodiversity

Science is a combination of gathering facts and making theories; neither can progress on its own. In the history of science, the laborious accumulation of facts is the dominant mode, not a novelty.

Peter Norvig

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-  [Most Active Editors](#)
-  [Most Active Reviewers](#)

[biodiversitydatajournal.com/browse_articles](#)[esauhs.org/archive/default.htm](#)[Ecology](#)[Yes](#)[Yes](#)[Yes](#)

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Publish with Scientific Data

Scientific Data is a peer-reviewed, open-access journal for descriptions of research datasets

Data Descriptor | 22 August 2017 | **OPEN**

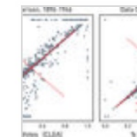
A new global anthropogenic heat estimation based on high-resolution nighttime light data



Wangming Yang, Yibo Luan [...] Xuefeng Cui

Data Descriptor | 15 August 2017 | **OPEN**

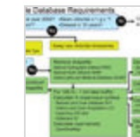
Spatiotemporal database of US congressional elections, 1896–2014



Levi John Wolf

Data Descriptor | 08 August 2017 | **OPEN**

Long-term chloride concentrations in North American and European freshwater lakes



Hilary A. Dugan, Jamie C. Summers [...] Kathleen C. Weathers

Search Scientific Data

All Subjects ▾



Realizando análises comparativas na nuvem



Arbor workflows

Visual workflows for phylogenetic comparative methods.

Analyze your data



Arbor is web-based software for carrying out phylogenetic comparative analyses of the tree of life.

Develop tools

Create your own functions and workflows

Teach

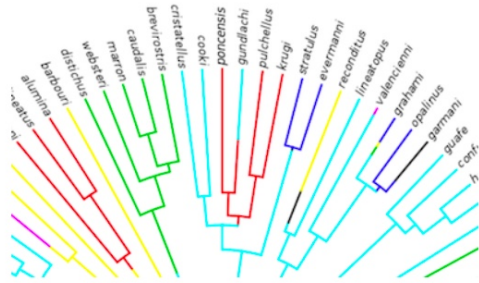
Use Arbor in the classroom

Get help

How to use Arbor

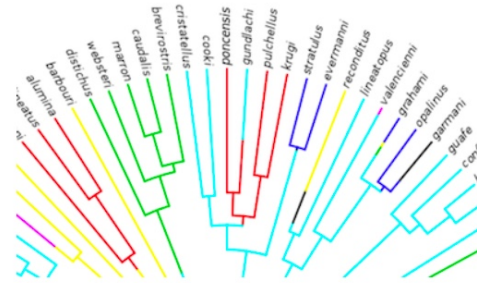
Arbor Apps

Comparative methods using a drag-and-drop interface



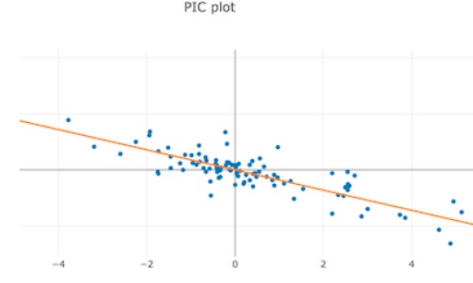
May 31, 2017

Plot tree using d3



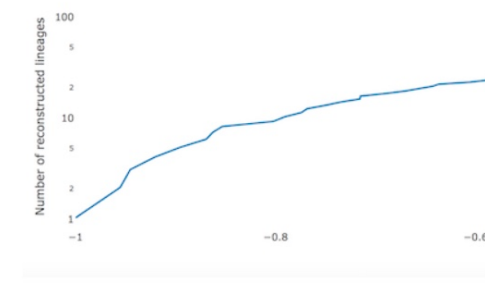
May 31, 2017

Stochastic character mapping using simmap



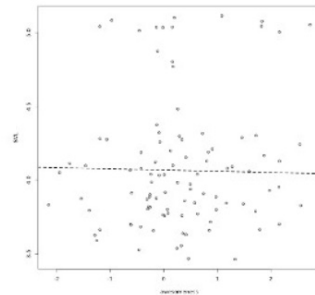
May 31, 2017

Phylogenetic independent contrasts



May 31, 2017

Lineage-through-time plots



May 30, 2017

PGLS - two variables only (for now!)



May 29, 2017

Ancestral state reconstruction

Reconstruct character evolution on a phylogenetic tree.



May 28, 2017

Phylogenetic signal

Tutorials

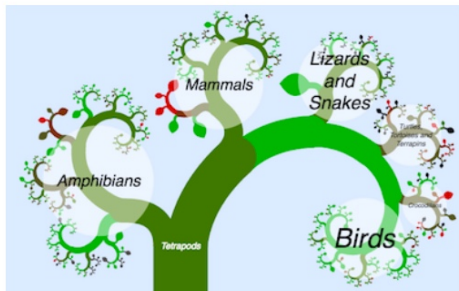
A collection of tutorials for Arbor.



January 08, 2017

Web services 2

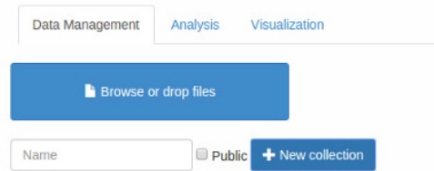
What are web services?



April 18, 2016

Megaphylogenies

What is a megaphylogeny?



January 06, 2017

Creating new collections and functions

How to make your own collection of Arbor functions and workflows



March 11, 2016

Web services 1

What are web services?



January 05, 2017

Using the Arbor webapp

How to run functions for comparative analyses in Arbor



March 03, 2016

Arbor apps

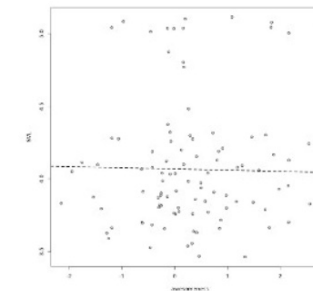
What are Arbor apps?



April 18, 2016

Timetrees

Creating timetrees



July 17, 2015

PGLS

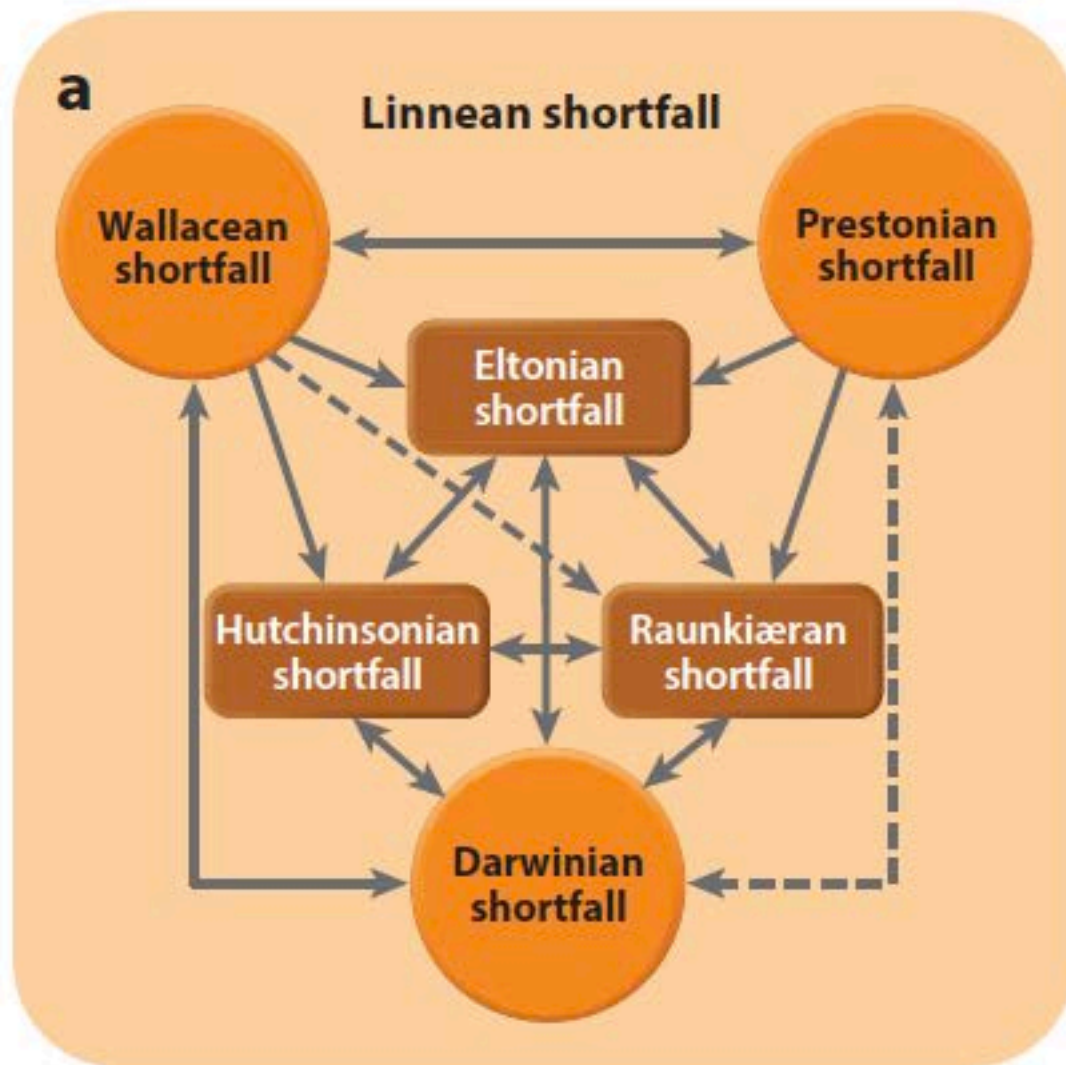
Use PGLS to test for a correlation

Desenho de estudos comparativos

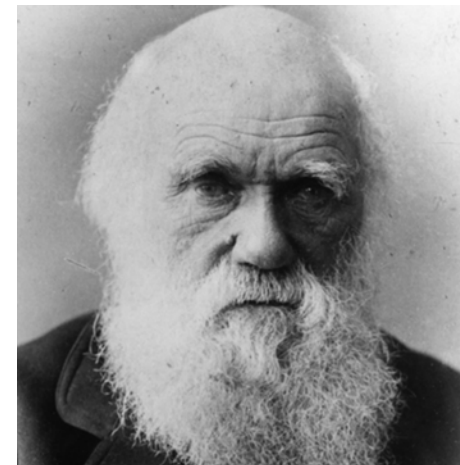
Como maximizar a amostragem e lidar com *mismatch*
de dados

Problemas mais comuns

- Falta completa de dados (filogenéticos e traits)
- **Descasamento (*mismatch*) entre disponibilidade de dados de traits e de filogenias**
- Dados existem mas não estão acessíveis
- Déficits de conhecimento sobre biodiversidade: Darwiniano e Raunkiaeriano
 - Dificultam ganho de conhecimento sobre evolução de atributos
 - Impedimento para estudos comparativos



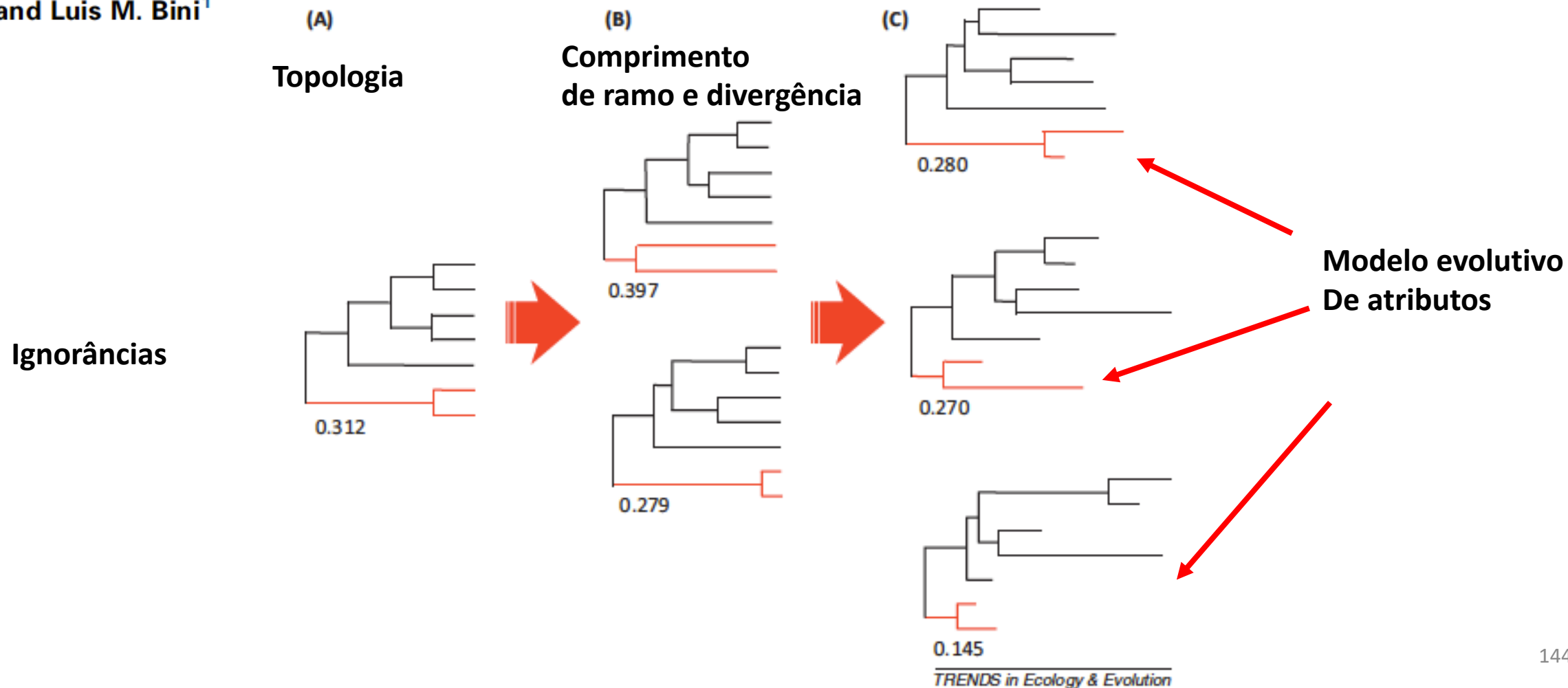
Déficit Raunkiaeriano
Falta de informações sobre
atributos das espécies



Déficit Darwiniano
Falta de informações sobre
relações entre espécies

Darwinian shortfalls in biodiversity conservation

José Alexandre F. Diniz-Filho¹, Rafael D. Loyola¹, Pasquale Raia², Arne O. Mooers³, and Luis M. Bini¹



Meu deus o que eu faço agora?
Meu projeto foi pras cucuias?
Perdi meu doutorado? Quem
poderá me socorrer?!...Oh oh!!



Meu deus
Meu proje
Perdi meu
poderá me

MIGA SUA LOCA



SIACALME

Pacotes phyndr e taxonlookup

Methods in Ecology and Evolution



British Ecological Society

Methods in Ecology and Evolution 2015

doi: 10.1111/2041-210X.12517

APPLICATION

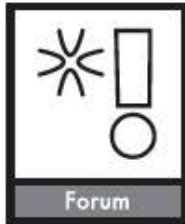
A simple approach for maximizing the overlap of phylogenetic and comparative data

Matthew W. Pennell^{1,2†}, Richard G. FitzJohn^{3†} and William K. Cornwell^{4,5†}

- Maximiza a amostragem trocando espécies na árvore, para os quais não há dados de atributos, por espécies "filogeneticamente equivalentes" sem alterar a estrutura da filogenia
 - Baseado em ranking taxonômico (ou só os nomes dos gêneros) ou uma árvore topológica
 - Alternativa a métodos de aleatorização ou de imputação de dados faltantes

Filling the gap in functional trait databases: use of ecological hypotheses to replace missing data

Simon Taugourdeau^{1,2,3}, Jean Villerd^{1,2}, Sylvain Plantureux^{1,2}, Olivier Huguenin-Elie³ & Bernard Amiaud^{4,5}



EDITOR'S
CHOICE

Ecography 37: 105–110, 2014

doi: 10.1111/j.1600-0587.2013.00528.x

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Subject Editor: Nathan J. Sanders. Accepted 26 July 2013

Phylogenetic imputation of plant functional trait databases

Nathan G. Swenson

Phylogenetic eigenvector maps: a framework to model and predict species traits

Guillaume Guénard^{1*}, Pierre Legendre¹ and Pedro Peres-Neto²



Genetics and Molecular Biology, 38, 3, 396–400 (2015)

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DOI: <http://dx.doi.org/10.1590/S1415-475738320140391>

Short Communication

The best of both worlds: Phylogenetic eigenvector regression and mapping

José Alexandre Felizola Diniz Filho, Fabricio Villalobos and Luis Mauricio Bini

APPLICATION

Rphylopars: fast multivariate phylogenetic comparative methods for missing data and within-species variation

Eric W. Goolsby^{1,2,*}, Jorn Bruggeman³ and Cécile Ané^{4,5}

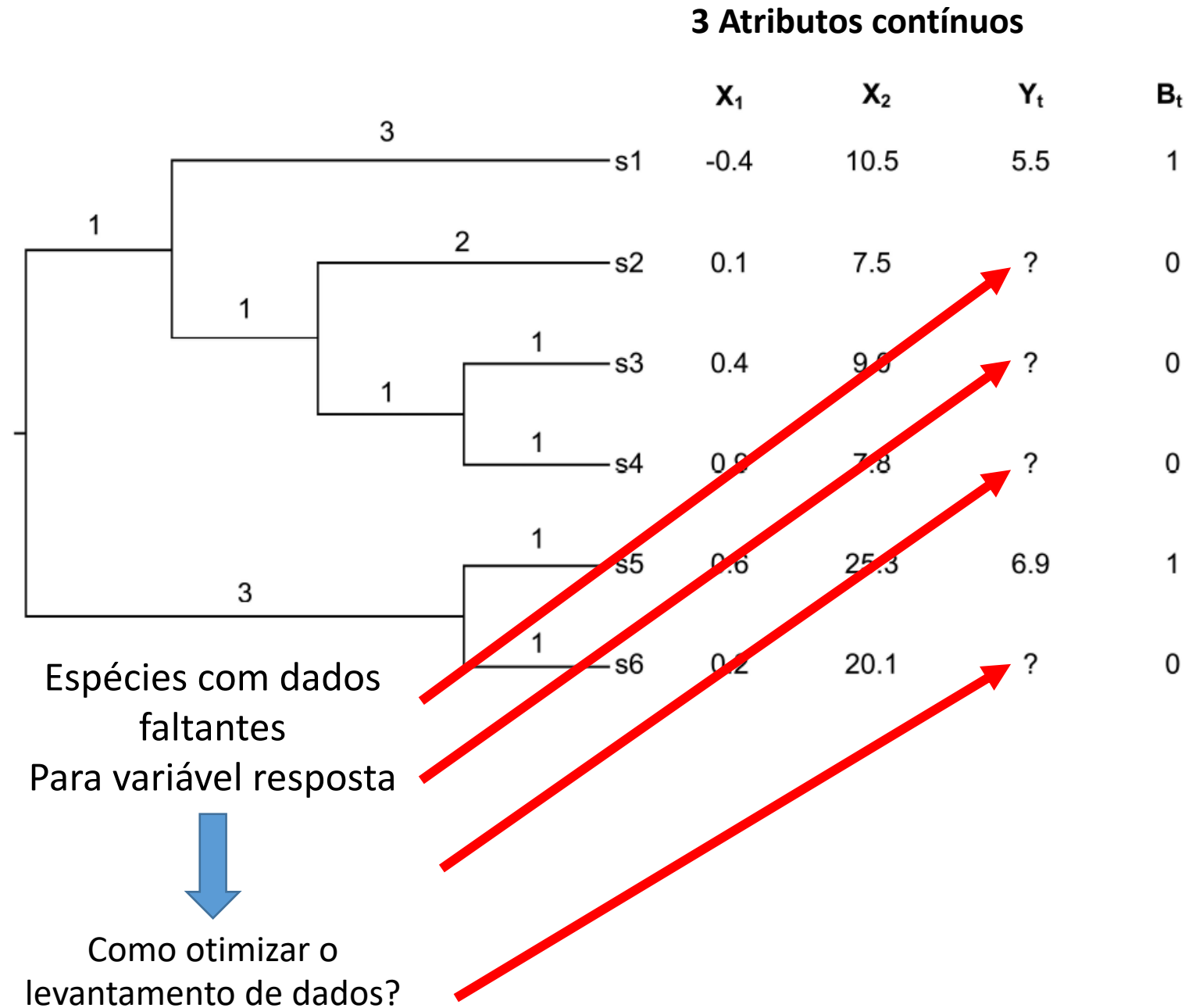
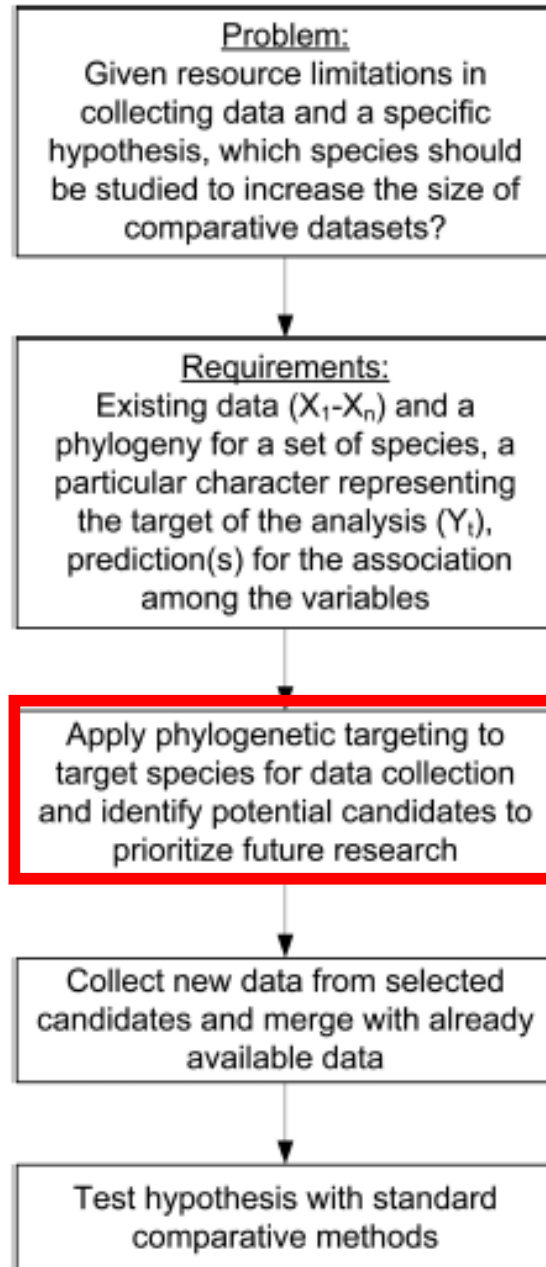
Imputation of missing data in life-history trait datasets: which approach performs the best?

Caterina Penone^{1*}, Ana D. Davidson^{2,3}, Kevin T. Shoemaker², Moreno Di Marco⁴, Carlo Rondinini⁴, Thomas M. Brooks⁵, Bruce E. Young³, Catherine H. Graham² and Gabriel C. Costa¹

Phylogenetic Targeting of Research Effort in Evolutionary Biology

Christian Arnold^{1,2,*} and Charles L. Nunn¹

Workflow PhyloTargeting



Como o Phylotargeting funciona?

- O que é preciso?
 - Identificação de variáveis resposta e preditoras
 - Hipótese
 - Filogenia
- Objetivo: Identificar espécies para as quais devem ser coletados dados da variável resposta (Y_t) usando informação de variáveis preditoras de espécies já estudadas
- Atribui um *score* pra cada par de espécies que incorpora filogenia e valores de variáveis relevantes para testar hipóteses, envolvendo especificamente as variáveis preditoras e resposta
- Espécies com scores altos devem ser preferidas para coleta de novos dados

Como o Phylotargeting funciona?

- Espécies que diferem mais na(s) variável(is) resposta são as preferidas, de maneira a ampliar o *range* de variação nesta variável, aumentando o poder do teste
- Assume que as variáveis estão relacionadas linearmente
- Permite variáveis discretas e contínuas
- Uma ou mais preditoras (covariáveis)
- Mais adequado pra desenhos e perguntas que envolvem modelos lineares (PGLS, correlação, ANOVA, PGLMM etc)

[Home](#)[Phylogenetic Targeting](#)[PhyloTargeting WebServer](#)[Help](#)

Welcome to the PhyloTargeting Website!



This website provides information about the phylogenetic targeting approach as described in **Arnold and Nunn 2010** (see the [Help](#) section for more details). We also provide a web-based implementation.

This project is supported by the NSF (for more details, see the [Project Description](#)).

<http://phyлотargeting.nunn-lab.org/index.html>

Data File Selection

Load data file

Selecionar Arquivo

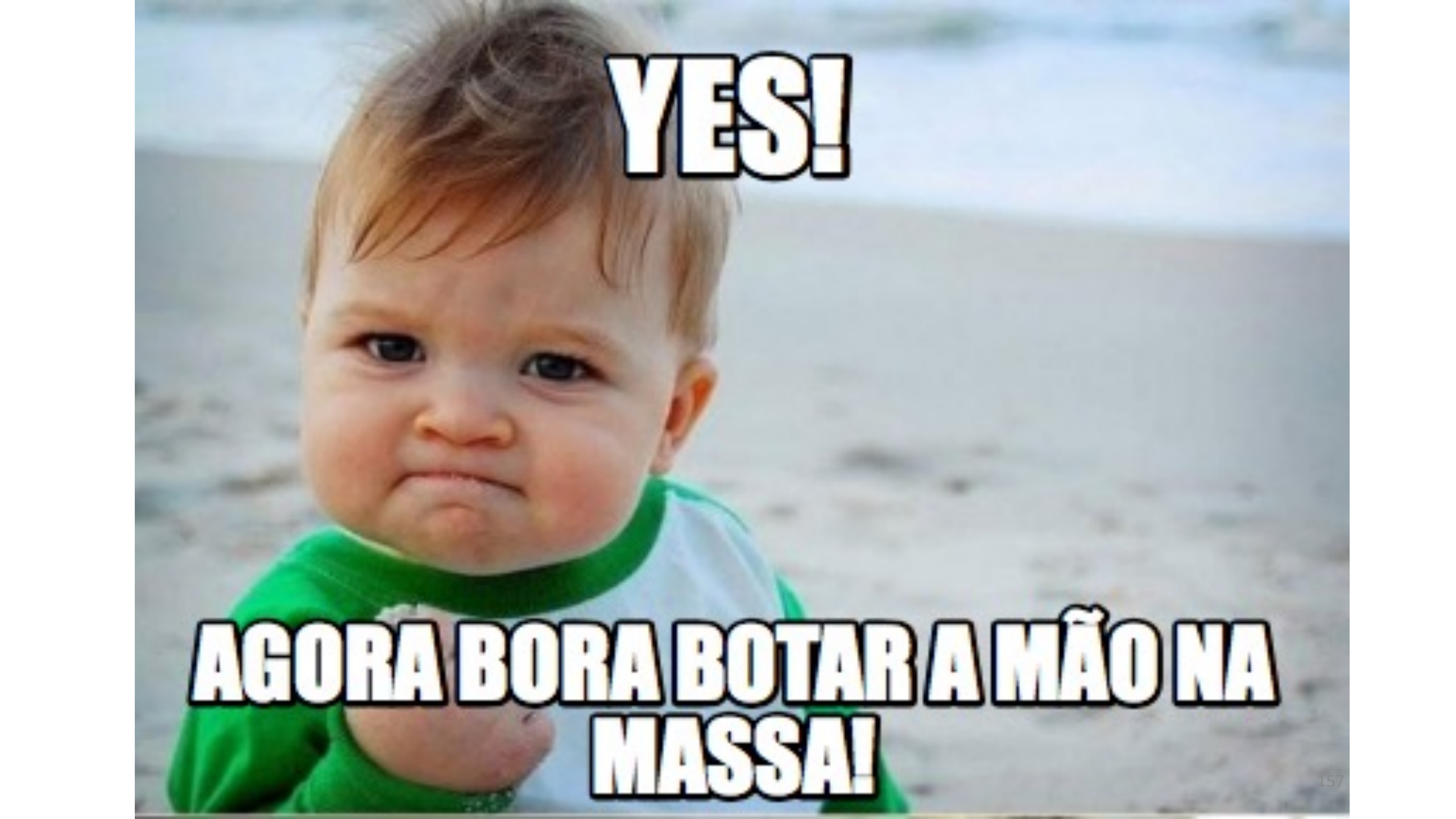
Upload

Test server with example dataset

Load stored session file and continue a previous analysis

Selecionar Arquivo

Load

A close-up photograph of a baby with light brown hair and blue eyes, wearing a green and white shirt. The baby has a determined, slightly pouting expression. The background is a blurred beach with sand and water.

YES!

**AGORA BORA BOTAR A MÃO NA
MASSA!**

Prática

- Introdução ao R
 - Importação e manipulação de dados
 - Importação e manipulação de $\neq$ formatos de filogenias
- Timetree.org
- Phyndr
- Phylotargeting
- rotl
- Rphylopars